L-UTSOI 102.9.0

User’s Manual

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Contents

Section 1

Introduction

L-UTSOI is a compact model dedicated to Fully-Depleted on Silicon-On-Insulator (FDSOI) technologies with low-doped channel, developed at CEA-LETI and previously named Leti-UTSOI [1]. The device architecture described by this model is illustrated in Figure 1.1.

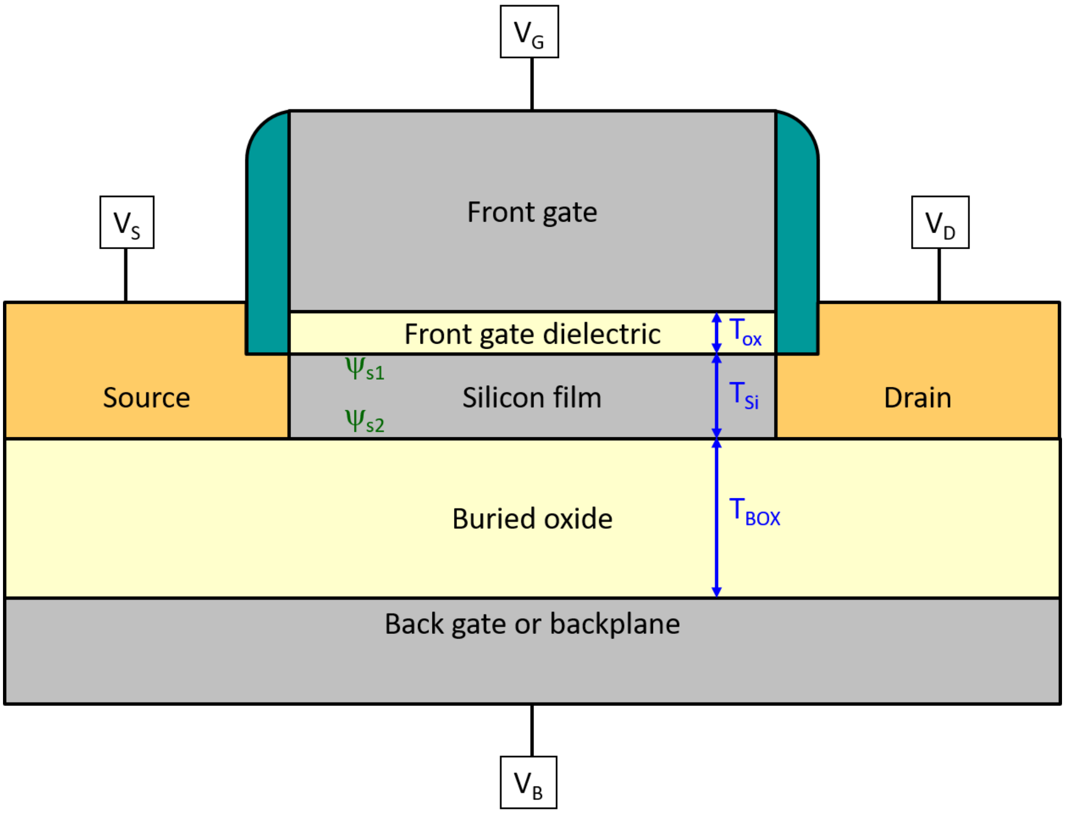

Figure 1.1: Schematic cross-section of a FDSOI transistor, describing the characteristic thicknesses and the surface potentials at the front (ψ_{s1}) and back (ψ_{s2}) interfaces.

1.1 Model structure

As described in Figure 1.2, the model structure is the same as in the different versions of Leti-UTSOI1 [2] [3] and, thus, similar to PSP [4] [5]. It is based on a hierarchical construction featuring two levels of parameter set:

- A local mode, in which the knowledge of the device geometry (channel length and width) is not needed. In this mode, the local parameter values are directly obtained from the model cards.
- A global mode, in which the local parameters are computed from the global model card and the device geometry through scaling laws (see paragraph 3.1). The so-computed local parameters are then used to compute the model equations. In this mode, some local parameters can also be modified according to the stress model (see paragraphs 3.2, 3.3).

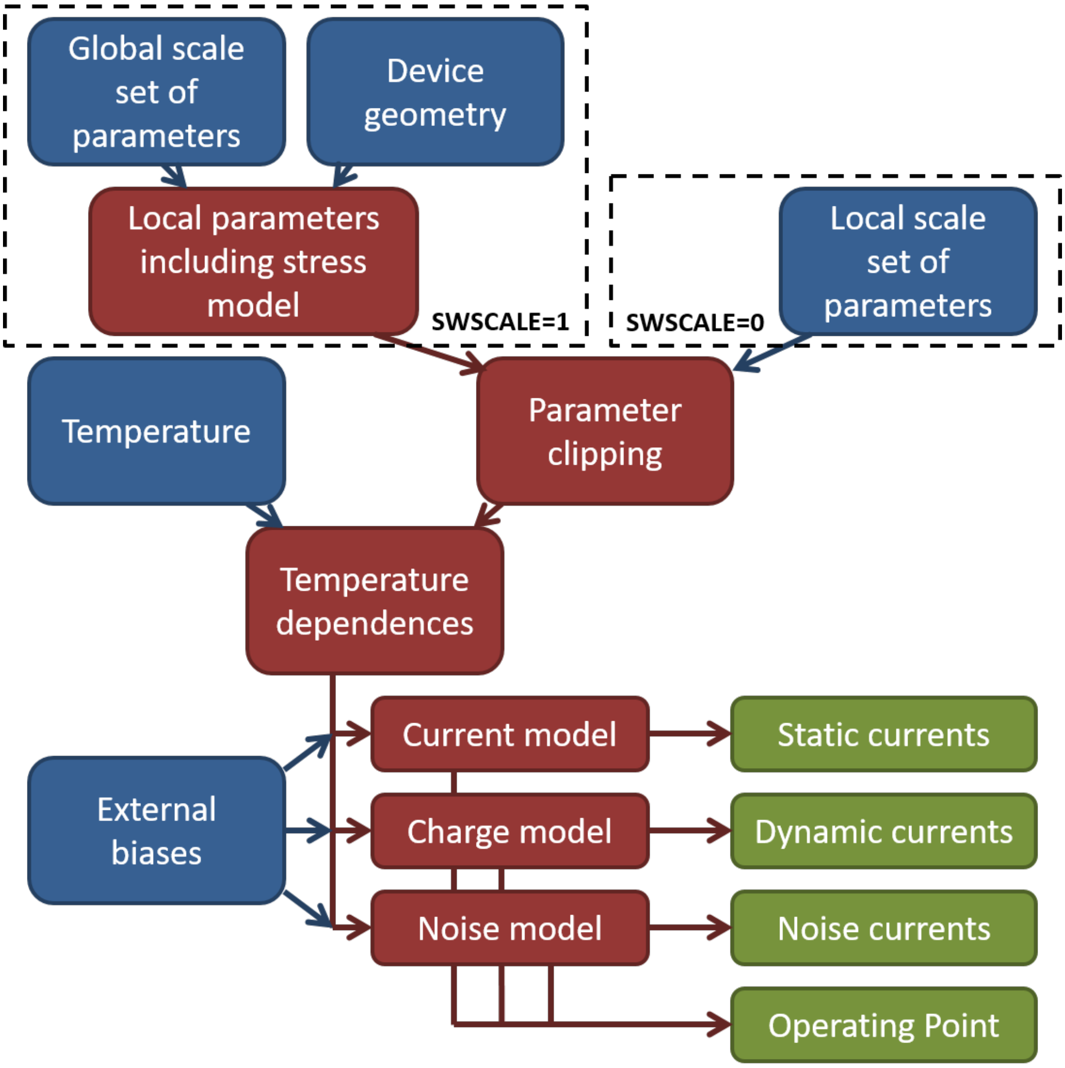

Figure 1.2: Model structure, showing the two scale levels of parameter set: local (**SWSCALE**=0) and global (**SWSCALE**=1).

1.2 Device physics

In terms of device physics, all the features included in Leti-UTSOI1 are also present in L-UTSOI, but L-UTSOI includes some additional ingredients.
First, L-UTSOI has been developed with the aim of being able to describe device operations with a strongly inverted back interface. Thus, an original analytical procedure has been developed to calculate the exact values of the surface potentials at front and back interfaces in all operating conditions (see Appendix A). Besides this surface potential calculation, new drain current (paragraph 4.3) and intrinsic charge (paragraph 4.9) models have also

been developed. It should be noticed that this new model core is valid not only for FDSOI technologies, but more generally for all Independent Double Gate device architectures. An overview of L-UTSOI can be found in [6] and detailed description of the model core in [7] [8] [9].
Second, in L-UTSOI is introduced the effect of backplane depletion (paragraph 4.2.2), computed through a bulk MOSFET like surface potential calculation. This effect can be activated or not through the value of the **SWSUB-DEP** flag. If the flag is set to 0, the backplane is assumed metallic, as in Leti-UTSOI1.
Third, L-UTSOI offers the possibility to define two different junctions at source and drain sides, as done in PSP. Junction related parameters are thus duplicated, the drain related parameters having the same name as their source side counterparts plus a final "D". This possible junction asymmetry can be activated or not thanks to the **SWJU-NASYM** flag. When it is de-activated, drain related parameters are ignored (see paragraph 3.6).
Finally, it should be noticed that the Leti-UTSOI1 possibility to switch from external to internal series resistance through the **SWRSMOD** flag is no longer included in L-UTSOI.
Note that, from version 2.1.0, the temperature node of the transistor (Tnode), that gives channel temperature elevation induced by self-heating, is accessible from the circuit netlist (i.e. declared as "inout"). So, from version 2.1.0, transistor instantiation accepts 5 nodes: drain, gate, source, bulk and internal temperature. Model evolutions from version 2.0.0 are detailed in part 9.

1.3 Output currents

Figure 1.3 describes how the quantities calculated afterwards in this documentation are sent as output currents in the different branches. Static currents are in blue, dynamic currents in red and noise currents in green. It should be noticed that a multiplication parameter $Mult_f$ = **MULT** × **NF** is applied to all the output quantities specified in Figure 1.3.

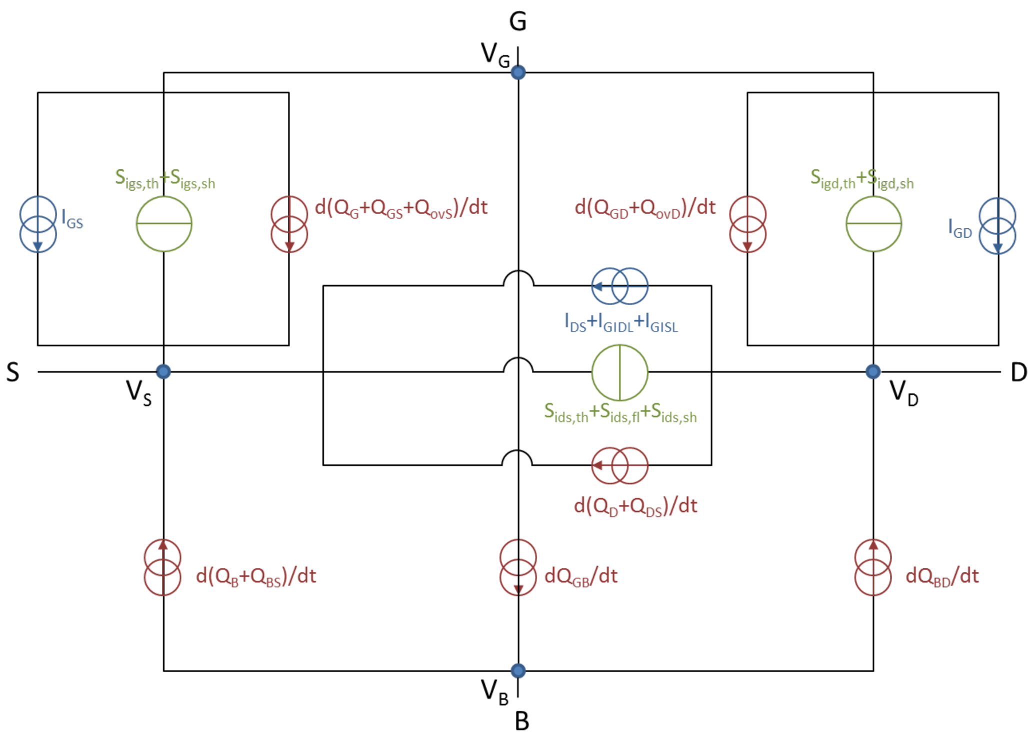

Figure 1.3: Description of the output currents in the device branches. All quantities are multiplied by $Mult_f$.

Section 2

Constants and model parameters

2.1 Constants

The following table details the signification and the values of the constants used in the model.

Symbol	Unit	Description	Value
k_B	J/K	Boltzmann constant	$1.3806488 \cdot 10^{-23}$
$\hbar$	$J.s$	Reduced Planck constant	$1.054571726 \cdot 10^{-34}$
q	C	Elementary unit charge	$1.602176565 \cdot 10^{-19}$
m_0	kg	Electron intrinsic mass	$9.10938291 \cdot 10^{-31}$
ϵ_{ox}	F/m	Permittivity of silicon dioxide (relative value is 3.9)	$3.45313 \cdot 10^{-11}$
ϵ_{Si}	F/m	Permittivity of silicon (relative value is 11.8)	$1.04479 \cdot 10^{-10}$
$E_{g0,Si}$	V	Bandgap voltage for silicon at 0K	1.170
α_{Si}	V/K	First bandgap temperature dependence for silicon	$4.730 \cdot 10^{-4}$
β_{Si}	K	Second bandgap temperature dependence for silicon	636.0
ϵ_{Ge}	F/m	Permittivity of germanium (relative value is 16.2)	$1.43438 \cdot 10^{-10}$
$E_{g0,Ge}$	V	Bandgap voltage for germanium at 0K	0.744
α_{Ge}	V/K	First bandgap temperature dependence for germanium	$4.774 \cdot 10^{-4}$
β_{Ge}	K	Second bandgap temperature dependence for germanium	235.0
C_G	$-$	Non linearity coefficient for SiGe bandgap	-0.4
$n_{i,fact,300}$	m^{-3}	Intrinsic concentration pre-factor for silicon at 300K	$4.05 \cdot 10^{25}$
QM_N	$V^{1/3}nm^{2/3}$	Constant for quantum confinement of electrons	1.27520989
QM_P	$V^{1/3}nm^{2/3}$	Constant for quantum confinement of holes	1.54120870

QM_N and QM_N are equal to $\left(\frac{9\cdot\pi\cdot\hbar}{4\cdot\sqrt{2\cdot q\cdot m_{conf}}}\right)^{2/3}$ with $m_{conf} = 0.918 \cdot m_0$ for electrons and $m_{conf} = 0.52 \cdot m_0$ for holes.

2.2 Model selection, effect activation switches and general parameter management

The table below details the parameters used for model selection, the flags used to activate different physical effects, the selection of the transistor type and the reference temperature. Except for **QMC**, the value of the flags is 0 (effect de-activated) or 1 (effect activated). **QMC** parameter is used to adjust quantum confinement effects and can take other values than 0 or 1.

Name	Unit	Description	Default	Min.	Max.
SWSCALE	–	Scale level: 0 = Local, 1 = Global	1	0	1
VERSION	–	Model version	102.80	–	–
SWSUBDEP	–	Flag for backplane depletion effect	0	0	1
SWIGATE	–	Flag for gate current model	0	0	1
SWGIDL	–	Flag for gate induced source/drain leakage model	0	0	1
SWSHE	–	Flag for self-heating effect	0	0	1
SWIGN	–	Flag for induced gate noise model	1	0	1
SWJUNASYM	–	Flag for source/drain junction asymmetry	0	0	1
SWIMPACT	–	Flag for impact ionization current	0	0	1
SWPDEP *	–	Flag for poly-depletion model	0	0	1
SWCRYO	–	Flag for cryogenic temperature simulation improvement	0	0	1
SWQMOD	–	Flag for separate charge calculation	0	0	1
SWEDGE	–	Flag for drain current of edge transistors	0	0	1
QMC	–	Quantum confinement coefficient	1.0	0.0	–
TYPE	–	Channel type: +1 = NMOS, -1 = PMOS	1	-1	1
TR *or* TREF	$^\circ C$	Temperature of parameter extraction	21.0	-273.0	–
TMAX	$^\circ C$	Maximum self-heating temperature elevation	150.0	0.0	–
DTEMP	K	Device temperature offset	0.0	–	–

* **Note on SWPDEP usage**: L-UTSOI features two gate material modes (in practice a threshold voltage shift appears between **SWPDEP** = 0 and 1):

- When **SWPDEP**=0, a metal gate is considered: Its workfunction is set by the value of **VFB** (= 0 corresponds to a midgap gate, < 0 corresponds to n-type gate and > 0 corresponds to p-type gate).
- When **SWPDEP**=1, a poly-Si gate is considered: The poly-Si doping of the same type as the source/drain (e.g. N-poly for nMOS) (The gate workfunction is consistently computed from the poly doping level **NP** and **VFB** is used to fine tune the threshold voltage).

Warning: Parameter **TYPE** defines the transistor type for the VerilogA version of the model. To define the transistor type when using an implemented version of L-UTSOI in a SPICE simulator, please refer to the corresponding simulator documentation.

2.3 Instance parameters

The following table describes the instance parameters of L-UTSOI.

Name	Unit	Description	Default	Min.	Max.
L	m	Drawn channel length	10^{-6}	10^{-9}	–
W	m	Drawn channel width	10^{-6}	$10^{-9}NF$	–
ASOURCE	m^2	Source region area	10^{-12}	0.0	–
ADRAIN	m^2	Drain region area	10^{-12}	0.0	–

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Name	Unit	Description	Default	Min.	Max.
PSOURCE	m	Source region perimeter	10^{-6}	0.0	–
PDRAIN	m	Drain region perimeter	10^{-6}	0.0	–
SA	m	Distance between active edge and poly at source side	0.0	0.0	–
SB	m	Distance between active edge and poly at drain side	0.0	0.0	–
SD	m	Distance between neighbouring fingers	0.0	0.0	–
NF	–	Number of fingers	1	1	–
MULT	–	Number of devices in parallel	1	0	–
MULT_I	–	Currents multiplication factor	1	0	–
MULT_Q	–	Charges multiplication factor	1	0	–
MULT_FN	–	Flicker noise multiplication factor	$MULT\_I$	0	–
DELVTO	V	Threshold voltage shift parameter	0.0	–	–
FACTUO	–	Low field mobility pre-factor	1.0	0.0	–
NGCON	–	Number of gate contacts	1	1	2
XGW	m	Distance from the gate contact to the channel edge	10^{-7}	–	–
NRS	–	Number of squares of source diffusion	0.0	–	–
NRD	–	Number of squares of drain diffusion	0.0	–	–

2.4 Scaling parameters

In the following table are given the geometrical parameters that link drawn, physical and electrical transistor dimensions.

Name	Unit	Description	Default	Min.	Max.
LVARO	m	Long channel difference between physical and drawn gate lengths	0.0	–	–
LVARL	–	Length dependence of physical to drawn gate length difference	0.0	–	–
LVARW	–	Width dependence of physical to drawn gate length difference	0.0	–	–
LAP	m	Effective channel length reduction per side	0.0	–	–
WVARO	m	Wide channel difference between physical and drawn active width	0.0	–	–
WVARL	–	Length dependence of physical to drawn active width difference	0.0	–	–
WVARW	–	Width dependence of physical to drawn active width difference	0.0	–	–
WOT	m	Effective channel width reduction per side	0.0	–	–
DLQ	m	Effective channel length additional offset for charge model	0.0	–	–
DWQ	m	Effective channel width additional offset for charge model	0.0	–	–

2.5 Stress model parameters

Besides the existing STI-based stress model, a new stress model dedicated to strained-FDSOI technologies has been introduced in L-UTSOI from version 2.2.0. Selection between the two models is done through **SWSTRESS** flag: 1 for STI-based model, 2 for strained-SOI model. Stress effect can be de-activated by setting the flag to 0.

Name	Unit	Description	Default	Min.	Max.
SWSTRESS	–	Stress model selection flag	1	0	2
SAREF	m	Reference distance between active edge and poly from one side	10^{-6}	10^{-9}	–
SBREF	m	Reference distance between active edge and poly from other side	10^{-6}	10^{-9}	–

2.5.1 Parameters for SWSTRESS=1

The next table describes the parameters used for the STI-stress model. This model comes originally from BSIM4.4 and has been slightly modified in PSP.

Name	Unit	Description	Default	Min.	Max.
WLOD	m	Width parameter	0.0	–	–
KUO	m	Mobility degradation/enhancement coefficient	0.0	–	–
KVSAT	–	Saturation velocity degradation/enhancement coefficient	0.0	-1.0	1.0
TKUO	–	Temperature dependence of KUO	0.0	–	–
LKUO	$m^{LLODKUO}$	Length dependence of KUO	0.0	–	–
WKUO	$m^{WLODKUO}$	Width dependence of KUO	0.0	–	–
PKUO	$m^{sumLODKUO}$	Cross-term dependence of KUO	0.0	–	–
LLODKUO	–	Length parameter for mobility stress effect	0.0	0.0	–
WLODKUO	–	Width parameter for mobility stress effect	0.0	0.0	–
KVTHO	Vm	Threshold voltage shift parameter	0.0	–	–
LKVTHO	$m^{LLODVTH}$	Length dependence of KVTHO	0.0	–	–
WKVTHO	$m^{WLODVTH}$	Width dependence of KVTHO	0.0	–	–
PKVTHO	$m^{sumLODVTH}$	Cross-term dependence of KVTHO	0.0	–	–
LLODVTH	–	Length parameter for threshold voltage stress effect	0.0	0.0	–
WLODVTH	–	Width parameter for threshold voltage stress effect	0.0	0.0	–
STETAO	m	ETAO shift factor related to threshold voltage change	0.0	–	–
LODETAO	–	ETAO shift modification factor	1.0	0.0	–

Note: sumLODKUO and sumLODVTH refer to LLODKUO+WLODKUO and LLODVTH+WLODVTH respectively.

2.5.2 Parameters for SWSTRESS=2

The next table describes the parameters used for the strained-SOI dedicated model.

Name	Unit	Description	Default	Min.	Max.
STRLAMBDA	m	Strain relaxation characteristic length	10^{-7}	10^{-9}	10^{-5}
STRALPHA	–	Strain relaxation asymmetry parameter	3.0	0.5	–
STRDVFBO	V	Threshold shift parameter	0.0	–	–
STRWDVFBO	–	Width dependence of threshold shift parameter	0.0	–	–
STRDCFL	–	DIBL variation parameter	0.0	–	–
STRRUO	–	Mobility degradation/enhancement coefficient	0.0	–	–
STRTRUO	–	Temperature dependence of STRRUO	0.0	–	–
STRRVSAT	–	Saturation velocity degradation/enhancement coefficient	0.0	–	–

2.6 Process parameters

In the following tables are given all the local model parameters, in bold. Each local parameter is immediately followed by its related global scale parameters, in italic.

Name	Unit	Description	Default	Min.	Max.
TOXE	m	Front gate equivalent oxide thickness (EOT)	$2\cdot10^{-9}$	$3\cdot10^{-10}$	10^{-6}
TOXEO	m	*Geometry independent global scale parameter for TOXE*	$2\cdot10^{-9}$	$3\cdot10^{-10}$	10^{-6}
TSI	m	Silicon or SiGe film thickness	10^{-8}	$3\cdot10^{-9}$	$2\cdot10^{-8}$
TSIO	m	*Geometry independent global scale parameter for TSI*	10^{-8}	$3\cdot10^{-9}$	$2\cdot10^{-8}$
XGE	–	Fraction of germanium content in the channel	0.0	0.0	1.0
XGEO	–	*Geometry independent global scale parameter for XGE*	0.0	0.0	1.0
TBOX	m	Back gate equivalent oxide thickness (EOT)	10^{-7}	$3\cdot10^{-10}$	10^{-6}
TBOXO	m	*Geometry independent global scale parameter for TBOX*	10^{-7}	$3\cdot10^{-10}$	10^{-6}
NCH	cm^{-3}	Thin film doping: positive = p-type, negative = n-type	0.0	0.0	10^{19}
NCHO	cm^{-3}	*Geometry independent global scale parameter for NCH*	0.0	0.0	10^{19}
NSUB	cm^{-3}	Backplane doping level: positive = p-type, negative = n-type	$3\cdot10^{18}$	10^{16}	10^{21}
NSUBO	cm^{-3}	*Geometry independent global scale parameter for NSUB*	$3\cdot10^{18}$	10^{16}	10^{21}
CT	–	Interface states factor	0.0	0.0	–

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Name	Unit	Description	Default	Min.	Max.
CTO	–	*Geometry independent global scale parameter for CT*	0.0	0.0	–
TOXP	m	Front gate physical oxide thickness	$2 \cdot 10^{-9}$	$3 \cdot 10^{-10}$	10^{-6}
TOXPO	m	*Geometry independent global scale parameter for TOXP*	$2 \cdot 10^{-9}$	$3 \cdot 10^{-10}$	10^{-6}
NOV	cm^{-3}	Effective doping level of overlap-LDD regions	10^{20}	10^{15}	10^{21}
NOVO	cm^{-3}	*Geometry independent global scale parameter for NOV*	10^{20}	10^{15}	10^{21}
NOVD	cm^{-3}	Effective doping level of overlap-LDD regions at drain side	10^{20}	10^{15}	10^{21}
NOVDO	cm^{-3}	*Geometry independent global scale parameter for NOVD*	10^{20}	10^{15}	10^{21}
VFB	V	Front gate workfunction referenced to Si midgap at TR	0.0	–	–
VFBO	V	*Long and wide channel value of VFB*	0.0	–	–
VFBL	V	*Channel length scaling parameter of VFB*	0.0	–	–
VFBLEXP	–	*Channel length scaling exponent of VFB*	2.0	–	–
VFBL2	–	*Second order channel length dependence of VFB*	0.0	–	–
VFBLEXP2	–	*Second order channel length scaling exponent of VFB*	2.0	–	–
VFBW	V	*Channel width scaling parameter of VFB*	0.0	–	–
VFBLW	V	*Channel area scaling parameter of VFB*	0.0	–	–
VFBB	V	Back gate workfunction offset at TR	0.0	–	–
VFBBO	V	*Long and wide channel value of VFBB*	0.0	–	–
VFBLBO	–	*Back to front interface asymmetry factor applied to VFBL*	0.0	–	–
STVFB	V/K	Temperature dependence of VFB and VFBB	0.0	–	–
STVFBO	V/K	*Long and wide channel value of STVFB*	0.0	–	–
STVFBL	–	*Channel length scaling parameter of STVFB*	0.0	–	–
STVFBW	–	*Channel width scaling parameter of STVFB*	0.0	–	–
STVFBLW	–	*Channel area scaling parameter of STVFB*	0.0	–	–
NP	cm^{-3}	Gate poly-silicon doping	10^{21}	10^{19}	10^{22}
NPO	cm^{-3}	*Geometry-independent gate poly-silicon doping*	10^{21}	–	–
NPL	–	*Length dependence of gate poly-silicon doping*	0.0	–	–

2.7 Gate to interface coupling parameters

Name	Unit	Description	Default	Min.	Max.
CICF	–	Long channel front interface coupling coefficient	1.0	0.1	10.0
CICFO	–	*Geometry independent global scale parameter for CICF*	1.0	0.1	10.0
CIC	–	Long channel back interface coupling coefficient	1.0	0.1	10.0
CICO	–	*Geometry independent global scale parameter for CIC*	1.0	0.1	10.0
PSCE	–	Short channel coupling attenuation parameter	0.0	0.0	5.0
PSCEL	–	*Channel length scaling parameter of PSCE*	0.0	–	–
PSCELEXP	–	*Channel length scaling exponent of PSCE*	2.0	–	–
PSCEW	–	*Channel width scaling parameter of PSCE*	0.0	–	–
PSCEB	–	Short channel back to front interface asymmetry factor	1.0	0.0	–
PSCEBO	–	*Geometry independent global scale parameter for PSCEB*	1.0	0.0	–
NSDDC	cm^{-3}	Source/drain effective doping level for DC model	10^{22}	10^{18}	10^{22}
NSDDCO	cm^{-3}	*Geometry independent global scale parameter for NSDDC*	10^{22}	10^{18}	10^{22}
PSCEDLB	–	Back bias dependence of short channel effect modulation	0.0	0.0	–
PSCEDLBO	–	*Geometry independent global scale parameter for PSCEDLB*	0.0	0.0	–
PNCE	–	Narrow channel effect on body factor	0.0	-1.0	1.0
PNCEW	–	*Channel width scaling parameter of PNCE*	0.0	–	–

2.8 Drain Induced Barrier Lowering parameters

Name	Unit	Description	Default	Min.	Max.
CF	–	DIBL parameter at TR	0.0	0.0	–
CFL	–	*Channel length scaling parameter of CF*	0.0	–	–
CFLEXP	–	*Channel length scaling exponent of CF*	2.0	–	–
CFW	–	*Channel width scaling parameter of CF*	0.0	–	–
CFB	–	DIBL back to front interface asymmetry factor	1.0	0.0	–
CFBO	–	*Geometry independent global scale parameter for CFB*	1.0	0.0	–
STCF	K^{-1}	Temperature dependence of CF	0.0	–	–
STCFL	K^{-1}	*Channel length scaling parameter for STCF*	0.0	–	–
CFD	V	Drain voltage dependence parameter of DIBL	0.2	0.05	–

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Name	Unit	Description	Default	Min.	Max.
CFDO	V	*Geometry independent global scale parameter for CFD*	0.2	0.05	–
CFDL	–	DIBL modulation due to Leff dependence on biases	0.0	–	–
CFDLL	–	*Channel length scaling parameter of CFDL*	0.0	–	–
CFDLW	–	*Channel width scaling parameter of CFDL*	0.0	–	–
CFDLB	–	Back bias dependence of DIBL modulation	0.0	0.0	–
CFDLBO	–	*Geometry independent global scale parameter for CFDLB*	0.0	0.0	–

2.9 Mobility parameters

Name	Unit	Description	Default	Min.	Max.
BETN	$m^2/V/s$	Front channel aspect ratio times low field mobility at TR	0.05	10^{-10}	–
UO	$m^2/V/s$	*Front channel low field mobility at TR*	0.05	0.0	–
FBET1	–	*First length dependence modulation of BETN*	0.0	–	–
FBET1W	–	*Width dependence of FBET1*	0.0	–	–
LP1	m	*First characteristic length of BETN scaling*	10^{-8}	10^{-10}	–
LP1W	–	*Width dependence of LP1*	0.0	–	–
FBET2	–	*Second length dependence modulation of BETN*	0.0	–	–
LP2	m	*Second characteristic length of BETN scaling*	10^{-8}	10^{-10}	–
BETW1	–	*First width dependence modulation of BETN*	0.0	–	–
BETW2	–	*Second width dependence modulation of BETN*	0.0	–	–
WBET	m	*Characteristic width of BETN scaling*	10^{-8}	10^{-10}	–
BETNB	–	Back channel over front channel low field mobility ratio	1.0	0.1	10.0
BETNBO	–	*Geometry independent global scale parameter for BETNB*	1.0	0.1	10.0
STBET	–	Temperature dependence exponent of BETN	1.5	–	–
STBETO	–	*Long and wide channel value of STBET*	1.5	–	–
STBETL	–	*Channel length scaling parameter of STBET*	0.0	–	–
STBETW	–	*Active width scaling parameter of STBET*	0.0	–	–
STBETLW	–	*Active area scaling parameter of STBET*	0.0	–	–
CS	–	Coulomb scattering parameter at TR	0.0	0.0	–
CSO	–	*Long and wide channel value of CS*	0.0	–	–
CSL	–	*Channel length scaling parameter of CS*	0.0	–	–
CSLEXP	–	*Channel length scaling exponent of CS*	1.0	–	–
CSW	–	*Channel width scaling parameter of CS*	0.0	–	–

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Name	Unit	Description	Default	Min.	Max.
CSLW	–	*Channel area scaling parameter of CS*	0.0	–	–
CSFI	–	Field dependence of Coulomb scattering at front interface	0.0	0.0	–
CSFIO	–	*Geometry independent global scale parameter for CSFI*	0.0	0.0	–
CSBI	–	Field dependence of Coulomb scattering at back interface	0.0	0.0	–
CSBIO	–	*Geometry independent global scale parameter for CSBI*	0.0	0.0	–
STCS	–	Temperature dependence exponent of CS	0.0	–	–
STCSO	–	*Long and wide channel value of STCS*	0.0	–	–
STCSL	–	*Channel length scaling parameter of STCS*	0.0	–	–
STCSW	–	*Channel width scaling parameter of STCS*	0.0	–	–
STCSLW	–	*Channel area scaling parameter of STCS*	0.0	–	–
THECS	–	Coulomb scattering exponent at TR	1.5	0.0	–
THECSO	–	*Geometry independent global scale parameter for THECS*	1.5	0.0	–
STTHECS	–	Temperature dependence exponent of THECS	0.0	–	–
STTHECSO	–	*Geometry independent global scale parameter for STTHECS*	0.0	–	–
CSTHR	–	Coulomb scattering threshold level	2.0	0.001	–
CSTHRO	–	*Geometry independent global scale parameter for CSTHR*	2.0	0.001	–
CSTHRB	–	Coulomb scattering threshold asymmetry parameter	1.0	0.1	–
CSTHRBO	–	*Geometry independent global scale parameter for CSTHRB*	1.0	0.1	–
MUE	cm/MV	High field mobility reduction coefficient at TR	0.0	0.0	–
MUEO	cm/MV	*Geometry independent global scale parameter for MUE*	0.0	0.0	–
STMUE	–	Temperature dependence exponent of MUE	0.0	–	–
STMUEO	–	*Geometry independent global scale parameter for STMUE*	0.0	–	–
THEMU	–	High field mobility reduction exponent at TR	1.5	0.0	–
THEMUO	–	*Geometry independent global scale parameter for THEMU*	1.5	0.0	–
STTHEMU	–	Temperature dependence exponent of THEMU	0.0	–	–
STTHEMUO	–	*Geometry independent global scale parameter for STTHEMU*	0.0	–	–

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Name	Unit	Description	Default	Min.	Max.
XCOR	–	High field mobility non universality factor at TR	0.0	–	–
XCORO	–	*Long and wide channel value of XCOR*	0.0	–	–
XCORL	–	*Channel length scaling parameter of XCOR*	0.0	–	–
XCORLEXP	–	*Channel length scaling exponent of XCOR*	1.0	–	–
XCORW	–	*Channel width scaling parameter of XCOR*	0.0	–	–
XCORLW	–	*Channel area scaling parameter of XCOR*	0.0	–	–
XCORB	–	Asymmetry term of non-universality factor	1.0	–	–
XCORBO	–	*Geometry independent global scale parameter for XCORB*	1.0	–	–
STXCOR	–	Temperature dependence exponent of XCOR	0.0	–	–
STXCORO	–	*Geometry independent global scale parameter for STXCOR*	0.0	–	–
FETA	–	Transverse effective field parameter	1.0	0.0	–
FETAO	–	*Geometry independent global scale parameter for FETA*	1.0	0.0	–

2.10 Series resistance parameters

Name	Unit	Description	Default	Min.	Max.
RS	Ω	Source/drain series resistance at TR	30.0	0.0	–
RSW1	Ω	*Source/drain series resistance for a WEN width at TR*	30.0	–	–
RSW2	–	*Second order width scaling parameter of RS*	0.0	–	–
RSIG	–	Source/drain extension resistance coefficient	0.0	0.0	–
RSIGO	–	*Geometry independent global scale parameter for RSIG*	0.0	0.0	–
STRS	–	Temperature dependence exponent of RS	0.0	–	–
STRSO	–	*Geometry independent global scale parameter for STRS*	0.0	–	–
RSG	–	Transverse electric field dependence of RS	0.0	−0.5	–
RSGO	–	*Geometry independent global scale parameter for RSG*	0.0	−0.5	–
RSB	–	Back bias dependence of RS	0.0	–	–
RSBO	–	*Geometry independent global scale parameter for RSB*	0.0	–	–

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Name	Unit	Description	Default	Min.	Max.
THERSG	–	Transverse electric field dependence exponent of RS	2.0	–	–
THERSGO	–	*Geometry independent global scale parameter for THERSG*	2.0	–	–

2.11 Velocity saturation parameters

Name	Unit	Description	Default	Min.	Max.
THESAT	V^{-1}	Velocity saturation parameter at TR	0.0	–	–
THESATO	s/m^2	*Long and wide channel parameter for THESAT*	0.0	–	–
THESATL	s/m^2	*Channel length scaling parameter of THESAT*	0.0	–	–
THESATLEXP	–	*Channel length scaling exponent of THESAT*	1.0	–	–
THESATW	–	*Channel width scaling parameter of THESAT*	0.0	–	–
THESATLW	–	*Channel area scaling parameter of THESAT*	0.0	–	–
STTHESAT	–	Temperature dependence exponent of THESAT	-0.1	–	–
STTHESATO	–	*Long and wide channel parameter for STTHESAT*	-0.1	–	–
STTHESATL	–	*Channel length scaling parameter of STTHESAT*	0.0	–	–
STTHESATW	–	*Channel width scaling parameter of STTHESAT*	0.0	–	–
STTHESATLW	–	*Channel area scaling parameter of STTHESAT*	0.0	–	–
THESATG	–	Front gate bias dependence of velocity saturation	0.0	-0.5	–
THESATGO	–	*Geometry independent global scale parameter for THESATG*	0.0	-0.5	–
THESATB	–	Back gate bias dependence of velocity saturation	0.0	-0.5	–
THESATBO	–	*Geometry independent global scale parameter for THESATB*	0.0	-0.5	–

2.12 Saturation and Channel Length Modulation parameters

Name	Unit	Description	Default	Min.	Max.
AX	–	Linear/saturation transition exponent	8.0	1.0	16.0
AXO	–	*Long and wide channel value of AX*	8.0	–	–
AXL	–	*Channel length scaling parameter of AX*	0.0	–	–
AXLEXP	–	*Channel length scaling exponent of AX*	1.0	–	–
AXL2	–	*Second order channel length scaling parameter of AX*	0.0	–	–
AXLEXP2	–	*Second order channel length scaling exponent of AX*	1.5	–	–

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Name	Unit	Description	Default	Min.	Max.
ALP	–	Channel length modulation pre-factor	0.0	0.0	–
ALPL1	–	*Channel length scaling parameter of ALP*	0.0	–	–
ALPLEXP	–	*Channel length scaling exponent of ALP*	1.0	–	–
ALPL2	–	*Second order channel length dependence of ALP*	0.0	0.0	–
ALPLEXP2	–	*Second order channel length scaling exponent of ALP*	2.0	–	–
ALPW	–	*Channel width scaling parameter of ALP*	0.0	–	–
ALP1	*V*	Channel length modulation enhancement above threshold	0.0	0.0	–
ALP1L1	*V*	*Channel length scaling parameter of ALP1*	0.0	–	–
ALP1LEXP	–	*Channel length scaling exponent of ALP1*	0.5	–	–
ALP1L2	–	*Second order channel length dependence of ALP1*	0.0	0.0	–
ALP1LEXP2	–	*Second order channel length scaling exponent of ALP1*	1.5	–	–
ALP1W	–	*Channel width scaling parameter of ALP1*	0.0	–	–
ALPB	–	Back bias dependence of channel length modulation	0.0	–	–
ALPBO	–	*Geometry independent global scale parameter for ALPB*	0.0	–	–
VP	*V*	Channel length modulation logarithm dependence factor	0.05	10^{-10}	–
VPO	*V*	*Geometry independent global scale parameter for VP*	0.05	10^{-10}	–
VPG	–	Tranverse field dependence of CLM logarithm factor	0.0	0.0	–
VPGO	–	*Geometry independent global scale parameter for VPG*	0.0	0.0	–

2.13 Gate current parameters

Name	Unit	Description	Default	Min.	Max.
GCO	–	Gate tunneling energy adjustment in inversion mode	0.0	−10.0	10.0
GCOO	–	*Geometry independent global scale parameter for GCO*	0.0	−10.0	10.0
IGINV	*A*	Gate to channel current pre-factor	0.0	0.0	–
IGINVLW	*A*	*IGINV value for a LEN.WEN area*	0.0	0.0	–
IGOVINV	*A*	Gate-overlap current pre-factor in inversion	0.0	0.0	–

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Name	Unit	Description	Default	Min.	Max.
IGOVINVW	*A*	*IGOVINV value for a WEN width*	0.0	0.0	–
IGOVINVD	*A*	Gate-overlap current pre-factor in inversion at drain side	0.0	0.0	–
IGOVINVDW	*A*	*IGOVINVD for a WEN width*	0.0	0.0	–
IGOVACC	*A*	Gate-overlap current pre-factor in accumulation	0.0	0.0	–
IGOVACCW	*A*	*IGOVACC value for a WEN width*	0.0	0.0	–
IGOVACCD	*A*	Gate-overlap current pre-factor in accumulation at drain side	0.0	0.0	–
IGOVACCDW	*A*	*IGOVACCD value for a WEN width*	0.0	0.0	–
STIG	–	Temperature dependence of all gate current pre-factors	0.0	–	–
STIGO	–	*Geometry independent global scale parameter for STIG*	0.0	–	–
GC2CH	–	Gate to channel current slope factor	0.375	0.0	10.0
GC2CHO	–	*Geometry independent global scale parameter for GC2CH*	0.375	0.0	10.0
GC3CH	–	Gate to channel current curvature factor	0.063	−2.0	2.0
GC3CHO	–	*Geometry independent global scale parameter for GC3CH*	0.063	−2.0	2.0
GC2OVINV	–	Gate-overlap current slope factor in inversion	0.375	0.0	10.0
GC2OVINVO	–	*Geometry independent global scale parameter for GC2OVINV*	0.375	0.0	10.0
GC3OVINV	–	Gate-overlap current curvature factor in inversion	0.063	−2.0	2.0
GC3OVINVO	–	*Geometry independent global scale parameter for GC3OVINV*	0.063	−2.0	2.0
GC2OVACC	–	Gate-overlap current slope factor in accumulation	0.375	0.0	10.0
GC2OVACCO	–	*Geometry independent global scale parameter for GC2OVACC*	0.375	0.0	10.0
GC3OVACC	–	Gate-overlap current curvature factor in accumulation	0.063	−2.0	2.0
GC3OVACCO	–	*Geometry independent global scale parameter for GC3OVACC*	0.063	−2.0	2.0
GCDOV	V^{-1}	High drain voltage dependence of overlap gate current	0.0	–	–
GCDOVL	V^{-1}	*GCDOV value for a LEN length*	0.0	–	–
GCVDOV	*V*	Threshold of high drain voltage effect on overlap gate current	1.0	–	–

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Name	Unit	Description	Default	Min.	Max.
GCVDOVO	*V*	*Geometry independent global scale parameter for GCVDOV*	1.0	–	–
CHIB	V	Tunneling barrier height	3.1	1.0	–
CHIBO	*V*	*Geometry independent global scale parameter for CHIB*	3.1	1.0	–
NIGINV	–	Gate tunneling slope adjustment in subthreshold regime	0.0	0.0	–
NIGINVO	–	*Geometry independent global scale parameter for NIGINV*	0.0	0.0	–
FNOVINV	A	Extra gate to overlap current pre-factor in inversion	0.0	0.0	–
FNOVINVW	*A*	*FNOVINV for a WEN width*	0.0	0.0	–
FNOVINVD	A	Extra gate to overlap current pre-factor in inversion at drain side	0.0	0.0	–
FNOVINVDW	*A*	*FNOVINVD for a WEN width*	0.0	0.0	–
GCOVINVFN	–	Extra gate current slope factor for overlap regions in inversion mode	0.2	0.1	10.0
GCOVINVFNO	–	*Geometry independent global scale parameter for GCOVINVFN*	0.2	0.1	10.0
STIGFN	–	Temperature dependence of extra gate to overlap current	0.0	–	–
STIGFNO	–	*Geometry independent global scale parameter for STIGFN*	0.0	–	–

2.14 Gate Induced Drain/Source Leakage (GIDL/GISL) parameters

Name	Unit	Description	Default	Min.	Max.
AGIDL	A/V^3	GIDL/GISL current pre-factor	0.0	0.0	–
AGIDLO	A/V^3	*Geometry independent global scale parameter for AGIDL*	0.0	–	–
AGIDLW	A/V^3	*Channel width scaling parameter of AGIDL*	0.0	–	–
AGIDLD	A/V^3	GIDL current pre-factor at drain side	0.0	0.0	–
AGIDLDO	A/V^3	*Geometry independent global scale parameter for AGIDLD*	0.0	–	–
AGIDLDW	A/V^3	*Channel width scaling parameter of AGIDLD*	0.0	–	–
BGIDL	V	GIDL/GISL probability factor at TR	41	0.0	–
BGIDLO	*V*	*Geometry independent global scale parameter for BGIDL*	41	0.0	–
BGIDLD	V	GIDL probability factor at TR at drain side	41	0.0	–

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Name	Unit	Description	Default	Min.	Max.
BGIDLDO	V	*Geometry independent global scale parameter for BGIDLD*	41	0.0	–
STBGIDL	V/K	Temperature dependence of BGIDL	0.0	–	–
STBGIDLO	V/K	*Geometry independent global scale parameter for STBGIDL*	0.0	–	–
STBGIDLD	V/K	Temperature dependence of BGIDLD	0.0	–	–
STBGIDLDO	V/K	*Geometry independent global scale parameter for STBGIDLD*	0.0	–	–
CGIDL	V^{-1}	Substrate bias dependence of GIDL/GISL	0.0	–	–
CGIDLO	V^{-1}	*Geometry independent global scale parameter for CGIDL*	0.0	–	–
CGIDLD	V^{-1}	Substrate bias dependence of GIDL at drain side	0.0	–	–
CGIDLDO	V^{-1}	*Geometry independent global scale parameter for CGIDLD*	0.0	–	–
DGIDL	V^{-1}	High longitudinal field dependence of GIDL/GISL	0.0	–	–
DGIDLO	V^{-1}	*Geometry independent global scale parameter for DGIDL*	0.0	–	–
DGIDLL	V^{-1}	*Channel length scaling parameter of DGIDL*	0.0	–	–
DGIDLD	V^{-1}	High longitudinal field dependence of GIDL at drain side	0.0	–	–
DGIDLDO	V^{-1}	*Geometry independent global scale parameter for DGIDLD*	0.0	–	–
DGIDLDL	V^{-1}	*Channel length scaling parameter of DGIDLD*	0.0	–	–

2.15 Edge transistor parameters

Name	Unit	Description	Default	Min.	Max.
WEDGE	m	*Electrical width of edge transistor per side*	10^{-8}	0	–
WEDGEW	m	*Main transistor width dependence of WEDGE*	0	0	–
CTEDGE	–	Interface states factor of edge transistors	0.0	0.0	–
CTEDGEO	–	*Geometry independent global scale parameter for CTEDGE*	0.0	0.0	–
VFBEDGE	V	Flat-band voltage of the front gate of edge transistors at TR	0.0	–	–
VFBEDGEO	V	*Long and wide channel value of VFBEDGE*	0.0	–	–
VFBEDGEL	V	*Channel length scaling parameter of VFBEDGE*	0.0	–	–
VFBEDGELEXP	–	*Channel length scaling exponent of VFBEDGE*	2.0	–	–

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Name	Unit	Description	Default	Min.	Max.
VFBEDGEW	V	*Channel width scaling parameter of VFBEDGE*	0.0	–	–
VFBEDGELW	V	*Channel area scaling parameter of VFBEDGE*	0.0	–	–
VFBBEDGE	V	Flat-band voltage of the back gate of edge transistors at TR	0.0	–	–
VFBBEDGEO	V	*Long and wide channel value of VFBBEDGE*	0.0	–	–
STVFBEDGE	V/K	Temperature dependence of VFBEDGE and VFBBEDGE	0.0	–	–
STVFBEDGEO	V/K	*Long and wide channel value of STVFBEDGE*	0.0	–	–
STVFBEDGEL	–	*Channel length scaling parameter of STVFBEDGE*	0.0	–	–
STVFBEDGEW	–	*Channel width scaling parameter of STVFBEDGE*	0.0	–	–
STVFBEDGELW	–	*Channel area scaling parameter of STVFBEDGE*	0.0	–	–
CICFEDGE	–	Long channel front interface coupling coefficient of edge transistors	1.0	0.1	10.0
CICFEDGEO	–	*Geometry independent global scale parameter for CICFEDGE*	1.0	0.1	10.0
CICEDGE	–	Long channel back interface coupling coefficient of edge transistors	1.0	0.1	10.0
CICEDGEO	–	*Geometry independent global scale parameter for CICEDGE*	1.0	0.1	10.0
PSCEEDGE	–	Short channel effect coefficient of edge transistors	0.0	0.0	5.0
PSCEEDGEL	–	*Channel length scaling parameter of PSCEEDGE*	0.0	–	–
PSCEEDGELEXP	–	*Channel length scaling exponent of PSCEEDGE*	2.0	–	–
PSCEEDGEW	–	*Channel width scaling parameter of PSCEEDGE*	0.0	–	–
PSCEBEDGE	–	Short channel back to front interface asymmetry factor of edge transistors	1.0	0.0	–
PSCEBEDGEO	–	*Geometry independent global scale parameter for PSCEEDGEB*	1.0	0.0	–
CFEDGE	–	DIBL parameter of edge transistors	0.0	0.0	–
CFEDGEL	–	*Channel length scaling parameter of CFEDGE*	0.0	–	–
CFEDGELEXP	–	*Channel length scaling exponent of CFEDGE*	2.0	–	–
CFEDGEW	–	*Channel width scaling parameter of CFEDGE*	0.0	–	–
CFBEDGE	–	DIBL back to front interface asymmetry factor of edge transistors	1.0	0.0	–
CFBEDGEO	–	*Geometry independent global scale parameter for CFBEDGE*	1.0	0.0	–
CFDEDGE	–	Drain voltage dependence parameter of DIBL of edge transistors	0.2	0.05	–
CFDEDGEO	–	*Geometry independent global scale parameter for CFEDGEB*	0.2	0.05	–
BETNEDGE	$m^2/V/s$	Front channel aspect ratio times zero-field mobility of edge transistors	0.05	10^{-10}	–

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Name	Unit	Description	Default	Min.	Max.
FBETEDGE	–	*Length dependence of edge transistor mobility*	0	–	–
LPEDGE	m	*Exponent for length dependence of edge transistor mobility*	10^{-8}	10^{-10}	–
BETEDGEW	m	*Width scaling coefficient of edge transistor mobility*	0	–	–
STBETEDGE	–	Temperature dependence of BETNEDGE	1.5	–	–
STBETEDGEO	–	*Long and wide channel value of STBETEDGE*	1.5	–	–
STBETEDGEL	–	*Channel length scaling parameter of STBETEDGE*	0.0	–	–
STBEEDGETW	–	*Active width scaling parameter of STBETEDGE*	0.0	–	–
STBETEDGELW	–	*Active area scaling parameter of STBETEDGE*	0.0	–	–

2.16 Impact Ionization Parameters

Name	Unit	Description	Default	Min.	Max.
A1	–	Impact ionization pre-factor	1.0	0.0	–
A1O	–	*Geometry independent global scale parameter for A1*	1.0	–	–
A1L	–	*Channel length scaling parameter of A1*	0.0	–	–
A1W	–	*Channel width scaling parameter of A1*	0.0	–	–
A2	–	Impact ionization exponent at **TR**	10.0	0.0	–
A2O	–	*Geometry independent global scale parameter for A2*	10.0	0.0	–
STA2	–	Temperature dependence of **A2**	0.0	–	–
STA2O	–	*Geometry independent global scale parameter for STA2*	0.0	–	–
A3	–	Saturation voltage dependence of impact ionization	1.0	0.0	–
A3O	–	*Geometry independent global scale parameter for A3*	1.0	–	–
A3L	–	*Channel length scaling parameter of A3*	0.0	–	–
A3W	–	*Channel width scaling parameter of A3*	0.0	–	–

2.17 Charge model parameters

Name	Unit	Description	Default	Min.	Max.
AREAQ	m^2	Effective channel area for intrinsic charge model	10^{-12}	10^{-18}	–

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Name	**Unit**	**Description**	**Default**	**Min.**	**Max.**
CGBOV	F	Oxide capacitance for gate to substrate overlap	0.0	0.0	–
CGBOVO	F	*Geometry independent global scale parameter for CGBOV*	0.0	–	–
CGBOVL	F	*Gate length scaling parameter of CGBOV*	0.0	–	–
NSDAC	cm^{-3}	Source/drain effective doping level for AC model	10^{22}	10^{18}	10^{22}
NSDACO	cm^{-3}	*Geometry independent global scale parameter for NSDAC*	10^{22}	10^{18}	10^{22}
FIF	–	Inner fringe capacitance pre-factor	0.0	0.0	–
FIFW	–	*FIF value for a WEN width*	0.0	0.0	–
FSCEAC	–	Short channel effect adjustment factor for charge model	0.0	0.0	–
FSCEACO	–	*Geometry independent global scale parameter for FSCEAC*	0.0	0.0	–
VFBAC	V	Front gate workfunction referenced to Si midgap at TR when SWQMOD=1	0.0	–	–
VFBACO	V	*Long and wide channel value of VFBAC*	0.0	–	–
VFBACL	V	*Channel length scaling parameter of VFBAC*	0.0	–	–
VFBACLEXP	–	*Channel length scaling exponent of VFBAC*	2.0	–	–
VFBACL2	–	*Second order channel length dependence of VFBAC*	0.0	–	–
VFBACLEXP2	–	*Second order channel length scaling exponent of VFBAC*	2.0	–	–
VFBACW	V	*Channel width scaling parameter of VFBAC*	0.0	–	–
VFBACLW	V	*Channel area scaling parameter of VFBAC*	0.0	–	–
VFBBAC	V	Back gate workfunction offset at TR when SWQMOD=1	0.0	–	–
VFBBACO	V	*Long and wide channel value of VFBBAC*	0.0	–	–
VFBLBACO	–	*Back to front interface asymmetry factor applied to VFBBAC*	0.0	–	–
PSCEAC	–	Short channel coupling attenuation parameter when SWQMOD=1	0.0	0.0	5.0
PSCEACL	–	*Channel length scaling parameter of PSCEAC*	0.0	–	–
PSCEACLEXP	–	*Channel length scaling exponent of PSCEAC*	2.0	–	–
PSCEACW	–	*Channel width scaling parameter of PSCEAC*	0.0	–	–
CFAC	–	DIBL parameter at TR when SWQMOD=1	0.0	–	–
CFACL	–	*Channel length scaling parameter of CFAC*	0.0	–	–
CFACLEXP	–	*Channel length scaling exponent of CFAC*	2.0	–	–
CFACW	–	*Channel width scaling parameter of CFAC*	0.0	–	–
THESATAC	V^{-1}	Velocity saturation parameter at TR when SWQMOD=1	0.0	–	–

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Name	Unit	Description	Default	Min.	Max.
THESATACO	s/m^2	*Long and wide channel parameter for THESATAC*	0.0	–	–
THESATACL	s/m^2	*Channel length scaling parameter of THESATAC*	0.0	–	–
THESATACLEXP	–	*Channel length scaling exponent of THESATAC*	1.0	–	–
THESATACW	–	*Channel width scaling parameter of THESATAC*	0.0	–	–
THESATACLW	–	*Channel area scaling parameter of THESATAC*	0.0	–	–
AXAC	–	Linear/saturation transition exponent when SWQMOD=1	8.0	1.0	16.0
AXACO	–	*Long and wide channel value of AXAC*	8.0	–	–
AXACL	–	*Channel length scaling parameter of AXAC*	0.0	–	–
AXACLEXP	–	*Channel length scaling exponent of AXAC*	1.0	–	–
AXACL2	–	*Second order channel length scaling parameter of AXAC*	0.0	–	–
AXACLEXP2	–	*Second order channel length scaling exponent of AXAC*	1.5	–	–
ALPAC	–	Channel length modulation pre-factor when SWQMOD=1	0.0	0.0	–
ALPACL1	–	*Channel length scaling parameter of ALPAC*	0.0	–	–
ALPACLEXP	–	*Channel length scaling exponent of ALPAC*	1.0	–	–
ALPACL2	–	*Second order channel length dependence of ALPAC*	0.0	–	–
ALPACLEXP2	–	*Second order channel length scaling exponent of ALPAC*	2.0	–	–
ALPACW	–	*Channel width scaling parameter of ALPAC*	0.0	–	–
COV	F	Overlap capacitance per side	0.0	0.0	–
LOVO	m	*Overlap length for gate/source-drain overlap capacitance*	0.0	0.0	–
COVD	F	Overlap capacitance at drain side	0.0	0.0	–
LOVDO	m	*Overlap length for gate/drain overlap capacitance*	0.0	0.0	–
COVDL	–	Overlap capacitance modulation due to Leff bias-dependence	0.0	–	–
COVDLO	–	*Wide channel parameter for COVDL*	0.0	–	–
COVDLW	–	*Channel width scaling parameter of COVDL*	0.0	–	–
COVDLB	–	Overlap capacitance modulation with back bias	0.0	–	–
COVDLBO	–	*Geometry independent global scale parameter for COVDLB*	0.0	–	–
DVFBOV	V	Overlap capacitance flat-band voltage adjustment	0.0	–	–
DVFBOVO	V	*Geometry independent global scale parameter for DVFBOV*	0.0	–	–
CFR	F	Outer fringe capacitance per side	0.0	0.0	–

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Name	Unit	Description	Default	Min.	Max.
CFRO	F	*Corner related outer fringe capacitance*	0.0	–	–
CFRW	F	*Outer fringe capacitance per side for a WEN width*	0.0	–	–
CFRD	F	Outer fringe capacitance at drain side	0.0	0.0	–
CFRDO	F	*Corner related outer fringe capacitance at drain side*	0.0	–	–
CFRDW	F	*Outer fringe capacitance at drain side for a WEN width*	0.0	–	–
CSD	F	Drain to source direct capacitance	$1.04\,10^{-18}$	0.0	–
CSDO	–	*Drain to source direct capacitance correction factor*	1.0	0.0	–
CSDBP	F/m	Drain/source to substrate perimeter capacitance	0.0	0.0	–
CSDBPO	F/m	*Geometry independent global scale parameter for CSDBP*	0.0	0.0	–

2.18 Self-heating parameters

Name	Unit	Description	Default	Min.	Max.
RTH	K/W	Thermal resistance	10^{4}	10^{-6}	–
RTHO	K/W	*Geometry independent global scale parameter for RTH*	10^{5}	–	–
RTHL	–	*Channel length scaling parameter of RTH and CTH*	1.5	–	–
RTHW	–	*Channel width scaling parameter of RTH and CTH*	3.0	–	–
RTHLW	–	*Channel area scaling parameter of RTH and CTH*	4.5	–	–
STRTH	–	Temperature dependence of RTH	0.0	–	–
STRTHO	–	*Geometry independent global scale parameter for STRTH*	0.0	–	–
CTH	J/K	Thermal capacitance	10^{-11}	0.0	–
CTHO	J/K	*Geometry independent global scale parameter for CTH*	10^{-12}	–	–
LAMBTHO	m	*Characteristic length of thermal coupling for multifinger devices*	10^{-7}	10^{-9}	–
FTHO	–	*First neighbour thermal coupling factor for multi-finger devices*	0.0	0.0	–

2.19 Noise model parameters

Name	Unit	Description	Default	Min.	Max.
FNT	–	Thermal noise coefficient	1.0	0.0	–
FNTO	–	*Geometry independent global scale parameter for FNT*	1.0	0.0	–
FNTEXC	–	Excess noise coefficient	0.0	0.0	–
FNTEXCL	–	*Channel length scaling pre-factor of FNTEXC*	0.0	0.0	–
FNTEXCLEXP	–	*Channel length scaling exponent of FNTEXC*	2.0	–	–
NFA	$V^{-1}m^{-4}$	First coefficient of flicker noise	$8 \cdot 10^{22}$	0.0	–
NFALW	$V^{-1}m^{-4}$	*Channel area scaling parameter of NFA*	$8 \cdot 10^{22}$	–	–
NFAW	$V^{-1}m^{-4}$	*Channel width scaling parameter of NFA*	0.0	–	–
NFB	$V^{-1}m^{-2}$	Second coefficient of flicker noise	$3 \cdot 10^{7}$	0.0	–
NFBLW	$V^{-1}m^{-2}$	*NFB value for a LEN.WEN area*	$3 \cdot 10^{7}$	0.0	–
NFC	V^{-1}	Third coefficient of flicker noise	0.0	0.0	–
NFCLW	V^{-1}	*NFC value for a LEN.WEN area*	0.0	0.0	–
NFE	–	Front interface transverse field effect coefficient	0.0	−1.0	1.0
NFEO	–	*Geometry independent global scale parameter for NFE*	0.0	−1.0	1.0
NFEB	–	Back interface transverse field effect coefficient	0.0	−1.0	1.0
NFEBO	–	*Geometry independent global scale parameter for NFEB*	0.0	−1.0	1.0
EF	–	Frequency dependence exponent of flicker noise	1.0	0.1	–
EFO	–	*Geometry independent global scale parameter for EF*	1.0	0.1	–

2.20 NQS model parameters

Name	Unit	Description	Default	Min.	Max.
KDRIFT	–	Drift component parameter of NQS effect	1.0	0.0	–
KDRIFTO	–	*Geometry independent Drift component parameter of NQS effect*	1.0	0.0	–
KDRIFTL	–	*Length dependence of KDRIFT*	0.0	–	–
KDIFF	–	Diffusion component parameter of NQS effect	1.0	0.0	–
KDIFFO	–	*Geometry independent Diffusion component parameter of NQS effect*	1.0	0.0	–
KDIFFL	–	*Length dependence of KDIFF*	0.0	–	–
FRACINV	–	Fraction of inversion charge for the second pole of the NQS transition	1.0	0.0	1.0

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Name	Unit	Description	Default	Min.	Max.
FRACINVO	–	*Geometry independent Fraction of inversion charge for the second pole of the NQS transition*	1.0	0.0	1.0
KFRACINV	–	Second pole frequency coefficient of the NQS transition	10^{-15}	10^{-15}	1.0
KFRACINVO	–	*Geometry independent Second pole frequency coefficient of the NQS transition*	10^{-15}	10^{-15}	1.0

2.21 Cryogenic temperature effect parameters (SWCRYO=1)

Name	Unit	Description	Default	Min.	Max.
TMIN	K	Minimum temperature	1.0	1.0	–
ATMIN	–	Minimum temperature slope	0.0	0.0	1.0
BTMIN	K^2	Minimum temperature smoothing parameter	10^{-3}	0.0	10^3

2.22 Parameters for parasitic resistances

Name	Unit	Description	Default	Min.	Max.
RG	Ω	Gate resistance R_{gate}	0	0	–
RGO	Ω	*Geometry independent Gate resistance* R_{gate}	0	–	–
RINT	$\Omega \cdot m^2$	*Contact resistance between silicide and poly*	0	–	–
RVPOLY	$\Omega \cdot m^2$	*Vertical poly resistance*	0	–	–
RSHG	$\Omega/\square$	*Gate electrode diffusion sheet resistance*	0	–	–
DLSIL	m	*Silicide extension over the physical gate length*	0	–	–
RSE	Ω	External source resistance R_{source}	0	0	–
RSH	$\Omega/\square$	*Sheet resistance of source diffusion*	0	0	–
RDE	Ω	External drain resistance R_{drain}	0	0	–
RSHD	$\Omega/\square$	*Sheet resistance of drain diffusion*	0	0	–
RWELL	Ω	Well resistance R_{well}	0	0	–
RWELLO	Ω	*Well resistance* R_{well}	0	0	–

Section 3

Geometrical dependences, stress effects and junction asymmetry

In this section, the global parameters are used to calculate local parameter values as a function of device geometry. These calculations includes geometrical scaling, modification of some parameters through the stress model and symmetrization of the source/drain junctions when the **SWJUNASYM** flag is set to 0.

3.1 Scaling equations

In this part are detailed the calculation of local parameters from the global parameter set. These calculations are carried out when the scaling flag **SWSCALE** is set to 1.

3.1.1 Effective dimensions

Introduction of the number of fingers

$$W_f = W/\mathbf{NF} \tag{3.1}$$

$$A_{source,f} = \mathbf{ASOURCE}/\mathbf{NF} \tag{3.2}$$

$$A_{drain,f} = \mathbf{ADRAIN}/\mathbf{NF} \tag{3.3}$$

$$P_{source,f} = \mathbf{PSOURCE}/\mathbf{NF} \tag{3.4}$$

$$P_{drain,f} = \mathbf{PDRAIN}/\mathbf{NF} \tag{3.5}$$

$$Mult_f = \mathbf{MULT} \cdot \mathbf{NF} \tag{3.6}$$

Effective length and width for current model

$$L_{EN} = 10^{-6} \tag{3.7}$$

$$W_{EN} = 10^{-6} \tag{3.8}$$

$$\Delta L_{PS} = \mathbf{LVARO} \cdot \left(1 + \mathbf{LVARL} \cdot \frac{L_{EN}}{L}\right) \cdot \left(1 + \mathbf{LVARW} \cdot \frac{W_{EN}}{W_f}\right) \tag{3.9}$$

$$\Delta W_{OD} = \mathbf{WVARO} \cdot \left(1 + \mathbf{WVARL} \cdot \frac{L_{EN}}{L}\right) \cdot \left(1 + \mathbf{WVARW} \cdot \frac{W_{EN}}{W_f}\right) \tag{3.10}$$

$$L_E = L + \Delta L_{PS} - 2 \cdot \mathbf{LAP} \tag{3.11}$$

$$W_E = W_f + \Delta W_{OD} - 2 \cdot \mathbf{WOT} \tag{3.12}$$

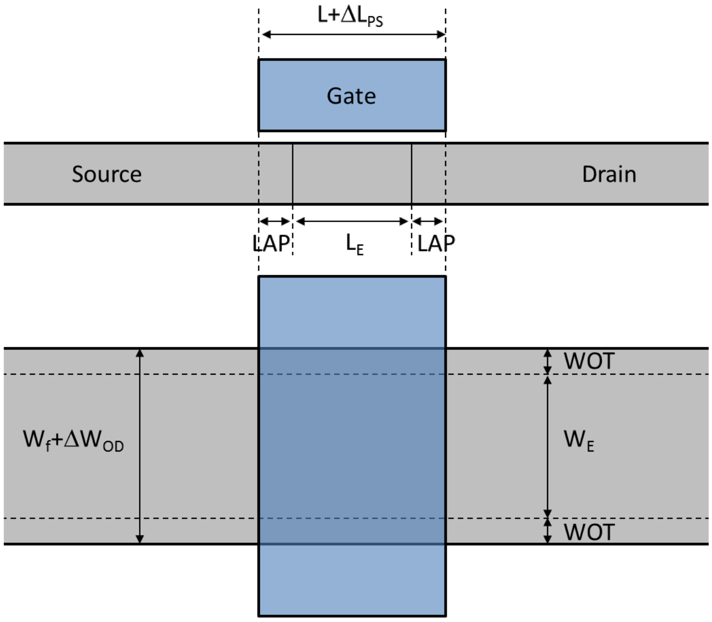

Figure 3.1: Description of transistor active and gate dimensions on a cross-section (top) and a top-view (bottom).

Effective length and width for charge model

$$L_{E,CV} = L + \Delta L_{PS} - 2 \cdot \mathbf{LAP} + \mathbf{DLQ} \tag{3.13}$$

$$W_{E,CV} = W_f + \Delta W_{OD} - 2 \cdot \mathbf{WOT} + \mathbf{DWQ} \tag{3.14}$$

Physical length and width for parasitic charges and self-heating

$$L_{phy} = L + \Delta L_{PS} \tag{3.15}$$

$$W_{phy} = W_f + \Delta W_{OD} \tag{3.16}$$

3.1.2 Process parameters

$$\mathbf{TOXE} = \mathbf{TOXEO} \tag{3.17}$$

$$\mathbf{TSI} = \mathbf{TSIO} \tag{3.18}$$

$$\mathbf{XGE} = \mathbf{XGEO} \tag{3.19}$$

$$\mathbf{TBOX} = \mathbf{TBOXO} \tag{3.20}$$

$$\mathbf{NCH} = \mathbf{NCHO} \tag{3.21}$$

$$\mathbf{NSUB} = \mathbf{NSUBO} \tag{3.22}$$

$$\mathbf{CT} = \mathbf{CTO} \tag{3.23}$$

$$\mathbf{TOXP} = \mathbf{TOXPO} \tag{3.24}$$

$$\mathbf{NOV} = \mathbf{NOVO} \tag{3.25}$$

$$\mathbf{NOVD} = \mathbf{NOVDO} \tag{3.26}$$

$$\mathbf{VFB} = \mathbf{VFBO} + \frac{\mathbf{VFBL} \cdot \left(\frac{L_{EN}}{L_E}\right)^{\mathbf{VFBLEXP}}}{1 + \mathbf{VFBL2} \cdot \left(\frac{L_{EN}}{L_E}\right)^{\mathbf{VFBLEXP2}}} + \mathbf{VFBW} \cdot \frac{W_{EN}}{W_E} + \mathbf{VFBLW} \cdot \frac{L_{EN}}{L_E} \cdot \frac{W_{EN}}{W_E} \tag{3.27}$$

$$\mathbf{VFBB} = \mathbf{VFBBO} + \mathbf{VFBLBO} \cdot \frac{\mathbf{TBOX}}{\mathbf{TOXE}} \cdot \frac{\mathbf{VFBL} \cdot \left(\frac{L_{EN}}{L_E}\right)^{\mathbf{VFBLEXP}}}{1 + \mathbf{VFBL2} \cdot \left(\frac{L_{EN}}{L_E}\right)^{\mathbf{VFBLEXP2}}} \tag{3.28}$$

$$\begin{aligned}\mathbf{STVFB} = \mathbf{STVFBO} \cdot \left(1 + \mathbf{STVFBL} \cdot \frac{L_{EN}}{L_E}\right) \cdot \\ \left(1 + \mathbf{STVFBW} \cdot \frac{W_{EN}}{W_E}\right) \cdot \left(1 + \mathbf{STVFBLW} \cdot \frac{L_{EN}}{L_E} \cdot \frac{W_{EN}}{W_E}\right)\end{aligned} \tag{3.29}$$

$$\mathbf{NP} = \mathbf{NPO} \cdot \mathrm{MAX}\left(10^{-6}, 1 + \mathbf{NPL} \cdot \frac{L_{EN}}{L_E}\right) \tag{3.30}$$

3.1.3 Gate to interface coupling parameters

$$\mathbf{CICF} = \mathbf{CICFO} \tag{3.31}$$

$$\mathbf{CIC} = \mathbf{CICO} \tag{3.32}$$

$$\lambda_{2D} = \sqrt{\frac{\epsilon_{Si} \cdot (1 - \mathbf{XGE}) + \epsilon_{Ge} \cdot \mathbf{XGE}}{\epsilon_{ox}} \cdot \mathbf{TSI} \cdot (\mathbf{TOXE} + 4 \cdot 10^{-10})} \tag{3.33}$$

$$\mathbf{PSCE} = 2 \cdot \mathbf{PSCEL} \cdot \left(\frac{\lambda_{2D}}{L_E}\right)^{\mathbf{PSCELEXP}} \cdot \left(1 + \mathbf{PSCEW} \cdot \frac{W_{EN}}{W_E}\right) \tag{3.34}$$

$$\mathbf{PSCEB} = \mathbf{PSCEBO} \tag{3.35}$$

$$\mathbf{NSDDC} = \mathbf{NSDDCO} \tag{3.36}$$

$$\mathbf{PSCEDLB} = \mathbf{PSCEDLBO} \tag{3.37}$$

$$\mathbf{PNCE} = \mathbf{PNCEW} \cdot \frac{W_{EN}}{W_E} \tag{3.38}$$

3.1.4 Drain Induced Barrier Lowering parameters

$$\mathbf{CF} = \mathbf{CFL} \cdot \left(\frac{\lambda_{2D}}{L_E}\right)^{\mathbf{CFLEXP}} \cdot \left(1 + \mathbf{CFW} \cdot \frac{W_{EN}}{W_E}\right) \tag{3.39}$$

$$\mathbf{CFB} = \mathbf{CFBO} \tag{3.40}$$

$$\mathbf{STCF} = \mathbf{STCFL} \cdot \left(\frac{\lambda_{2D}}{L_E}\right)^{\mathbf{CFLEXP}} \cdot \left(1 + \mathbf{CFW} \cdot \frac{W_{EN}}{W_E}\right) \tag{3.41}$$

$$\mathbf{CFD} = \mathbf{CFDO} \tag{3.42}$$

$$\mathbf{CFDL} = \frac{\mathbf{CFDLL} \cdot \frac{L_{EN}}{L_E}}{\max\left(1 + \mathbf{CFDLW} \cdot \frac{W_{EN}}{W_E}, 10^{-3}\right)} \tag{3.43}$$

$$\mathbf{CFDLB} = \mathbf{CFDLBO} \tag{3.44}$$

3.1.5 Mobility parameters

$$L_{P1eff} = \mathbf{LP1} \cdot \max\left(1 + \mathbf{LP1W} \cdot \frac{W_{EN}}{W_E}, 10^{-3}\right) \tag{3.45}$$

$$G_{PE} = \max\left(1 + \mathbf{FBET1} \cdot \left(1 + \mathbf{FBET1W} \cdot \frac{W_{EN}}{W_E}\right) \cdot \frac{1 - \exp\left(-L_E/L_{P1eff}\right)}{L_E/L_{P1eff}} + \mathbf{FBET2} \cdot \frac{1 - \exp\left(-L_E/L_{P2}\right)}{L_E/L_{P2}}, 10^{-6}\right) \tag{3.46}$$

$$G_{WE} = \max\left(1 + \mathbf{BETW1} \cdot \frac{W_{EN}}{W_E} + \mathbf{BETW2} \cdot \frac{W_{EN}}{W_E} \cdot \ln\left(1 + \frac{W_E}{\mathbf{WBET}}\right), 10^{-6}\right) \tag{3.47}$$

$$G_E = \mathbf{UO} \cdot \frac{G_{WE}}{G_{PE}} \tag{3.48}$$

Note that, unlike in Leti-UTSOI1, **UO** is integrated into G_E. This changes the signification of the **THESAT** related global parameters, as detailed in the velocity saturation parameters part.

$$\mathbf{BETN} = G_E \cdot \frac{W_E}{L_E} \tag{3.49}$$

$$\mathbf{BETNB} = \mathbf{BETNBO} \tag{3.50}$$

$$\mathbf{STBET} = \mathbf{STBETO} \cdot \left(1 + \mathbf{STBETL} \cdot \frac{L_{EN}}{L_E}\right) \cdot \left(1 + \mathbf{STBETW} \cdot \frac{W_{EN}}{W_E}\right) \cdot \left(1 + \mathbf{STBETLW} \cdot \frac{L_{EN}}{L_E} \cdot \frac{W_{EN}}{W_E}\right) \tag{3.51}$$

$$\mathbf{CS} = \left(\mathbf{CSO} + \mathbf{CSL} \cdot \left(\frac{L_{EN}}{L_E}\right)^{\mathbf{CSLEXP}}\right) \cdot \left(1 + \mathbf{CSW} \cdot \frac{W_{EN}}{W_E}\right) \cdot \left(1 + \mathbf{CSLW} \cdot \frac{L_{EN}}{L_E} \cdot \frac{W_{EN}}{W_E}\right) \tag{3.52}$$

$$\mathbf{CSFI} = \mathbf{CSFIO} \tag{3.53}$$

$$\mathbf{CSBI} = \mathbf{CSBIO} \tag{3.54}$$

$$\mathbf{STCS} = \mathbf{STCSO} \cdot \left(1 + \mathbf{STCSL} \cdot \frac{L_{EN}}{L_E}\right) \cdot \left(1 + \mathbf{STCSW} \cdot \frac{W_{EN}}{W_E}\right) \cdot \left(1 + \mathbf{STCSLW} \cdot \frac{L_{EN}}{L_E} \cdot \frac{W_{EN}}{W_E}\right) \tag{3.55}$$

$$\mathbf{THECS} = \mathbf{THECSO} \tag{3.56}$$

$$\mathbf{STTHECS} = \mathbf{STTHECSO} \tag{3.57}$$

$$\mathbf{CSTHR} = \mathbf{CSTHRO} \tag{3.58}$$

$$\mathbf{CSTHRB} = \mathbf{CSTHRBO} \tag{3.59}$$

$$\mathbf{MUE} = \mathbf{MUEO} \tag{3.60}$$

$$\mathbf{STMUE} = \mathbf{STMUEO} \tag{3.61}$$

$$\mathbf{THEMU} = \mathbf{THEMUO} \tag{3.62}$$

$$\mathbf{STTHEMU} = \mathbf{STTHEMUO} \tag{3.63}$$

$$\mathbf{XCOR} = \left(\mathbf{XCORO} + \mathbf{XCORL} \cdot \left(\frac{L_{EN}}{L_E}\right)^{\mathbf{XCORLEXP}}\right) \cdot \left(1 + \mathbf{XCORW} \cdot \frac{W_{EN}}{W_E}\right) \cdot \left(1 + \mathbf{XCORLW} \cdot \frac{L_{EN}}{L_E} \cdot \frac{W_{EN}}{W_E}\right) \tag{3.64}$$

$$\mathbf{XCORB} = \mathbf{XCORBO} \tag{3.65}$$

$$\mathbf{STXCOR} = \mathbf{STXCORO} \tag{3.66}$$

$$\mathbf{FETA} = \mathbf{FETAO} \tag{3.67}$$

3.1.6 Series resistance parameters

$$\mathbf{RS} = \mathbf{RSW1} \cdot \frac{W_{EN}}{W_E} \cdot \left(1 + \mathbf{RSW2} \cdot \frac{W_{EN}}{W_E}\right) \tag{3.68}$$

$$\mathbf{RSIG} = \mathbf{RSIGO} \tag{3.69}$$

$$\mathbf{STRS} = \mathbf{STRS} \tag{3.70}$$

$$\mathbf{RSG} = \mathbf{RSGO} \tag{3.71}$$

$$\mathbf{THERSG} = \mathbf{THERSGO} \tag{3.72}$$

$$\mathbf{RSB} = \mathbf{RSBO} \tag{3.73}$$

3.1.7 Velocity saturation parameters

$$\mathbf{THESAT} = G_E \cdot \left(\mathbf{THESATO} + \mathbf{THESATL} \cdot \left(\frac{L_{EN}}{L_E}\right)^{\mathbf{THESATLEXP}}\right) \cdot \left(1 + \mathbf{THESATW} \cdot \frac{W_{EN}}{W_E}\right) \cdot \left(1 + \mathbf{THESATLW} \cdot \frac{L_{EN}}{L_E} \cdot \frac{W_{EN}}{W_E}\right) \quad (3.74)$$

Since **UO** has been included in G_E (see (3.48)), **THESATL** is, in L-UTSOI, closely related to the saturation velocity itself.

$$\mathbf{STTHESAT} = \mathbf{STTHESATO} \cdot \left(1 + \mathbf{STTHESATL} \cdot \frac{L_{EN}}{L_E}\right) \cdot \left(1 + \mathbf{STTHESATW} \cdot \frac{W_{EN}}{W_E}\right) \cdot \left(1 + \mathbf{STTHESATLW} \cdot \frac{L_{EN}}{L_E} \cdot \frac{W_{EN}}{W_E}\right) \quad (3.75)$$

$$\mathbf{THESATG} = \mathbf{THESATGO} \quad (3.76)$$

$$\mathbf{THESATB} = \mathbf{THESATBO} \quad (3.77)$$

3.1.8 Saturation and Channel Length Modulation parameters

$$\mathbf{AX} = \frac{\mathbf{AXO}}{1 + \mathbf{AXL} \cdot \left(\frac{L_{EN}}{L_E}\right)^{\mathbf{AXLEXP}} \Big/ \left(1 + \mathbf{AXL2} \cdot \left(\frac{L_{EN}}{L_E}\right)^{\mathbf{AXLEXP2}}\right)} \quad (3.78)$$

$$\mathbf{ALP} = \frac{\mathbf{ALPL1} \cdot \left(\frac{L_{EN}}{L_E}\right)^{\mathbf{ALPLEXP}}}{1 + \mathbf{ALPL2} \cdot \left(\frac{L_{EN}}{L_E}\right)^{\mathbf{ALPLEXP2}}} \cdot \left(1 + \mathbf{ALPW} \cdot \frac{W_{EN}}{W_E}\right) \quad (3.79)$$

$$\mathbf{ALP1} = \frac{\mathbf{ALP1L1} \cdot \left(\frac{L_{EN}}{L_E}\right)^{\mathbf{ALP1LEXP}}}{1 + \mathbf{ALP1L2} \cdot \left(\frac{L_{EN}}{L_E}\right)^{\mathbf{ALP1LEXP2}}} \cdot \left(1 + \mathbf{ALP1W} \cdot \frac{W_{EN}}{W_E}\right) \quad (3.80)$$

$$\mathbf{ALPB} = \mathbf{ALPBO} \quad (3.81)$$

$$\mathbf{VP} = \mathbf{VPO} \quad (3.82)$$

$$\mathbf{VPG} = \mathbf{VPGO} \quad (3.83)$$

3.1.9 Gate current parameters

$$\mathbf{GCO} = \mathbf{GCOO} \quad (3.84)$$

$$\mathbf{IGINV} = \mathbf{IGINVLW} \cdot \frac{L_E}{L_{EN}} \cdot \frac{W_E}{W_{EN}} \quad (3.85)$$

$$\mathbf{IGOVINV} = \mathbf{IGOVINVW} \cdot \frac{W_E}{W_{EN}} \quad (3.86)$$

$$\mathbf{IGOVINVD} = \mathbf{IGOVINVDW} \cdot \frac{W_E}{W_{EN}} \tag{3.87}$$

$$\mathbf{IGOVACC} = \mathbf{IGOVACCW} \cdot \frac{W_E}{W_{EN}} \tag{3.88}$$

$$\mathbf{IGOVACCD} = \mathbf{IGOVACCDW} \cdot \frac{W_E}{W_{EN}} \tag{3.89}$$

Notice that, unlike in Leti-UTSOI1, gate currents in overlap regions are not linked with **LOVO** in L-UTSOI.

$$\mathbf{STIG} = \mathbf{STIGO} \tag{3.90}$$

$$\mathbf{GC2CH} = \mathbf{GC2CHO} \tag{3.91}$$

$$\mathbf{GC3CH} = \mathbf{GC3CHO} \tag{3.92}$$

$$\mathbf{GC2OVINV} = \mathbf{GC2OVINVO} \tag{3.93}$$

$$\mathbf{GC3OVINV} = \mathbf{GC3OVINVO} \tag{3.94}$$

$$\mathbf{GC2OVACC} = \mathbf{GC2OVACCO} \tag{3.95}$$

$$\mathbf{GC3OVACC} = \mathbf{GC3OVACCO} \tag{3.96}$$

$$\mathbf{GCDOV} = \mathbf{GCDOVL} \cdot \frac{L_{EN}}{L_E} \tag{3.97}$$

$$\mathbf{GCVDOV} = \mathbf{GCVDOVO} \tag{3.98}$$

$$\mathbf{CHIB} = \mathbf{CHIBO} \tag{3.99}$$

$$\mathbf{NIGINV} = \mathbf{NIGINVO} \tag{3.100}$$

$$\mathbf{FNOVINV} = \mathbf{FNOVINVW} \cdot \frac{W_E}{W_{EN}} \tag{3.101}$$

$$\mathbf{FNOVINVD} = \mathbf{FNOVINVDW} \cdot \frac{W_E}{W_{EN}} \tag{3.102}$$

$$\mathbf{GCOVINVFN} = \mathbf{GCOVINFNO} \tag{3.103}$$

$$\mathbf{STIGFN} = \mathbf{STIGFNO} \tag{3.104}$$

3.1.10 Gate Induced Drain/Source Leakage (GIDL/GISL) parameters

$$\mathbf{AGIDL} = \mathbf{AGIDLO} + \mathbf{AGIDLW} \cdot \frac{W_E}{W_{EN}} \tag{3.105}$$

$$\mathbf{AGIDLD} = \mathbf{AGIDLDO} + \mathbf{AGIDLDW} \cdot \frac{W_E}{W_{EN}} \tag{3.106}$$

Notice that, as for gate currents, GIDL/GISL currents are not linked with **LOVO** in L-UTSOI.

$$\mathbf{BGIDL} = \mathbf{BGIDLO} \tag{3.107}$$

$$\mathbf{BGIDLD} = \mathbf{BGIDLDO} \tag{3.108}$$

$$\mathbf{STBGIDL} = \mathbf{STBGIDLO} \tag{3.109}$$

$$\mathbf{STBGIDLD} = \mathbf{STBGIDLDO} \tag{3.110}$$

$$\mathbf{CGIDL} = \mathbf{CGIDLO} \tag{3.111}$$

$$\mathbf{CGIDLD} = \mathbf{CGIDLDO} \tag{3.112}$$

$$\mathbf{DGIDL} = \mathbf{DGIDLO} + \mathbf{DGIDLL} \cdot \frac{L_{EN}}{L_E} \tag{3.113}$$

$$\mathbf{DGIDLD} = \mathbf{DGIDLDO} + \mathbf{DGIDLDL} \cdot \frac{L_{EN}}{L_E} \tag{3.114}$$

3.1.11 Edge Transistor Parameters

$$W_{\mathrm{E,edge}} = 2 \cdot \mathbf{WEDGE} + \mathbf{WEDGEW} \cdot W_E \tag{3.115}$$

$$\mathbf{CTEDGE} = \mathbf{CTEDGEO} \tag{3.116}$$

$$\begin{aligned}\mathbf{VFBEDGE} = \mathbf{VFBEDGEO} + \mathbf{VFBEDGEL} \cdot \left(\frac{L_{EN}}{L_E}\right)^{\mathbf{VFBEDGELEXP}} \\ + \mathbf{VFBEDGEW} \cdot \frac{W_{EN}}{W_E} + \mathbf{VFBEDGELW} \cdot \frac{L_{EN}}{L_E} \cdot \frac{W_{EN}}{W_E}\end{aligned} \tag{3.117}$$

$$\mathbf{VFBBEDGE} = \mathbf{VFBBEDGEO} \tag{3.118}$$

$$\begin{aligned}\mathbf{STVFBEDGE} = \mathbf{STVFBEDGEO} \cdot \left(1 + \mathbf{STVFBEDGEL} \cdot \frac{L_{EN}}{L_E}\right) \cdot \\ \left(1 + \mathbf{STVFBEDGEW} \cdot \frac{W_{EN}}{W_E}\right) \cdot \left(1 + \mathbf{STVFBEDGELW} \cdot \frac{L_{EN}}{L_E} \cdot \frac{W_{EN}}{W_E}\right)\end{aligned} \tag{3.119}$$

$$\mathbf{CICFEDGE} = \mathbf{CICFEDGEO} \tag{3.120}$$

$$\mathbf{CICEDGE} = \mathbf{CICEDGEO} \tag{3.121}$$

$$\mathbf{PSCEEDGE} = 2 \cdot \mathbf{PSCEEDGEL} \cdot \left(\frac{\lambda_{2D}}{L_E}\right)^{\mathbf{PSCEEDGELEXP}} \cdot \left(1 + \mathbf{PSCEEDGEW} \cdot \frac{W_{EN}}{W_E}\right) \tag{3.122}$$

$$\mathbf{PSCEBEDGE} = \mathbf{PSCEBEDGEO} \tag{3.123}$$

$$\mathbf{CFEDGE} = \mathbf{CFEDGEL} \cdot \left(\frac{\lambda_{2D}}{L_E}\right)^{\mathbf{CFEDGELEXP}} \cdot \left(1 + \mathbf{CFEDGEW} \cdot \frac{W_{EN}}{W_E}\right) \tag{3.124}$$

$$\mathbf{CFBEDGE} = \mathbf{CFBEDGEO} \tag{3.125}$$

$$\mathbf{CFDEDGE} = \mathbf{CFDEDGEO} \tag{3.126}$$

$$G_{\mathrm{PE,edge}} = 1 + \mathbf{FBETEDGE} \cdot \frac{\mathbf{LPEDGE}}{L_E} \cdot \left[1 - \exp\left(-\frac{L_E}{\mathbf{LPEDGE}}\right)\right] \tag{3.127}$$

$$\mathbf{BETNEDGE} = \frac{\mathbf{UO}}{G_{\mathrm{PE,edge}}} \cdot \frac{W_{\mathrm{E,edge}}}{L_E} \cdot \left(1 + \mathbf{BETEDGEW} \cdot \frac{W_{EN}}{W_E}\right) \tag{3.128}$$

$$\mathbf{STBETEDGE} = \mathbf{STBETEDGEO} + \mathbf{STBETEDGEL} \cdot \frac{L_{EN}}{L_E} + \mathbf{STBETEDGEW} \cdot \frac{W_{EN}}{W_E} + \mathbf{STBETEDGELW} \cdot \frac{L_{EN}}{L_E} \cdot \frac{W_{EN}}{W_E} \tag{3.129}$$

3.1.12 Impact Ionization Parameters

$$\mathbf{A1} = \mathbf{A1O} \cdot \left(1 + \mathbf{A1L} \cdot \frac{L_{EN}}{L_E}\right) \cdot \left(1 + \mathbf{A1W} \cdot \frac{W_{EN}}{W_E}\right) \tag{3.130}$$

$$\mathbf{A2} = \mathbf{A2O} \tag{3.131}$$

$$\mathbf{STA2} = \mathbf{STA2O} \tag{3.132}$$

$$\mathbf{A3} = \mathbf{A3O} \cdot \left(1 + \mathbf{A3L} \cdot \frac{L_{EN}}{L_E}\right) \cdot \left(1 + \mathbf{A3W} \cdot \frac{W_{EN}}{W_E}\right) \tag{3.133}$$

3.1.13 Charge model parameters

$$\mathbf{AREAQ} = L_{E,CV} \cdot W_{E,CV} \tag{3.134}$$

$$\mathbf{CGBOV} = \mathbf{CGBOVO} + \mathbf{CGBOVL} \cdot \frac{L_{phy}}{L_{EN}} \tag{3.135}$$

$$\mathbf{NSDAC} = \mathbf{NSDACO} \tag{3.136}$$

$$\mathbf{FIF} = \mathbf{FIFW} \cdot \frac{W_{E,CV}}{W_{EN}} \tag{3.137}$$

$$\mathbf{FSCEAC} = \mathbf{FSCEACO} \tag{3.138}$$

$$\mathbf{VFBAC} = \mathbf{VFBACO} + \frac{\mathbf{VFBACL} \cdot \left(\frac{L_{EN}}{L_E}\right)^{\mathbf{VFBACLEXP}}}{1 + \mathbf{VFBACL2} \cdot \left(\frac{L_{EN}}{L_E}\right)^{\mathbf{VFBACLEXP2}}} + \mathbf{VFBACW} \cdot \frac{W_{EN}}{W_E} + \mathbf{VFBACLW} \cdot \frac{L_{EN}}{L_E} \cdot \frac{W_{EN}}{W_E} \tag{3.139}$$

$$\mathbf{VFBBAC} = \mathbf{VFBBACO} + \mathbf{VFBLBACO} \cdot \frac{\mathbf{TBOX}}{\mathbf{TOXE}} \cdot \frac{\mathbf{VFBACL} \cdot \left(\frac{L_{EN}}{L_E}\right)^{\mathbf{VFBACLEXP}}}{1 + \mathbf{VFBACL2} \cdot \left(\frac{L_{EN}}{L_E}\right)^{\mathbf{VFBACLEXP2}}} \tag{3.140}$$

$$\mathbf{PSCEAC} = 2 \cdot \mathbf{PSCEACL} \cdot \left(\frac{\lambda_{2D}}{L_E}\right)^{\mathbf{PSCEACLEXP}} \cdot \left(1 + \mathbf{PSCEACW} \cdot \frac{W_{EN}}{W_E}\right) \tag{3.141}$$

$$\mathbf{CFAC} = \mathbf{CFACL} \cdot \left(\frac{\lambda_{2D}}{L_E}\right)^{\mathbf{CFACLEXP}} \cdot \left(1 + \mathbf{CFACW} \cdot \frac{W_{EN}}{W_E}\right) \tag{3.142}$$

$$\begin{aligned}\mathbf{THESATAC} = G_E \cdot \left(\mathbf{THESATACO} + \mathbf{THESATACL} \cdot \left(\frac{L_{EN}}{L_E}\right)^{\mathbf{THESATACLEXP}}\right) \cdot \\ \left(1 + \mathbf{THESATACW} \cdot \frac{W_{EN}}{W_E}\right) \cdot \left(1 + \mathbf{THESATACLW} \cdot \frac{L_{EN}}{L_E} \cdot \frac{W_{EN}}{W_E}\right)\end{aligned} \tag{3.143}$$

$$\mathbf{AXAC} = \frac{\mathbf{AXACO}}{1 + \mathbf{AXACL} \cdot \left(\frac{L_{EN}}{L_E}\right)^{\mathbf{AXACLEXP}} \Big/ \left(1 + \mathbf{AXACL2} \cdot \left(\frac{L_{EN}}{L_E}\right)^{\mathbf{AXACLEXP2}}\right)} \tag{3.144}$$

$$\mathbf{ALPAC} = \frac{\mathbf{ALPACL1} \cdot \left(\frac{L_{EN}}{L_E}\right)^{\mathbf{ALPACLEXP}}}{1 + \mathbf{ALPACL2} \cdot \left(\frac{L_{EN}}{L_E}\right)^{\mathbf{ALPACLEXP2}}} \cdot \left(1 + \mathbf{ALPACW} \cdot \frac{W_{EN}}{W_E}\right) \tag{3.145}$$

$$\mathbf{COV} = \epsilon_{ox} \cdot \frac{W_{E,CV}}{\mathbf{TOXE}} \cdot \mathbf{LOVO} \tag{3.146}$$

$$\mathbf{COVD} = \epsilon_{ox} \cdot \frac{W_{E,CV}}{\mathbf{TOXE}} \cdot \mathbf{LOVDO} \tag{3.147}$$

$$\mathbf{COVDL} = \frac{\mathbf{COVDLO}}{\max\left(1 + \mathbf{COVDLW} \cdot \frac{W_{EN}}{W_{E,CV}}, 10^{-3}\right)} \tag{3.148}$$

$$\mathbf{COVDLB} = \mathbf{COVDLBO} \tag{3.149}$$

$$\mathbf{DVFBOV} = \mathbf{DVFBOVO} \tag{3.150}$$

$$\mathbf{CFR} = \mathbf{CFRO} + \mathbf{CFRW} \cdot \frac{W_{phy}}{W_{EN}} \tag{3.151}$$

$$\mathbf{CFRD} = \mathbf{CFRDO} + \mathbf{CFRDW} \cdot \frac{W_{phy}}{W_{EN}} \tag{3.152}$$

$$\mathbf{CSD} = \mathbf{CSDO} \cdot (\epsilon_{Si} \cdot (1 - \mathbf{XGE}) + \epsilon_{Ge} \cdot \mathbf{XGE}) \cdot \mathbf{TSI} \cdot \frac{W_E}{L_E} \tag{3.153}$$

$$\mathbf{CSDBP} = \mathbf{CSDBPO} \tag{3.154}$$

3.1.14 Self-heating parameters

$$z_{th} = \exp\left(-\frac{\mathbf{SD} + L}{\mathbf{LAMBTHO}}\right) \tag{3.155}$$

$$n_{th} = 1 + 2 \cdot \mathbf{FTHO} \cdot z_{th} \cdot \frac{1 - z_{th} - \left(1 - {z_{th}}^{\mathbf{NF}}\right)/\mathbf{NF}}{(1 - z_{th})^2} \tag{3.156}$$

$$\mathbf{RTH} = \frac{\mathbf{RTHO} \cdot n_{th}}{1 + \mathbf{RTHL} \cdot \frac{L_{G,TH}}{L_{EN}} + \mathbf{RTHW} \cdot \frac{W_{G,TH}}{W_{EN}} + \mathbf{RTHLW} \cdot \frac{L_{G,TH}}{L_{EN}} \cdot \frac{W_{G,TH}}{W_{EN}}} \tag{3.157}$$

$$\mathbf{STRTH} = \mathbf{STRTHO} \tag{3.158}$$

$$\mathbf{CTH} = \frac{\mathbf{CTHO}}{n_{th}} \cdot \left(1 + \mathbf{RTHL} \cdot \frac{L_{G,TH}}{L_{EN}} + \mathbf{RTHW} \cdot \frac{W_{G,TH}}{W_{EN}} + \mathbf{RTHLW} \cdot \frac{L_{G,TH}}{L_{EN}} \cdot \frac{W_{G,TH}}{W_{EN}}\right) \tag{3.159}$$

3.1.15 Noise model parameters

$$\mathbf{FNT} = \mathbf{FNTO} \tag{3.160}$$

$$\mathbf{FNTEXC} = \mathbf{FNTEXCL} \cdot \mathbf{BETN}^2 \cdot \left(\frac{W_{EN}}{W_E}\right)^2 \cdot \left(\frac{L_{EN}}{L_E}\right)^{\mathbf{FTNEXCLEXP}-2} \tag{3.161}$$

$$\mathbf{NFA} = \mathbf{NFALW} \cdot \frac{L_{EN}}{L_E} \cdot \frac{W_{EN}}{W_E} + \mathbf{NFAW} \cdot \frac{W_{EN}}{W_E} \tag{3.162}$$

$$\mathbf{NFB} = \mathbf{NFBLW} \cdot \frac{L_{EN}}{L_E} \cdot \frac{W_{EN}}{W_E} \tag{3.163}$$

$$\mathbf{NFC} = \mathbf{NFCLW} \cdot \frac{L_{EN}}{L_E} \cdot \frac{W_{EN}}{W_E} \tag{3.164}$$

$$\mathbf{EF} = \mathbf{EFO} \tag{3.165}$$

$$\mathbf{NFE} = \mathbf{NFEO} \tag{3.166}$$

$$\mathbf{NFEB} = \mathbf{NFEBO} \tag{3.167}$$

3.2 Stress model for SWSTRESS $= 1$

In this paragraph are reported the modifications brought to mobility, saturation velocity and threshold voltage parameters when the STI-stress model is selected (**SWSTRESS** $= 1$).

3.2.1 Effective SA/SB related parameters

$$R_A = \frac{1}{\mathbf{NF}} \cdot \sum_{i=0}^{\mathbf{NF}-1} \frac{1}{\mathbf{SA} + L/2 + i(\mathbf{SD} + L)} \tag{3.168}$$

$$R_B = \frac{1}{\mathbf{NF}} \cdot \sum_{i=0}^{\mathbf{NF}-1} \frac{1}{\mathbf{SB} + L/2 + i(\mathbf{SD} + L)} \tag{3.169}$$

$$R_{A,ref} = \frac{1}{\mathbf{SAREF} + L/2} \tag{3.170}$$

$$R_{B,ref} = \frac{1}{\mathbf{SBREF} + L/2} \tag{3.171}$$

3.2.2 Modification of mobility related parameters

$$K_{u0} = \left(1 + \frac{\mathbf{LKUO}}{(L+\Delta L_{PS})^{\mathbf{LLODKUO}}} + \frac{\mathbf{WKUO}}{(W_f+\Delta W_{OD}+\mathbf{WLOD})^{\mathbf{WLODKUO}}} + \frac{\mathbf{PKUO}}{(L+\Delta L_{PS})^{\mathbf{LLODKUO}} \cdot (W_f+\Delta W_{OD}+\mathbf{WLOD})^{\mathbf{WLODKUO}}}\right) \cdot \left(1 + \mathbf{TKUO} \cdot \left(\frac{T_{KC}}{T_{KR}} - 1\right)\right) \tag{3.172}$$

$$\rho_\beta = \frac{\mathbf{KUO}}{K_{u0}} \cdot (R_A + R_B) \tag{3.173}$$

$$\rho_{\beta,ref} = \frac{\mathbf{KUO}}{K_{u0}} \cdot (R_{A,ref} + R_{B,ref}) \tag{3.174}$$

$$\mathbf{BETN} = \frac{1+\rho_\beta}{1+\rho_{\beta,ref}} \cdot \mathbf{BETN}_{ref} \tag{3.175}$$

$$\mathbf{THESAT} = \frac{1+\rho_\beta}{1+\rho_{\beta,ref}} \cdot \frac{1+\mathbf{KVSAT}\cdot\rho_{\beta,ref}}{1+\mathbf{KVSAT}\cdot\rho_\beta} \cdot \mathbf{THESAT}_{ref} \tag{3.176}$$

$$\mathbf{THESATAC} = \frac{1+\rho_\beta}{1+\rho_{\beta,ref}} \cdot \frac{1+\mathbf{KVSAT}\cdot\rho_{\beta,ref}}{1+\mathbf{KVSAT}\cdot\rho_\beta} \cdot \mathbf{THESATAC}_{ref} \tag{3.177}$$

3.2.3 Modification of threshold voltage related parameters

$$K_{vth0} = 1 + \frac{\mathbf{LKVTHO}}{(L+\Delta L_{PS})^{\mathbf{LLODVTH}}} + \frac{\mathbf{WKVTHO}}{(W_f+\Delta W_{OD}+\mathbf{WLOD})^{\mathbf{WLODVTH}}} + \frac{\mathbf{PKVTHO}}{(L+\Delta L_{PS})^{\mathbf{LLODVTH}} \cdot (W_f+\Delta W_{OD}+\mathbf{WLOD})^{\mathbf{WLODVTH}}} \tag{3.178}$$

$$\Delta R = R_A + R_B - R_{A,ref} - R_{B,ref} \tag{3.179}$$

$$\mathbf{VFB} = \mathbf{VFB}_{ref} + \frac{\mathbf{KVTHO}}{K_{vth0}} \cdot \Delta R \tag{3.180}$$

$$\mathbf{VFBB} = \mathbf{VFBB}_{ref} + \frac{\mathbf{KVTHO}}{K_{vth0}} \cdot \Delta R \tag{3.181}$$

$$\mathbf{CF} = \mathbf{CF}_{ref} + \frac{\mathbf{STETAO}}{K_{vth0}{}^{\mathbf{LODETAO}}} \cdot \Delta R \tag{3.182}$$

$$\mathbf{VFBAC} = \mathbf{VFBAC}_{ref} + \frac{\mathbf{KVTHO}}{K_{vth0}} \cdot \Delta R \tag{3.183}$$

$$\mathbf{VFBBAC} = \mathbf{VFBBAC}_{ref} + \frac{\mathbf{KVTHO}}{K_{vth0}} \cdot \Delta R \tag{3.184}$$

$$\mathbf{CFAC} = \mathbf{CFAC}_{ref} + \frac{\mathbf{STETAO}}{K_{vth0}{}^{\mathbf{LODETAO}}} \cdot \Delta R \tag{3.185}$$

3.3 Stress model for SWSTRESS $= 2$

In this paragraph are reported the modifications brought to mobility, saturation velocity and threshold voltage parameters when the strained-SOI dedicated stress model is selected (**SWSTRESS** $= 2$).

3.3.1 Effective SA/SB related parameters

$$R_A(i) = \left(1 - \exp\left(-\frac{\mathbf{SA} + L/2 + i(\mathbf{SD} + L)}{\mathbf{STRLAMBDA}}\right)\right)^{-\mathbf{STRALPHA}} \tag{3.186}$$

$$R_B(i) = \left(1 - \exp\left(-\frac{\mathbf{SB} + L/2 + (\mathbf{NF} - 1 - i) \cdot (\mathbf{SD} + L)}{\mathbf{STRLAMBDA}}\right)\right)^{-\mathbf{STRALPHA}} \tag{3.187}$$

$$g = 1 - \frac{1}{\mathbf{NF}} \cdot \sum_{i=0}^{\mathbf{NF}-1} \left(\frac{2}{R_A(i) + R_B(i)}\right)^{1/\mathbf{STRALPHA}} \tag{3.188}$$

$$R_{A,ref} = \left(1 - \exp\left(-\frac{\mathbf{SAREF} + L/2}{\mathbf{STRLAMBDA}}\right)\right)^{-\mathbf{STRALPHA}} \tag{3.189}$$

$$R_{B,ref} = \left(1 - \exp\left(-\frac{\mathbf{SBREF} + L/2}{\mathbf{STRLAMBDA}}\right)\right)^{-\mathbf{STRALPHA}} \tag{3.190}$$

$$g_{ref} = 1 - \left(\frac{2}{R_{A,ref} + R_{B,ref}}\right)^{1/\mathbf{STRALPHA}} \tag{3.191}$$

3.3.2 Modification of mobility related parameters

$$r_{uo} = \frac{\mathbf{STRRUO}}{1 + \mathbf{STRTRUO} \cdot \left(\frac{T_{KC}}{T_{KR}} - 1\right)} \tag{3.192}$$

$$\rho_\beta = r_{uo} \cdot g \tag{3.193}$$

$$\rho_{\beta,ref} = r_{uo} \cdot g_{ref} \tag{3.194}$$

$$\mathbf{BETN} = \frac{1 + \rho_\beta}{1 + \rho_{\beta,ref}} \cdot \mathbf{BETN}_{ref} \tag{3.195}$$

$$\mathbf{THESAT} = \frac{1 + \rho_\beta}{1 + \rho_{\beta,ref}} \cdot \frac{1 + \mathbf{STRRVSAT} \cdot \rho_{\beta,ref}}{1 + \mathbf{STRRVSAT} \cdot \rho_\beta} \cdot \mathbf{THESAT}_{ref} \tag{3.196}$$

$$\mathbf{THESATAC} = \frac{1 + \rho_\beta}{1 + \rho_{\beta,ref}} \cdot \frac{1 + \mathbf{STRRVSAT} \cdot \rho_{\beta,ref}}{1 + \mathbf{STRRVSAT} \cdot \rho_\beta} \cdot \mathbf{THESATAC}_{ref} \tag{3.197}$$

3.3.3 Modification of threshold voltage related parameters

$$K_{vth0} = 1 + \mathbf{STRWDVFBO} \cdot \frac{W_f + \Delta W_{OD} + \mathbf{WLOD}}{W_{EN}} \tag{3.198}$$

$$\mathbf{VFB} = \mathbf{VFB}_{ref} + \frac{\mathbf{STRDVFBO}}{K_{vth0}} \cdot (g - g_{ref}) \tag{3.199}$$

$$\mathbf{VFBB} = \mathbf{VFBB}_{ref} + \frac{\mathbf{STRDVFBO}}{K_{vth0}} \cdot (g - g_{ref}) \tag{3.200}$$

$$\mathbf{CF} = \mathbf{CF}_{ref} + \mathbf{STRDCFL} \cdot \left(\frac{\lambda_{2D}}{L_E}\right)^{\mathbf{CFLEXP}} \cdot \left(1 + \mathbf{CFW} \cdot \frac{W_{EN}}{W_E}\right) \cdot (g - g_{ref}) \tag{3.201}$$

$$\mathbf{VFBAC} = \mathbf{VFBAC}_{ref} + \frac{\mathbf{STRDVFBO}}{K_{vth0}} \cdot (g - g_{ref}) \tag{3.202}$$

$$\mathbf{VFBBAC} = \mathbf{VFBBAC}_{ref} + \frac{\mathbf{STRDVFBO}}{K_{vth0}} \cdot (g - g_{ref}) \tag{3.203}$$

$$\mathbf{CFAC} = \mathbf{CFAC}_{ref} + \mathbf{STRDCFL} \cdot \left(\frac{\lambda_{2D}}{L_E}\right)^{\mathbf{CFLEXP}} \cdot \left(1 + \mathbf{CFW} \cdot \frac{W_{EN}}{W_E}\right) \cdot (g - g_{ref}) \tag{3.204}$$

3.4 Asymmetric junctions

After the calculations described in paragraphs 3.1, 3.2 and 3.3, local parameters are clipped according to the min/max values given in the parameters tables.

Then, if the switch parameter **SWJUNASYM** is equal to 0, drain and source junctions are assumed symmetrical and all parameters related to the drain junction are overwritten by their source side counterparts.

If **SWJUNASYM** $= 0$:

$$\mathbf{NOVD} = \mathbf{NOV} \tag{3.205}$$

$$\mathbf{IGOVINVD} = \mathbf{IGOVINV} \tag{3.206}$$

$$\mathbf{IGOVACCD} = \mathbf{IGOVACC} \tag{3.207}$$

$$\mathbf{AGIDLD} = \mathbf{AGIDL} \tag{3.208}$$

$$\mathbf{BGIDLD} = \mathbf{BGIDL} \tag{3.209}$$

$$\mathbf{STBGIDLD} = \mathbf{STBGIDL} \tag{3.210}$$

$$\mathbf{CGIDLD} = \mathbf{CGIDL} \tag{3.211}$$

$$\mathbf{DGIDLD} = \mathbf{DGIDL} \tag{3.212}$$

$$\mathbf{COVD} = \mathbf{COV} \tag{3.213}$$

$$\mathbf{CFRD} = \mathbf{CFR} \tag{3.214}$$

If **SWJUNASYM** $= 1$, the drain side related parameters are kept unchanged. Notice that, if some of the drain side parameters are not specified in the model card, they take their default value and *not* their source side counterpart value.

3.5 NQS parameters

$$\mathbf{KDRIFT} = \mathbf{KDRIFTO} + \mathbf{KDRIFTL} \cdot \frac{L_{EN}}{L} \tag{3.215}$$

$$\mathbf{KDIFF} = \mathbf{KDIFFO} + \mathbf{KDIFFL} \cdot \frac{L_{EN}}{L} \tag{3.216}$$

$$\mathbf{FRACINV} = \mathbf{FRACINVO} \tag{3.217}$$

$$\mathbf{KFRACINV} = \mathbf{KFRACINVO} \tag{3.218}$$

3.6 Parasitic resistances

L-UTSOI model contains a network of parasitic elements: a gate resistance, two diffusion resistances for source and drain, and well resistances. The parasitic resistance model is similar to the PSP model [4]. The complete circuit is shown in Figure 3.2.

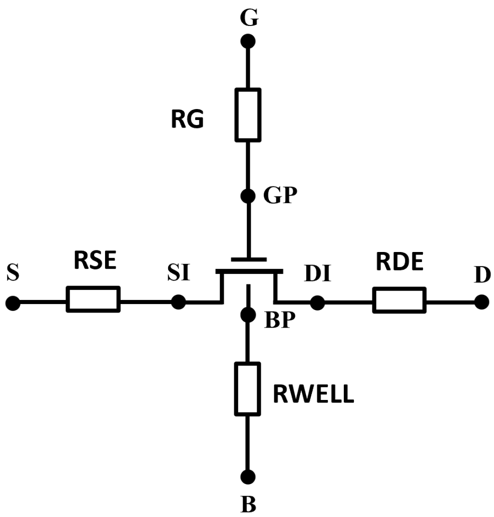

Figure 3.2: Parasitics circuit

Note: The resistance equations are calculated when **SWSCALE** $= 1$.

$$L_{\mathrm{f}} = L + \Delta L_{\mathrm{PS}} \tag{3.219}$$

$$L_{\mathrm{sil,f}} = L_{\mathrm{f}} + \mathbf{DLSIL} \tag{3.220}$$

$$W_{\mathrm{E,f}} = W_{\mathrm{f}} + \Delta W_{\mathrm{OD}} \tag{3.221}$$

$$X_{\mathrm{GWE}} = \mathbf{XGW} - 0.5 \cdot \Delta W_{\mathrm{OD}} \tag{3.222}$$

$$\mathbf{RG} = \mathbf{RGO} + \frac{1}{\mathbf{NF}} \cdot \left[\frac{\mathbf{RSHG} \cdot \left(\frac{W_{\mathrm{E,f}}}{3 \cdot \mathbf{NGCON}} + X_{\mathrm{GWE}} \right)}{\mathbf{NGCON} \cdot L_{\mathrm{sil,f}}} + \frac{\mathbf{RINT} + \mathbf{RVPOLY}}{W_{\mathrm{E,f}} \cdot L_{\mathrm{f}}} \right] \tag{3.223}$$

$$\mathbf{RSE} = \mathbf{NRS} \cdot \mathbf{RSH} \tag{3.224}$$

$$\mathbf{RDE} = \mathbf{NRD} \cdot \mathbf{RSHD} \tag{3.225}$$

$$\mathbf{RWELL} = \mathbf{RWELLO} \tag{3.226}$$

Section 4

Model equations

In this part are detailed all the equations of the model. The complete calculations of the surface potentials are given in Appendix A and Appendix B, and some useful functions are described in Appendix C.

4.1 Internal parameters including temperature dependences

In this section are calculated bias independent internal parameters. These calculations include temperature dependences and are carried out from local scale parameters. Local scale parameters are in capital characters and bold font.

4.1.1 Transistor temperature

$$T_{KR} = 273.15 + \mathbf{TR} \tag{4.1}$$

$$t_{kd} = 273.15 + T_{Ambient} + \mathbf{DTEMP} \tag{4.2}$$

$$\begin{cases} T_{KD} = \text{MAX\_FUNC}(t_{kd}, \mathbf{TMIN} + \mathbf{ATMIN} \cdot t_{kd}, \mathbf{BTMIN}) & \text{if } \mathbf{SWCRYO} = 1 \\ T_{KD} = \text{MAX\_FUNC}(t_{kd}, 1.0, 10^{-3}) & \text{else} \end{cases} \tag{4.3}$$

As in Leti-UTSOI1, a temperature node is used to compute the elevation of the channel temperature with respect to the circuit temperature due to self-heating when the **SWSHE** flag is set to 1 (see paragraph 4.10 for details). This temperature elevation is given by ΔT_C. The channel temperature T_{KC} is thus given by:

$$\begin{cases} T_{KC} = T_{KD} + \Delta T_C & \text{if } \mathbf{SWSHE} = 1 \\ T_{KC} = T_{KD} & \text{else} \end{cases} \tag{4.4}$$

$$\Delta T = T_{KC} - T_{KR} \tag{4.5}$$

$$\phi_{T0} = \frac{k_B \cdot T_{KC}}{q} \tag{4.6}$$

$$\begin{cases} x_{satmax} = \text{MAX\_FUNC}\left(10 \cdot q/(k_B \cdot T_{KD}), 600, 0.01\right) & \text{if } \mathbf{SWCRYO} = 1 \\ x_{satmax} = 600 & \text{else} \end{cases} \tag{4.7}$$

4.1.2 Local process parameters

$$\epsilon_{ch} = \epsilon_{Si} \cdot (1 - \mathbf{XGE}) + \epsilon_{Ge} \cdot \mathbf{XGE} \tag{4.8}$$

$$E_{g,Si} = E_{g0,Si} - \frac{\alpha_{Si} \cdot {T_{KC}}^2}{\beta_{Si} + T_{KC}} \tag{4.9}$$

$$E_{g,Ge} = E_{g0,Ge} - \frac{\alpha_{Ge} \cdot {T_{KC}}^2}{\beta_{Ge} + T_{KC}} \tag{4.10}$$

$$\delta E_g = (E_{g,Ge} - E_{g,Si} + C_G \cdot (1 - \mathbf{XGE})) \cdot \mathbf{XGE} \tag{4.11}$$

$$E_g = E_{g,Si} + \delta E_g \tag{4.12}$$

$$n_{eff} = \frac{n_{i,fact,300}}{1 + \sqrt{10 \cdot \mathbf{XGE}}} \cdot \left(\frac{T_{KC}}{300}\right)^{3/2\,*} \tag{4.13}$$

$$\delta V_{FB,ch} = 0.05 \cdot \mathbf{XGE} - \delta E_g/2 \tag{4.14}$$

$$\delta V_{FB1,NCH} = \frac{q \cdot \mathbf{NCH} \cdot 10^6 \cdot \mathbf{TSI}}{2 \cdot \epsilon_{ox}} \cdot (\mathbf{TOXE} + 4 \cdot 10^{-10} \cdot \mathbf{QMC}) \tag{4.15}$$

$$\delta V_{FB2,NCH} = \frac{q \cdot \mathbf{NCH} \cdot 10^6 \cdot \mathbf{TSI}}{2 \cdot \epsilon_{ox}} \cdot (\mathbf{TBOX} + 4 \cdot 10^{-10} \cdot \mathbf{QMC}) \tag{4.16}$$

[*] **Note on $\mathbf{n_{eff}}$:** Since the 102.7 release, n_{eff} refers to the goemetric mean of the conduction and valence band effective density of states $\sqrt{N_c \cdot N_v}$, and not to the intrinsic carrier concentration n_i as in previous releases. This modification results from the potentiel reference shift introduced to allow low temperature simulations.

4.1.3 Interface coupling internal parameters

$$\begin{cases} C'_{ox1} = \frac{\epsilon_{ox}}{\mathbf{TOXE}} & \text{if } \mathbf{PNCE} \leq 0 \\ C'_{ox1} = \frac{\epsilon_{ox}}{\mathbf{TOXE}} \cdot (1 + \mathbf{PNCE}) & \text{else} \end{cases} \tag{4.17}$$

$$\begin{cases} C'_{ox2} = \frac{\epsilon_{ox}}{\mathbf{TBOX}} \cdot (1 - \mathbf{PNCE}) & \text{if } \mathbf{PNCE} \leq 0 \\ C'_{ox2} = \frac{\epsilon_{ox}}{\mathbf{TBOX}} & \text{else} \end{cases} \tag{4.18}$$

$$C'_{Si0} = \frac{\epsilon_{ch}}{\mathbf{TSI}} \tag{4.19}$$

$$\phi_T = \phi_{T0} \cdot \left(1 + \mathbf{CT} \cdot \frac{T_{KR}}{T_{KC}}\right) \tag{4.20}$$

$$k_{1,1D} = \frac{C'_{ox1}}{C'_{Si0}} \tag{4.21}$$

$$k_{2,1D} = \frac{C'_{ox2}}{C'_{Si0}} \tag{4.22}$$

$$k_{eq,1D} = \frac{1}{1 + 1/k_{1,1D} + 1/k_{2,1D}} \tag{4.23}$$

$$PSCE_1 = \mathbf{PSCE} \tag{4.24}$$

$$PSCE_2 = \mathbf{PSCEB} \cdot \mathbf{PSCE} \cdot \mathbf{TBOX}/\mathbf{TOXE} \tag{4.25}$$

$$f_{A0} = \frac{2 \cdot q \cdot n_{eff} \cdot \epsilon_{ch}}{\phi_T} \tag{4.26}$$

$$x_{th,1D} = \ln\left(\frac{{C'_{Si0}}^2}{2 \cdot f_{A0}}\right) \tag{4.27}$$

$$x_{SD,dep} = \frac{q \cdot \mathbf{NSDDC} \cdot 10^6 \cdot \mathbf{TSI}}{2 \cdot (C'_{ox1} + C'_{ox2}) \cdot \phi_T} \tag{4.28}$$

4.1.4 Drain Induced Barrier Lowering internal parameters

$$CF_1 = \mathbf{CF} + \mathbf{STCF} \cdot \Delta T \tag{4.29}$$

$$CF_2 = \mathbf{CF} \cdot \mathbf{CFB} \cdot \frac{\mathbf{TBOX}}{\mathbf{TOXE}} + \mathbf{STCF} \cdot \Delta T \tag{4.30}$$

$$x_{d0} = \frac{\mathbf{CFD}}{\phi_T} \tag{4.31}$$

4.1.5 Backplane internal parameters

Since the backplane is located under the buried oxide of the device, it is assumed that its temperature is that of the environment. Thus, the self-heating effect is not included for the corresponding parameters and temperature T_{KD} is considered.

$$n_{eff,sub} = n_{i,fact,300} \cdot \left(\frac{T_{KD}}{300}\right)^{3/2} \cdot \exp\left(-\frac{\delta E_g}{2(k_B \cdot T_{KD}/q)}\right) \tag{4.32}$$

$$G_{f,sub} = \frac{\sqrt{2 \cdot q \cdot \epsilon_{Si} \cdot \mathbf{NSUB} \cdot 10^6}}{C'_{ox2} \cdot \sqrt{k_B \cdot T_{KD}/q}} \tag{4.33}$$

$$\xi_{sub} = 1 + \frac{G_{f,sub}}{\sqrt{2}} \tag{4.34}$$

$$\xi_{mrg,sub} = 10^{-5} \cdot \xi_{sub} \tag{4.35}$$

$$x_{b,sub} = \ln\left(\frac{\mathbf{NSUB} \cdot 10^6}{n_{eff,sub}}\right) + \frac{E_g}{2 \cdot (k_B \cdot T_{KD}/q)} \tag{4.36}$$

$$x_{n,sub} = 2 \cdot x_{b,sub} \tag{4.37}$$

$$\begin{cases} type_{sub} = +1 & \text{if } \mathbf{NSUB} \geq 0 \\ type_{sub} = -1 & \text{else} \end{cases} \tag{4.38}$$

4.1.6 Gate poly-depletion parameters

When **SWPDEP** $= 1$, Gate poly-depletion is activated.

$$n_{eff,poly} = n_{i,fact,300} \cdot \left(\frac{T_{KC}}{300}\right)^{3/2} + \frac{E_g}{2 \cdot k_B \cdot T_{KC}/q} \cdot \exp\left(-\frac{E_{g,Si}}{2 \cdot k_B \cdot T_{KC}/q}\right) \tag{4.39}$$

$$\delta V_{FB1,PDEP} = \phi_{T0} \cdot \left(\ln\left(\frac{\mathbf{NP} \cdot 10^6}{n_{eff,poly}}\right) + \frac{E_g}{2 \cdot k_B \cdot T_{KC}/q}\right) \tag{4.40}$$

$$K_p = \frac{\mathbf{TOXE}}{\epsilon_{ox}} \cdot \sqrt{2 \cdot q \cdot \epsilon_{Si} \cdot \mathbf{NP} \cdot 10^6} \tag{4.41}$$

4.1.7 Quantum mechanical correction internal parameters

When **QMC** > 0, quatum mechanical correctionq are activated. In this case:

$$\begin{cases} e_{min} = \text{MAX\_FUNC}(15 \cdot 198/T_{KD}, 15, 10^{-6}) & \text{if } \mathbf{SWCRYO} = 1 \\ e_{min} = 15 & \text{else} \end{cases} \tag{4.42}$$

$$\begin{cases} \delta V_{FB,qm} = \frac{0.409618895}{10^{18} \cdot \mathbf{TSI}^2} & \text{if } \mathbf{TYPE} = +1\ (NMOS) \\ \delta V_{FB,qm} = \frac{0.723134895}{10^{18} \cdot \mathbf{TSI}^2} & \text{if } \mathbf{TYPE} = -1\ (PMOS) \end{cases} \tag{4.43}$$

$$\begin{cases} qq = \frac{0.4 \cdot QM_N \cdot \mathbf{QMC}}{(10^{18} \cdot \mathbf{TSI}^2 \cdot \phi_T)^{1/3}} & \text{if } \mathbf{TYPE} = +1\ (NMOS) \\ qq = \frac{0.4 \cdot QM_P \cdot \mathbf{QMC}}{(10^{18} \cdot \mathbf{TSI}^2 \cdot \phi_T)^{1/3}} & \text{if } \mathbf{TYPE} = -1\ (PMOS) \end{cases} \tag{4.44}$$

Else $\delta V_{FB,qm} = 0$ and $qq = 0$.

Thus, the flatband voltages become:

$$\begin{aligned} V_{FB1} = \mathbf{TYPE} \cdot (\mathbf{VFB} + \mathbf{STVFB} \cdot \Delta T + \delta V_{FB,ch} + \delta V_{FB1,NCH}) \\ + \delta V_{FB,qm} - \delta V_{FB1,PDEP} + \mathbf{DELVTO} \end{aligned} \tag{4.45}$$

$$\begin{cases} V_{FB2} & = \mathbf{TYPE} \cdot (\mathbf{VFBB} + \mathbf{STVFB} \cdot \Delta T + \delta V_{FB,ch} + \delta V_{FB2,NCH}) \\ & + \delta V_{FB,qm} & \text{if } \mathbf{SWSUBDEP} = 0 \\ V_{FB2} & = \mathbf{TYPE} \cdot (\mathbf{VFBB} + \mathbf{STVFB} \cdot \Delta T + \delta V_{FB,ch} + \delta V_{FB2,NCH} \\ & + type_{sub} \cdot x_{b,sub} \cdot (k_B \cdot T_{KD}/q)) + \delta V_{FB,qm} & \text{else} \end{cases} \tag{4.46}$$

4.1.8 Mobility internal parameters

$$\beta_{N1} = \mathbf{BETN} \cdot \left(\frac{T_{KR}}{T_{KC}}\right)^{\mathbf{STBET}} \tag{4.47}$$

$$\beta_{N2} = \mathbf{BETN} \cdot \mathbf{BETNB} \cdot \left(\frac{T_{KR}}{T_{KC}}\right)^{\mathbf{STBET}} \tag{4.48}$$

$$\mu_E = \mathbf{MUE} \cdot \left(\frac{T_{KR}}{T_{KC}}\right)^{\mathbf{STMUE}} \tag{4.49}$$

$$\theta_\mu = \mathbf{THEMU} \cdot \left(\frac{T_{KR}}{T_{KC}}\right)^{\mathbf{STTHEMU}} \tag{4.50}$$

$$C_S = \mathbf{CS} \cdot \left(\frac{T_{KR}}{T_{KC}}\right)^{\mathbf{STCS}} \tag{4.51}$$

$$\theta_{CS} = \mathbf{THECS} \cdot \left(\frac{T_{KR}}{T_{KC}}\right)^{\mathbf{STTHECS}} \tag{4.52}$$

$$X_{cor} = \mathbf{XCOR} \cdot \left(\frac{T_{KR}}{T_{KC}}\right)^{\mathbf{STXCOR}} \tag{4.53}$$

$$f_{\mu E} = 10^{-8} \cdot \frac{\phi_T}{\textbf{TSI}} \cdot \mu_E \tag{4.54}$$

$$q_{i1th,CS} = 0.5 \cdot \textbf{CSTHR} \tag{4.55}$$

$$q_{i2th,CS} = q_{i1th,CS} \cdot \textbf{CSTHRB} \tag{4.56}$$

$$\begin{cases} \eta_\mu = 1/2 \cdot \textbf{FETA} & \text{if } \textbf{TYPE} = +1\ (NMOS) \\ \eta_\mu = 1/3 \cdot \textbf{FETA} & \text{if } \textbf{TYPE} = -1\ (PMOS) \end{cases} \tag{4.57}$$

4.1.9 Series resistance internal parameters

$$R_S = \textbf{RS} \cdot \left(\frac{T_{KR}}{T_{KC}}\right)^{\textbf{STRS}} \tag{4.58}$$

$$f_{RS} = 2 \cdot \phi_T \cdot R_S \tag{4.59}$$

4.1.10 Velocity saturation internal parameters

$$\theta_{sat} = \textbf{FACTUO} \cdot \textbf{THESAT} \cdot \left(\frac{T_{KR}}{T_{KC}}\right)^{\textbf{STTHESAT}+\textbf{STBET}} \tag{4.60}$$

Notice the presence of the **FACTUO** pre-factor and of **STBET** in the temperature exponent. This contributes to decouple velocity saturation parameters from low longitudinal field mobility ones, and makes in particular **STTHESAT** the temperature exponent of the saturation velocity itself.

$$f_{vsat} = \phi_T \cdot \theta_{sat} \tag{4.61}$$

4.1.11 Channel Length Modulation internal parameters

$$\gamma_{AX} = \left(2^{16/\textbf{AX}} - 1\right)^{3/8} - 1 \tag{4.62}$$

$$f_{alp1} = \frac{\textbf{ALP1}}{\phi_T} \tag{4.63}$$

4.1.12 Gate current internal parameters

$$I_{G,inv} = \textbf{IGINV} \cdot \left(\frac{T_{KC}}{T_{KR}}\right)^{\textbf{STIG}} \tag{4.64}$$

$$I_{G,ovinv} = \textbf{IGOVINV} \cdot \left(\frac{T_{KC}}{T_{KR}}\right)^{\textbf{STIG}} \tag{4.65}$$

$$I_{G,ovinvD} = \textbf{IGOVINVD} \cdot \left(\frac{T_{KC}}{T_{KR}}\right)^{\textbf{STIG}} \tag{4.66}$$

$$I_{G,ovacc} = \textbf{IGOVACC} \cdot \left(\frac{T_{KC}}{T_{KR}}\right)^{\textbf{STIG}} \tag{4.67}$$

$$I_{G,ovaccD} = \textbf{IGOVACCD} \cdot \left(\frac{T_{KC}}{T_{KR}}\right)^{\textbf{STIG}} \tag{4.68}$$

$$I_{G,fnovinv} = \textbf{FNOVINV} \cdot \left(\frac{T_{KC}}{T_{KR}}\right)^{\textbf{STIGFN}} \tag{4.69}$$

$$I_{G,fnovinvD} = \textbf{FNOVINVD} \cdot \left(\frac{T_{KC}}{T_{KR}}\right)^{\textbf{STIGFN}} \tag{4.70}$$

$$B_{ch} = \frac{4}{3} \cdot \frac{\sqrt{2 \cdot q \cdot m_0 \cdot \textbf{CHIB}}}{\hbar} \cdot \textbf{TOXP} \tag{4.71}$$

$$B_{ov} = \frac{4}{3} \cdot \frac{\sqrt{2 \cdot q \cdot m_0 \cdot \textbf{CHIB}}}{\hbar} \cdot \textbf{TOXP} \tag{4.72}$$

Note that, B_{ch} and B_{ov} are equal in L-UTSOI, since the Leti-UTSOI1 parameter **TOXOV** has been removed.

$$\begin{cases} G_{CQ,ch} = 0 & \text{if } \textbf{GC3CH} \geq 0 \\ G_{CQ,ch} = -0.495 \cdot \textbf{GC2CH}/\textbf{GC3CH} & \text{else} \end{cases} \tag{4.73}$$

$$\begin{cases} G_{CQ,ovinv} = 0 & \text{if } \textbf{GC3OVINV} \geq 0 \\ G_{CQ,ovinv} = -0.495 \cdot \textbf{GC2OVINV}/\textbf{GC3OVINV} & \text{else} \end{cases} \tag{4.74}$$

$$\begin{cases} G_{CQ,ovacc} = 0 & \text{if } \textbf{GC3OVACC} \geq 0 \\ G_{CQ,ovacc} = -0.495 \cdot \textbf{GC2OVACC}/\textbf{GC3OVACC} & \text{else} \end{cases} \tag{4.75}$$

$$\alpha_b = \frac{E_g}{2} \tag{4.76}$$

$$D_{ch} = \phi_T \cdot \textbf{GCO} \tag{4.77}$$

$$D_{ov} = \phi_{T0} \cdot \textbf{GCO} \tag{4.78}$$

$$n_{iginv} = \frac{1}{1 + \textbf{NIGINV} \cdot E_g/(2 \cdot \phi_T)} \tag{4.79}$$

4.1.13 Gate Induced Drain/Source Leakage (GIDL/GISL) internal parameters

$$A_{GIDL} = \frac{4 \cdot 10^{-18}}{\textbf{TOXP}^2} \cdot \textbf{AGIDL} \tag{4.80}$$

$$A_{GIDLD} = \frac{4 \cdot 10^{-18}}{\textbf{TOXP}^2} \cdot \textbf{AGIDLD} \tag{4.81}$$

$$B_{GIDL} = 5 \cdot 10^8 \cdot \textbf{TOXP} \cdot \textbf{BGIDL} \cdot (1 + \textbf{STBGIDL} \cdot \Delta T) \tag{4.82}$$

$$B_{GIDLD} = 5 \cdot 10^8 \cdot \textbf{TOXP} \cdot \textbf{BGIDLD} \cdot (1 + \textbf{STBGIDLD} \cdot \Delta T) \tag{4.83}$$

4.1.14 Edge transistor parameters

$$\phi_{T,edge} = \phi_{T0,edge} \cdot \left(1 + \textbf{CTEDGE} \cdot \frac{T_{KR}}{T_{KC}}\right) \tag{4.84}$$

$$f_{A0,edge} = \frac{2 \cdot q \cdot n_{eff} \cdot \epsilon_{ch}}{\phi_{T,edge}} \tag{4.85}$$

$$PSCE_{1,edge} = \textbf{PSCEEDGE} \tag{4.86}$$

$$PSCE_{2,edge} = \textbf{PSCEBEDGE} \cdot \textbf{PSCEEDGE} \cdot \frac{\textbf{TBOX}}{\textbf{TOXE}} \tag{4.87}$$

$$CF_{1,edge} = \textbf{CFEDGE} \tag{4.88}$$

$$CF_{2,edge} = \textbf{CFEDGE} \cdot \textbf{CFBEDGE} \cdot \frac{\textbf{TBOX}}{\textbf{TOXE}} \tag{4.89}$$

$$V_{FB1,edge} = \textbf{TYPE} \cdot (\textbf{VFBEDGE} + \textbf{STVFBEDGE} \cdot \Delta T + \delta V_{FB,ch} + \delta V_{FB1,NCH}) \\ + \delta V_{FB,qm} - \delta V_{FB1,PDEP} + \textbf{DELVTO} \tag{4.90}$$

$$V_{FB2,edge} = \textbf{TYPE} \cdot (\textbf{VFBBEDGE} + \textbf{STVFBEDGE} \cdot \Delta T + \delta V_{FB,ch} \\ + \delta V_{FB2,NCH}) + \delta V_{FB,qm} \tag{4.91}$$

$$\beta_{N,edge} = \textbf{FACTUO} \cdot \textbf{BETNEDGE} \cdot \left(\frac{T_{KR}}{T_{KC}}\right)^{\textbf{STBETEDGE}} \tag{4.92}$$

4.1.15 Impact ionization internal parameters

$$a_2 = \textbf{A2} \cdot \left(\frac{T_{KC}}{T_{KR}}\right)^{\textbf{STA2}} \tag{4.93}$$

4.1.16 Charge model internal parameters

$$f_{area} = \phi_T \cdot \textbf{AREAQ} \tag{4.94}$$

$$f_{SD,inner} = \frac{q \cdot \textbf{NSDAC} \cdot 10^6}{4 \cdot \epsilon_{ch} \cdot \phi_T} \tag{4.95}$$

$$x_{SD} = \ln\left(\frac{\textbf{NSDAC} \cdot 10^6}{n_{eff}}\right) \tag{4.96}$$

$$f_{if} = 1.25 \cdot 10^{-6} \cdot \phi_T \cdot \textbf{FIF} \tag{4.97}$$

$$PSCE_{1,ac} = \textbf{PSCEAC} \tag{4.98}$$

$$PSCE_{2,ac} = \textbf{PSCEB} \cdot \textbf{PSCEAC} \cdot \textbf{TBOX}/\textbf{TOXE} \tag{4.99}$$

$$CF_{1,ac} = \textbf{CFAC} + \textbf{STCF} \cdot \Delta T \tag{4.100}$$

$$CF_{2,ac} = \textbf{CFAC} \cdot \textbf{CFB} \cdot \frac{\textbf{TBOX}}{\textbf{TOXE}} + \textbf{STCF} \cdot \Delta T \tag{4.101}$$

$$V_{FB1,ac} = \textbf{TYPE} \cdot (\textbf{VFBAC} + \textbf{STVFBAC} \cdot \Delta T + \delta V_{FB,ch} + \delta V_{FB1,NCH}) \\ + \delta V_{FB,qm} - \delta V_{FB1,PDEP} + \textbf{DELVTO} \tag{4.102}$$

$$\begin{cases} V_{FB2,ac} & = \textbf{TYPE} \cdot (\textbf{VFBBAC} + \textbf{STVFBAC} \cdot \Delta T + \delta V_{FB,ch} + \delta V_{FB2,NCH}) \\ & + \delta V_{FB,qm} & \text{if } \textbf{SWSUBDEP} = 0 \\ V_{FB2,ac} & = \textbf{TYPE} \cdot (\textbf{VFBBAC} + \textbf{STVFBAC} \cdot \Delta T + \delta V_{FB,ch} + \delta V_{FB2,NCH} \\ & + type_{sub} \cdot x_{b,sub} \cdot (k_B \cdot T_{KD}/q)) + \delta V_{FB,qm} & \text{else} \end{cases} \quad (4.103)$$

$$\theta_{sat,ac} = \textbf{FACTUO} \cdot \textbf{THESATAC} \cdot \left(\frac{T_{KR}}{T_{KC}}\right)^{\textbf{STTHESAT}+\textbf{STBET}} \quad (4.104)$$

$$f_{vsat,ac} = \phi_T \cdot \theta_{sat,ac} \quad (4.105)$$

$$\gamma_{AX,ac} = \left(2^{16/\textbf{AXAC}} - 1\right)^{3/8} - 1 \quad (4.106)$$

4.1.17 Self-heating internal parameters

$$R_{th} = \textbf{RTH} \cdot \left(\frac{T_{KR}}{T_{KC}}\right)^{\textbf{STRTH}} \quad (4.107)$$

4.1.18 Noise model internal parameters

$$n_T = 4 \cdot k_B \cdot T_{KC} \cdot \textbf{FNT} \quad (4.108)$$

$$n_{T0} = 4 \cdot k_B \cdot T_{KD} \cdot \textbf{FNT} \quad (4.109)$$

$$f_{exc} = 10^{12} \cdot m_0 \cdot \textbf{FNTEXC} \quad (4.110)$$

4.2 Intrinsic surface potentials and charges

This part is dedicated to the calculation of the surface potentials and charges of MOSFET intrinsic part. Since this calculation preserves the formal symmetry between front and back interfaces, it is valid not only for Ultra-Thin Body and Buried oxide FDSOI transistors, but also for Independent Double Gate MOSFETs.
The input voltages used in the model are V_{GS}, V_{DS} and V_{SB} in nMOSFET and positive V_{DS} configuration (i.e. sign of external voltages is reversed for pMOSFET, and source/drain are interchanged in case of negative V_{DS}), from which V_{SD}, V_{GD}, V_{DB} and V_{GB} are computed. Then, dimensionless quantities are defined from these voltages.
Depending on the value of the parameter SWQMOD, the surface potentials and the charges (at source- and drain-side of the channel) and associated computations, i.e., Eq. (4.113) to Eq. (4.382), may be evaluated twice: once for the dc-characteristics and a second time for the ac-characteristics of the model. Thus, during the charge model calculation, if SWQMOD=1 then $PSCE_1$, $PSCE_2$, CF_1, CF_2, V_{FB1}, V_{FB2}, θ_{sat}, f_{vsat} and γ_{AX} are replaced by $PSCE_{1,ac}$, $PSCE_{2,ac}$, $CF_{1,ac}$, $CF_{2,ac}$, $V_{FB1,ac}$, $V_{FB2,ac}$, $\theta_{sat,ac}$, $f_{vsat,ac}$ and $\gamma_{AX,ac}$, respectively.

4.2.1 Terminal voltage conditioning

$$x_d = \frac{V_{DS}}{\phi_T} \quad (4.111)$$

$$x_{dsx} = \frac{\sqrt{{V_{DS}}^2 + 0.01} - 0.1}{\phi_T} \quad (4.112)$$

$$x_{g10} = \frac{V_{GS} - V_{FB1}}{\phi_T} - \frac{x_d - x_{dsx}}{2} - \frac{E_g}{2 \cdot \phi_{T0}} \tag{4.113}$$

$$x_{g20} = \frac{-V_{SB} - V_{FB2}}{\phi_T} - \frac{x_d - x_{dsx}}{2} - \frac{E_g}{2 \cdot \phi_{T0}} \tag{4.114}$$

$$x_{g20shift} = x_{g20} + \frac{E_g}{2 \cdot \phi_{T0}} \tag{4.115}$$

4.2.2 Backplane depletion

Depletion of the backplane is accounted for through the calculation of an effective backplane bias x_{g2eff}. If backplane depletion is not activated (i.e. **SWSUBDEP**=0), then x_{g2eff} is simply given by:

$$x_{g2eff} = x_{g20} \tag{4.116}$$

On the contrary, if backplane depletion is activated (i.e. **SWSUBDEP**=1), then the computation of x_{g2eff} uses a PSP-like surface potential calculation sequence. Physically, the inversion charge screens the gate control to the back plane depletion. To mimic this effect, we propose an approximated value of back channel surface potential using similar initial guess (corresponding to x_2 in Appendix A.2 where k_1, k_2, A_0, $diff_{min}$, $x_{1,WI0}$, $x_{2,WI0}$ calculation are adapted here). The detailed calculation of x_{g2eff} is:

$$x_{g1int} = \textbf{TYPE} \cdot type_{sub} \cdot x_2 \tag{4.117}$$

$$x_{g2int} = \textbf{TYPE} \cdot type_{sub} \cdot x_{g20} \tag{4.118}$$

$$x_{gbint} = x_{g1int} - x_{g2int} \tag{4.119}$$

If $|x_{gbint}| \leq \xi_{mrg,sub}$:

$$\delta x_{g2,sub} = \frac{x_{gbint}}{\xi_{sub}} \cdot \left(1 + \frac{x_{gbint} \cdot G_{f,sub} \cdot (1 - \exp(-x_{n,sub}))}{6 \cdot \sqrt{2} \cdot {\xi_{sub}}^2}\right) \tag{4.120}$$

Else if $x_{gbint} < -\xi_{mrg,sub}$:

$$y_g = -x_{gbint} \tag{4.121}$$

$$y_{sub} = 1.25 \cdot \frac{y_g}{\xi_{sub}} \tag{4.122}$$

$$\eta = \frac{y_{sub} + 10 - \sqrt{(y_{sub} - 6)^2 + 64}}{2} \tag{4.123}$$

$$a = (y_g - \eta)^2 + {G_{f,sub}}^2 \cdot (\eta + 1) \tag{4.124}$$

$$c = 2 \cdot (y_g - \eta) - {G_{f,sub}}^2 \tag{4.125}$$

$$\tau = -\eta + \ln\left(a/{G_{f,sub}}^2\right) \tag{4.126}$$

$$y_0 = \sigma_3(a, c, \tau, \eta) \tag{4.127}$$

$$\Delta_0 = \exp(y_0) \tag{4.128}$$

$$\chi_0 = \frac{{y_0}^2}{2 + {y_0}^2} \tag{4.129}$$

$$\chi_1 = \frac{4 \cdot y_0}{(2 + {y_0}^2)^2} \tag{4.130}$$

$$\chi_2 = \frac{8 - 12 \cdot {y_0}^2}{(2 + {y_0}^2)^3} \tag{4.131}$$

$$p_C = 2 \cdot (y_g - y_0) + {G_{f,sub}}^2 \cdot (\Delta_0 - 1 + \exp(-x_{n,sub}) \cdot (1 - \chi_1 - 1/\Delta_0)) \tag{4.132}$$

$$q_C = (y_g - y_0)^2 + {G_{f,sub}}^2 \cdot (y_0 - \Delta_0 + 1 + \exp(-x_{n,sub}) \cdot (1 + \chi_0 - 1/\Delta_0 - y_0)) \tag{4.133}$$

$$\delta x_{g2,sub} = -y_0 - \frac{2 \cdot q_C}{p_C + \sqrt{{p_C}^2 - 2 \cdot q_C \cdot (2 - {G_{f,sub}}^2 \cdot (\Delta_0 + \exp(-x_{n,sub}) \cdot (1/\Delta_0 - \chi_2)))}} \tag{4.134}$$

Else:

$$x_{g1} = 1.25 + 0.7324648775 \cdot G_{f,sub} \tag{4.135}$$

$$\frac{1}{\widetilde{x}_{g1}} = \frac{1.25 \cdot \xi_{sub}/x_{g1} - 1}{x_{g1}} \tag{4.136}$$

$$\bar{x} = \frac{x_{gbint}}{\xi_{sub}} \cdot \left(1 + \frac{x_{gbint}}{\widetilde{x}_{g1}}\right) \tag{4.137}$$

$$w = 1 - \exp(-\bar{x}) \tag{4.138}$$

$$x_1 = x_{gbint} + {G_{f,sub}}^2/2 - G_{f,sub} \cdot \sqrt{x_{gbint} + {G_{f,sub}}^2/4 - w} \tag{4.139}$$

$$b_x = x_{n,sub} + 3 \tag{4.140}$$

$$\eta = \text{MIN\_FUNC}(x_1, b_x, 5) - \frac{b_x - \sqrt{{b_x}^2 + 5}}{2} \tag{4.141}$$

$$\chi_0 = \frac{\eta^2}{2 + \eta^2} \tag{4.142}$$

$$\chi_1 = \frac{4 \cdot \eta}{(2 + \eta^2)^2} \tag{4.143}$$

$$\chi_2 = \frac{8 - 12 \cdot \eta^2}{(2 + \eta^2)^3} \tag{4.144}$$

$$a = (x_{gbint} - \eta)^2 - {G_{f,sub}}^2 \cdot (\exp(-\eta) + \eta - 1 - \exp(-x_{n,sub}) \cdot (\eta + 1 + \chi_0)) \tag{4.145}$$

$$b = 1 - {G_{f,sub}}^2 \cdot \frac{\exp(-\eta) - \chi_2 \cdot \exp(-x_{n,sub})}{2} \tag{4.146}$$

$$c = 2 \cdot (x_{gbint} - \eta) + {G_{f,sub}}^2 \cdot (1 - \exp(-\eta) - \exp(-x_{n,sub}) \cdot (1 + \chi_1)) \tag{4.147}$$

$$\tau = x_{n,sub} - \eta + \ln\left(\frac{a}{{G_{f,sub}}^2}\right) \tag{4.148}$$

$$x_0 = \sigma_2(a, b, c, \tau, \eta) \tag{4.149}$$

$$\Delta_0 = \exp(x_0) \tag{4.150}$$

$$\chi'_0 = \frac{{x_0}^2}{2 + {x_0}^2} \tag{4.151}$$

$$\chi'_1 = \frac{4 \cdot x_0}{(2 + {x_0}^2)^2} \tag{4.152}$$

$$\chi'_2 = \frac{8 - 12 \cdot {x_0}^2}{(2 + {x_0}^2)^3} \tag{4.153}$$

$$p_C = 2 \cdot (x_{gbint} - x_0) + {G_{f,sub}}^2 \cdot (1 - 1/\Delta_0 + \exp(-x_{n,sub}) \cdot (\Delta_0 - 1 - \chi'_1)) \tag{4.154}$$

$$q_C = (x_{gbint} - x_0)^2 - {G_{f,sub}}^2 \cdot (x_0 - 1 + 1/\Delta_0 + \exp(-x_{n,sub}) \cdot (\Delta_0 - 1 - \chi'_0 - x_0)) \tag{4.155}$$

$$\delta x_{g2,sub} = x_0 + \frac{2 \cdot q_C}{p_C + \sqrt{{p_C}^2 - 2 \cdot q_C \cdot (2 - {G_{f,sub}}^2 \cdot (1/\Delta_0 + \exp(-x_{n,sub}) \cdot (\Delta_0 - \chi'_2)))}} \tag{4.156}$$

Finally:

$$x_{g2eff} = \mathbf{TYPE} \cdot type_{sub} \cdot (x_{g2int} + \delta x_{g2,sub}) \tag{4.157}$$

4.2.3 Quantum mechanical correction in subthreshold regime

To properly account for quantum confinement when the effect is activated (**QMC**>0), the effective geometry of the device is modified. In the first correction detailed here, it is assumed that there's no charge in the channel. Thus, strictly speaking, this correction is only valid in the subthreshold regime. A second correction will be brought afterwards to account properly for quantum confinement also in the strong inversion regime.

If **QMC** > 0:

$$e_1 = \text{MAX\_FUNC}\left(k_{eq,1D} \cdot (x_{g10} - x_{g2eff}), e_{min}, {e_{min}}^2\right) \tag{4.158}$$

$$e_2 = \text{MAX\_FUNC}\left(-k_{eq,1D} \cdot (x_{g10} - x_{g2eff}), e_{min}, {e_{min}}^2\right) \tag{4.159}$$

$$C'_{Si} = \frac{C'_{Si0}}{1 - qq \cdot \left({e_1}^{-1/3} + {e_2}^{-1/3}\right)} \tag{4.160}$$

$$k_{1,1D,QM} = k_{1,1D} \cdot \frac{1 - qq \cdot \left({e_1}^{-1/3} + {e_2}^{-1/3}\right)}{1 + k_{1,1D} \cdot {e_1}^{-1/3} qq} \tag{4.161}$$

$$k_{2,1D,QM} = k_{2,1D} \cdot \frac{1 - qq \cdot \left({e_1}^{-1/3} + {e_2}^{-1/3}\right)}{1 + k_{2,1D} \cdot {e_2}^{-1/3} qq} \tag{4.162}$$

$$k_{eq,1D,QM} = \frac{1}{1 + 1/k_{1,1D,QM} + 1/k_{2,1D,QM}} \tag{4.163}$$

$$t_{ox1,fact} = 1 + k_{1,1D,QM} \cdot {e_1}^{-1/3} \cdot qq \tag{4.164}$$

$$t_{ox2,fact} = 1 + k_{2,1D,QM} \cdot {e_2}^{-1/3} \cdot qq \tag{4.165}$$

Else:

$$C'_{Si} = C'_{Si0} \tag{4.166}$$

$$k_{1,1D,QM} = k_{1,1D} \tag{4.167}$$

$$k_{2,1D,QM} = k_{2,1D} \tag{4.168}$$

$$k_{eq,1D,QM} = k_{eq,1D} \tag{4.169}$$

$$t_{ox1,fact} = 1 \tag{4.170}$$

$$t_{ox2,fact} = 1 \tag{4.171}$$

4.2.4 Short channel effects

$$\delta x_{WI,1D} = k_{eq,1D,QM} \cdot (x_{g10} - x_{g2eff}) \tag{4.172}$$

$$\begin{cases} x_{WI,1D} = x_{g10} - \frac{\delta x_{WI,1D}}{k_{1,1D,QM}} + \ln\left(\frac{1+\exp(-\delta x_{WI,1D})}{2}\right) & \text{if } \delta x_{WI,1D} > 0 \\ x_{WI,1D} = x_{g2eff} + \frac{\delta x_{WI,1D}}{k_{2,1D,QM}} + \ln\left(\frac{1+\exp(\delta x_{WI,1D})}{2}\right) & \text{else} \end{cases} \tag{4.173}$$

$$x_{1D} = \text{MIN\_FUNC}(x_{WI,1D}, x_{th,1D}, 4.0) \tag{4.174}$$

$$\delta l_{eff} = \sqrt{1 + 2 \cdot \frac{x_{th,1D} - x_{1D}}{x_{SD,dep}}} - 1 \tag{4.175}$$

$$x_{edge} = x_{1D} + x_{SD,dep} \cdot \delta l_{eff} \tag{4.176}$$

$$c_{sce1} = \frac{1}{1 + PSCE_1 \cdot \text{MAX\_FUNC}(1 + \textbf{PSCEDLB} \cdot x_{g20shift}, 0.5, 0.01)} \tag{4.177}$$

$$c_{sce2} = \frac{1}{1 + PSCE_2 \cdot \text{MAX\_FUNC}(1 + \textbf{PSCEDLB} \cdot x_{g20shift}, 0.5, 0.01)} \tag{4.178}$$

$$\delta x_{g1,DIBL} = 2 \cdot CF_1 \cdot x_{d0} \left(\sqrt{1 + x_{dsx}/x_{d0}} - 1\right) \cdot (1 + \textbf{CFDL} \cdot \delta l_{eff}) \cdot (1 + \textbf{CFDLB} \cdot x_{g20shift}) \tag{4.179}$$

$$\delta x_{g2,DIBL} = 2 \cdot CF_2 \cdot x_{d0} \left(\sqrt{1 + x_{dsx}/x_{d0}} - 1\right) \cdot (1 + \textbf{CFDL} \cdot \delta l_{eff}) \cdot (1 + \textbf{CFDLB} \cdot x_{g20shift}) \tag{4.180}$$

$$x_{g1} = (x_{g10} - x_{edge} + \delta x_{g1,DIBL}) \cdot c_{sce1} + x_{edge} + \frac{x_d - x_{dsx}}{2} \tag{4.181}$$

$$x_{g2} = (x_{g2eff} - x_{edge} + \delta x_{g2,DIBL}) \cdot c_{sce2} + x_{edge} + \frac{x_d - x_{dsx}}{2} \tag{4.182}$$

$$x_{g1x} = \text{MIN\_FUNC}(x_{g2} + \textbf{CICF} \cdot (x_{g1} - x_{g2}), x_{satmax}, 0.01) \tag{4.183}$$

$$x_{g2x} = \text{MIN\_FUNC}(x_{g1} + \textbf{CIC} \cdot (x_{g2} - x_{g1}), x_{satmax}, 0.01) \tag{4.184}$$

4.2.5 Interface coupling in subthreshold regime

$$k_1 = \frac{k_{1,1D,QM}}{c_{sce1}} \tag{4.185}$$

$$k_2 = \frac{k_{2,1D,QM}}{c_{sce2}} \tag{4.186}$$

$$k_{eq} = \frac{1}{1 + 1/k_1 + 1/k_2} \tag{4.187}$$

$$A_0 = \frac{f_{A0}}{{C'_{Si}}^2} \tag{4.188}$$

$$\delta x_{th} = \ln\left(\frac{1 + k_1}{1 + k_2}\right) \tag{4.189}$$

$$diff_{min} = 2 \cdot \delta x_{th} \cdot \frac{\exp(\delta x_{th}) + 1}{\exp(\delta x_{th}) - 1} \tag{4.190}$$

$$\delta x_{WI} = k_{eq} \cdot (x_{g1x} - x_{g2x}) \tag{4.191}$$

$$x_{1,WI0} = x_{g1x} - \delta x_{WI}/k_1 \tag{4.192}$$

$$x_{2,WI0} = x_{g2x} + \delta x_{WI}/k_2 \tag{4.193}$$

4.2.6 Inversion charge and related quantities at source side

First, the gate charge density at the source side q_{1S}, normalized to $k_1 \cdot C'_{Si} \cdot \phi_T$, is computed by a call to the "CHARGE_DENSITY" function described in Appendix A:

$$q_{1S} = \text{CHARGE\_DENSITY}(x_{g1x}, x_{g2x}, 0) \tag{4.194}$$

Then, inversion and back gate charge densities, normalized to $C'_{Si} \cdot \phi_T$ and $k_2 \cdot C'_{Si} \cdot \phi_T$ respectively, are calculated:

$$A_{e1S} = A_0 \cdot \exp(x_{g1x} - q_{1S}) \tag{4.195}$$

$$f_{qsqS} = {k_1}^2 \cdot {q_{1S}}^2 - A_{e1S} \tag{4.196}$$

If $f_{qsqS} < -0.005$:

$$f_{qctS} = \sqrt{|f_{qsqS}|} \cdot \cot\left(\sqrt{|f_{qsqS}|}/2\right) \tag{4.197}$$

$$f_{shS} = -\frac{f_{qsqS}}{\sin\left(\sqrt{|f_{qsqS}|}/2\right)^2} \tag{4.198}$$

$$f_{lnS} = \ln(f_{shS}) \tag{4.199}$$

Else if $f_{qsqS} > 0.005$:

$$\zeta_S = \exp\left(-\sqrt{|f_{qsqS}|}\right) \tag{4.200}$$

$$f_{qctS} = \sqrt{|f_{qsqS}|} \cdot \frac{1+\zeta_S}{1-\zeta_S} \tag{4.201}$$

$$f_{shS} = \frac{4 \cdot f_{qsqS}}{1-\zeta_S \cdot (2-\zeta_S)} \cdot \zeta_S \tag{4.202}$$

$$f_{lnS} = \ln\left(\frac{4 \cdot f_{qsqS}}{1-\zeta_S \cdot (2-\zeta_S)}\right) - \sqrt{|f_{qsqS}|} \tag{4.203}$$

Else:

$$f_{qctS} = 2 + \frac{f_{qsqS}}{6} \cdot \left(1 - \frac{f_{qsqS}}{60} \cdot \left(1 - \frac{f_{qsqS}}{42}\right)\right) \tag{4.204}$$

$$f_{shS} = 4 - \frac{f_{qsqS}}{3} \cdot \left(1 - \frac{f_{qsqS}}{20} \cdot \left(1 - \frac{5 f_{qsqS}}{126}\right)\right) \tag{4.205}$$

$$f_{lnS} = \ln(f_{shS}) \tag{4.206}$$

If $(1.01 \cdot k_1 \cdot q_{1S} + f_{qctS}) \leq 0$:

$$q_{iS} = \frac{k_1 \cdot q_{1S} - f_{qctS}}{1 - f_{shS}/A_{e1S}} \tag{4.207}$$

$$q_{2S} = \frac{q_{iS} - k_1 \cdot q_{1S}}{k_2} \tag{4.208}$$

Else if $(A_{e1S} \cdot k_1 \cdot q_{1S}) < \left(0.9 \cdot {k_1}^2 \cdot {q_{1S}}^2 \cdot (k_1 \cdot q_{1S} + f_{qctS})\right)$:

$$q_{iS} = \frac{A_{e1S}}{k_1 \cdot q_{1S} + f_{qctS}} \tag{4.209}$$

$$q_{2S} = \frac{q_{iS} - k_1 \cdot q_{1S}}{k_2} \tag{4.210}$$

Else:

$$q_{2S} = x_{g2x} - x_{g1x} + q_{1S} + 2 \cdot \ln(k_1 \cdot q_{1S} + f_{qctS}) - f_{lnS} \tag{4.211}$$

$$q_{iS} = k_1 \cdot q_{1S} + k_2 \cdot q_{2S} \tag{4.212}$$

Finally, some required quantities, including the drift electrostatic potential, are computed:

$$A_{e2S} = A_0 \cdot \exp(x_{g2x} - q_{2S}) \tag{4.213}$$

If $q_{iS} > 10^{-6}$:

$$b_{1S} = A_{e1S}/k_1 \tag{4.214}$$

$$b_{2S} = A_{e2S}/k_2 \tag{4.215}$$

$$a_{1S} = b_{1S} + 2 \cdot k_1 \cdot q_{1S} \tag{4.216}$$

$$a_{2S} = b_{2S} + 2 \cdot k_2 \cdot q_{2S} \tag{4.217}$$

$$\Sigma_S = 2 \cdot q_{iS} + b_{1S} + b_{2S} \tag{4.218}$$

$$\begin{cases} Z_S = \frac{-4 \cdot f_{qsqS} \cdot \Sigma_S}{q_{iS} \cdot (a_{1S} \cdot a_{2S} + 2 \cdot a_{1S} \cdot (q_{2S}+2) + 2 \cdot a_{2S} \cdot (q_{1S}+2))} & \text{if } |f_{qsqS}| > 0.005 \\ Z_S = \frac{A_{e1S} \cdot A_{e2S} \cdot \Sigma_S}{q_{iS} \cdot (a_{1S} \cdot A_{e1S} + a_{2S} \cdot A_{e2S} + a_{1S} \cdot a_{2S} \cdot q_{iS} \cdot \Gamma_S)} & \text{else} \\ \text{with } \Gamma_S = 1 + \frac{q_{iS}}{6} \cdot \left(1 - \frac{f_{qsqS}}{30} \cdot \left(1 - \frac{f_{qsqS}}{28} \cdot \left(1 - \frac{f_{qsqS}}{30}\right)\right)\right) & \end{cases} \tag{4.219}$$

Finally:

$$x_{driftS} = \ln(q_{iS}) \tag{4.220}$$

4.2.7 Mobility attenuation and series resistance at source side

Front and back transverse effective fields, normalized to $C'_{Si}\phi_T/\epsilon_{ch}$:

$$e_{surf1S} = 2 \cdot ln(1 + \exp(k_1 \cdot q_{1S}/2)) \tag{4.221}$$

$$e_{surf2S} = 2 \cdot \ln(1 + \exp(k_2 \cdot q_{2S}/2)) \tag{4.222}$$

$$e_{cpl1S} = e_{surf2S} - k_2 \cdot q_{2S} \tag{4.223}$$

$$e_{cpl2S} = e_{surf1S} - k_1 \cdot q_{1S} \tag{4.224}$$

$$e_{eff1S} = \eta_\mu \cdot e_{surf1S} + (1 - \eta_\mu) \cdot e_{cpl1S} \tag{4.225}$$

$$e_{eff2S} = \eta_\mu \cdot e_{surf2S} + (1 - \eta_\mu) \cdot e_{cpl2S} \tag{4.226}$$

Non-universality correction factor:

$$f_{corS} = \frac{\text{MAX\_FUNC}(1 + X_{cor} \cdot (e_{cpl1S} + \textbf{XCORB} \cdot e_{cpl2S}), 0, 0.01)}{\text{MAX\_FUNC}(1 + 0.2 \cdot X_{cor} \cdot (e_{cpl1S} + \textbf{XCORB} \cdot e_{cpl2S}), 0, 0.01)} \tag{4.227}$$

Coulomb scattering term:

$$q_{i1S} = \frac{e_{surf1S}}{e_{surf1S} + e_{surf2S}} \cdot q_{iS} \tag{4.228}$$

$$q_{i2S} = \frac{e_{surf2S}}{e_{surf1S} + e_{surf2S}} \cdot q_{iS} \tag{4.229}$$

$$G_{CSS} = \frac{C_S \cdot (1 + \textbf{CSFI} \cdot e_{cpl1S} + \textbf{CSBI} \cdot e_{cpl2S})}{(1 + q_{i1S}/q_{i1th,CS} + q_{i2S}/q_{i2th,CS})^{\theta_{CS}}} \tag{4.230}$$

Series resistance term:

$$\begin{cases} f_{RSGS} = 1 - \textbf{RSG} \cdot q_{iS}{}^{\textbf{THERSG}} & \text{if } \textbf{RSG} < 0 \\ f_{RSGS} = 1/(1 + \textbf{RSG} \cdot q_{iS}{}^{\textbf{THERSG}}) & \text{else} \end{cases} \tag{4.231}$$

$$G_{RSS} = f_{RS} \cdot C'_{Si} \cdot (q_{iS} \cdot f_{RSGS} + \textbf{RSIG}) \cdot \text{MAX\_FUNC}(1 - \textbf{RSB} \cdot x_{g20shift}, 0, 0.01) \tag{4.232}$$

Total mobility degradation term, including high field mobility effect:

$$G_{mob1S} = 1 + (f_{\mu E} \cdot e_{eff1S})^{\theta_\mu} + G_{CSS} + \textbf{FACTUO} \cdot \beta_{N1} \cdot G_{RSS} \tag{4.233}$$

$$G_{mob2S} = 1 + (f_{\mu E} \cdot e_{eff2S})^{\theta_\mu} + G_{CSS} + \textbf{FACTUO} \cdot \beta_{N2} \cdot G_{RSS} \tag{4.234}$$

$$c_{1S} = e_{surf1S} \cdot \beta_{N1} \tag{4.235}$$

$$c_{2S} = e_{surf2S} \cdot \beta_{N2} \tag{4.236}$$

$$G_{mobS} = f_{corS} \cdot \frac{c_{1S} + c_{2S}}{c_{1S}/G_{mob1S} + c_{2S}/G_{mob2S}} \tag{4.237}$$

4.2.8 Drain saturation voltage, including velocity saturation effect

Derivative of inversion charge versus drift potential at the onset of saturation:

If $\delta x_{WI} > 0.007$:

$$s_1 = \frac{\delta x_{WI}}{1 - \exp(-\delta x_{WI})} \tag{4.238}$$

$$s_2 = \exp(-\delta x_{WI}) \cdot s_1 \tag{4.239}$$

$$\delta x_\infty = \ln\left(\frac{A_0}{2 \cdot q_{iS} \cdot s_1}\right) + x_{1,WI0} \tag{4.240}$$

$$\hat{q}_{1\infty} = \frac{-\delta x_{WI}}{k_{eq} \cdot (1 - s_1 - \delta x_{WI}/k_2)} \tag{4.241}$$

$$\hat{q}_{2\infty} = \frac{\delta x_{WI}}{k_{eq} \cdot (1 - s_2 - \delta x_{WI}/k_1)} \tag{4.242}$$

$$d_\infty = \frac{\delta x_{WI}}{(s_2/k_2 + 1/2)/\hat{q}_{2\infty} - (s_1/k_1 + 1/2)/\hat{q}_{1\infty}} \tag{4.243}$$

Else if $\delta x_{WI} < -0.007$:

$$s_2 = \frac{\delta x_{WI}}{\exp(\delta x_{WI}) - 1} \tag{4.244}$$

$$s_1 = \exp(\delta x_{WI}) \cdot s_2 \tag{4.245}$$

$$\delta x_\infty = \ln\left(\frac{A_0}{2 \cdot q_{iS} \cdot s_2}\right) + x_{2,WI0} \tag{4.246}$$

$$\hat{q}_{1\infty} = \frac{-\delta x_{WI}}{k_{eq} \cdot (1 - s_1 - \delta x_{WI}/k_2)} \tag{4.247}$$

$$\hat{q}_{2\infty} = \frac{\delta x_{WI}}{k_{eq} \cdot (1 - s_2 - \delta x_{WI}/k_1)} \tag{4.248}$$

$$d_\infty = \frac{\delta x_{WI}}{(s_2/k_2 + 1/2)/\hat{q}_{2\infty} - (s_1/k_1 + 1/2)/\hat{q}_{1\infty}} \tag{4.249}$$

Else:

$$s_1 = 1 + \frac{\delta x_{WI}}{2} + \frac{{\delta x_{WI}}^2}{12} \tag{4.250}$$

$$s_2 = 1 - \frac{\delta x_{WI}}{2} + \frac{{\delta x_{WI}}^2}{12} \tag{4.251}$$

$$\hat{q}_{1\infty} = \frac{1}{k_{eq} \cdot (1/2 + 1/k_2 + \delta x_{WI}/12)} \tag{4.252}$$

$$\hat{q}_{2\infty} = \frac{1}{k_{eq} \cdot (1/2 + 1/k_1 - \delta x_{WI}/12)} \tag{4.253}$$

$$\delta x_{\infty} = \ln\left(\frac{A_0}{2 \cdot q_{iS} \cdot (1 - {\delta x_{WI}}^2/24)}\right) + \frac{x_{1,WI0} + x_{2,WI0}}{2} \tag{4.254}$$

$$d_{\infty} = \frac{-12}{4 - 3 \cdot k_{eq} + 12 \cdot k_{eq}/(k_1 \cdot k_2) + (k_{eq}/k_1 - k_{eq}/k_2) \cdot \delta x_{WI} + (1/15 - k_{eq}/12) \cdot {\delta x_{WI}}^2} \tag{4.255}$$

Calculation of δx_{sat} and q_{iDsat}:

If $q_{iS} > 10^{-6}$:

$$w_{sat1S} = \frac{100 \cdot e_{surf1S}}{100 + e_{surf1S}} \tag{4.256}$$

$$\begin{cases} sat_{fact1S} = 1/(1 - w_{sat1S} \cdot \textbf{THESATG}) & \text{if } \textbf{THESATG} < 0 \\ sat_{fact1S} = 1 + w_{sat1S} \cdot \textbf{THESATG} & \text{else} \end{cases} \tag{4.257}$$

$$w_{sat2S} = \frac{100 \cdot e_{surf2S}}{100 + e_{surf2S}} \tag{4.258}$$

$$\begin{cases} sat_{fact2S} = 1/(1 - w_{sat2S} \cdot \textbf{THESATB}) & \text{if } \textbf{THESATB} < 0 \\ sat_{fact2S} = 1 + w_{sat2S} \cdot \textbf{THESATB} & \text{else} \end{cases} \tag{4.259}$$

$$Z_{iS} = \frac{Z_S \cdot \Sigma_S}{a_{1S} \cdot a_{2S}} - \frac{1}{q_{iS}} \cdot \left(\frac{A_{e1S}}{a_{1S}} + \frac{A_{e2S}}{a_{2S}}\right) \tag{4.260}$$

$$d_S = \frac{Z_{iS} \cdot q_{iS}}{Z_{iS} + 1} \tag{4.261}$$

$$\delta x_i = \text{MAX\_FUNC}\left(\frac{q_{iS} + d_{\infty} \cdot \delta x_{\infty}}{d_{\infty} - d_S}, 0, 10^{-6}\right) \tag{4.262}$$

$$\gamma_{sat} = \frac{f_{vsat}}{G_{mobS}} \cdot \frac{sat_{fact1S} + sat_{fact2S}}{2} \tag{4.263}$$

$$v_S = 1 - q_{iS}/d_S \tag{4.264}$$

$$v_D = 1 + \delta x_{\infty} \tag{4.265}$$

$$w_D = \left(\frac{2 \cdot d_S - q_{iS}}{d_{\infty}} - 2 - \delta x_{\infty}\right) \cdot \delta x_i \tag{4.266}$$

$$p_S = 2/{\gamma_{sat}}^2 \tag{4.267}$$

$$q_S = p_S \cdot v_S \tag{4.268}$$

$$p_D = p_S + w_D \tag{4.269}$$

$$q_D = p_S \cdot v_D \tag{4.270}$$

$$r_{cS} = \sqrt{q_S{}^2 + 4 \cdot p_S{}^3/27} \tag{4.271}$$

$$r_{cD} = \sqrt{q_D{}^2 + 4 \cdot p_D{}^3/27} \tag{4.272}$$

$$\delta x_{satS} = \left(\frac{r_{cS} + q_S}{2}\right)^{1/3} - \left(\frac{r_{cS} - q_S}{2}\right)^{1/3} \tag{4.273}$$

$$\delta x_{satD} = \left(\frac{r_{cD} + q_D}{2}\right)^{1/3} - \left(\frac{r_{cD} - q_D}{2}\right)^{1/3} \tag{4.274}$$

$$\delta x_{sat} = 0.94 \cdot \text{MAX\_FUNC}\left(\delta x_{satS}, \delta x_{satD}, 10 \cdot (d_S - d_\infty)^2\right) \tag{4.275}$$

$$q_{iDsatS} = q_{iS} + d_S \cdot \delta x_{sat} \tag{4.276}$$

$$q_{iDsatD} = d_\infty \cdot (\delta x_{sat} - \delta x_\infty) \tag{4.277}$$

$$q_{iDsat} = \text{MAX\_FUNC}\left(q_{iDsatS}, q_{iDsatD}, 36 \cdot (d_S - d_\infty)^2\right) \tag{4.278}$$

Else:

$$d_S = d_\infty \tag{4.279}$$

$$\delta x_{sat} = 0.94 \cdot (1 + \delta x_\infty) \tag{4.280}$$

$$q_{iDsat} = \frac{q_{iS}}{2} + d_\infty \left(\delta x_{sat} - \frac{\delta x_\infty}{2}\right) \tag{4.281}$$

Normalized saturation and effective drain voltages:

$$x_{nDS,0} = \delta x_{sat} + \ln\left(\frac{q_{iS}}{0.5 + \ln(1 + \exp(q_{iDsat} - 0.5))}\right) \tag{4.282}$$

$$x_{nDS,1} = 6 + \ln(1 + \exp(x_{nDS,0} - 6)) \tag{4.283}$$

$$x_{nDS,sat} = x_{satmax} - \ln(1 + \exp(x_{satmax} - x_{nDS,1})) \tag{4.284}$$

$$x_{Deff} = \frac{x_D}{\left((1 + \gamma_{AX}(x_D/x_{nDS,sat})^4)^{8/3} + (x_D/x_{nDS,sat})^{16}\right)^{1/16}} \tag{4.285}$$

4.2.9 Inversion charge and related quantities at drain side

Gate charge density at drain side:

$$q_{1D} = \text{CHARGE\_DENSITY}(x_{g1x}, x_{g2x}, x_{Deff}) \tag{4.286}$$

Then, inversion and back gate charge densities, normalized to $C'_{Si} \cdot \phi_T$ and $k_2 \cdot C'_{Si} \cdot \phi_T$ respectively, are calculated:

$$A_{e1D} = A_0 \cdot \exp(x_{g1x} - q_{1D} - x_{Deff}) \tag{4.287}$$

$$f_{qsqD} = {k_1}^2 \cdot {q_{1D}}^2 - A_{e1D} \tag{4.288}$$

If $f_{qsqD} < -0.005$:

$$f_{qctD} = \sqrt{|f_{qsqD}|} \cdot \cot\left(\sqrt{|f_{qsqD}|}/2\right) \tag{4.289}$$

$$f_{shD} = -\frac{f_{qsqD}}{\sin\left(\sqrt{|f_{qsqD}|}/2\right)^2} \tag{4.290}$$

$$f_{lnD} = \ln(f_{shD}) \tag{4.291}$$

Else if $f_{qsqD} > 0.005$:

$$\zeta_D = \exp\left(-\sqrt{|f_{qsqD}|}\right) \tag{4.292}$$

$$f_{qctD} = \sqrt{|f_{qsqD}|} \cdot \frac{1+\zeta_D}{1-\zeta_D} \tag{4.293}$$

$$f_{shD} = \frac{4 \cdot f_{qsqD}}{1 - \zeta_D \cdot (2-\zeta_D)} \cdot \zeta_D \tag{4.294}$$

$$f_{lnD} = \ln\left(\frac{4 \cdot f_{qsqD}}{1 - \zeta_D \cdot (2-\zeta_D)}\right) - \sqrt{|f_{qsqD}|} \tag{4.295}$$

Else:

$$f_{qctD} = 2 + \frac{f_{qsqD}}{6} \cdot \left(1 - \frac{f_{qsqD}}{60} \cdot \left(1 - \frac{f_{qsqD}}{42}\right)\right) \tag{4.296}$$

$$f_{shD} = 4 - \frac{f_{qsqD}}{3} \cdot \left(1 - \frac{f_{qsqD}}{20} \cdot \left(1 - \frac{5 f_{qsqD}}{126}\right)\right) \tag{4.297}$$

$$f_{lnD} = \ln(f_{shD}) \tag{4.298}$$

If $(1.01 \cdot k_1 \cdot q_{1D} + f_{qctD}) \leq 0$:

$$q_{iD} = \frac{k_1 \cdot q_{1D} - f_{qctD}}{1 - f_{shD}/A_{e1D}} \tag{4.299}$$

$$q_{2D} = \frac{q_{iD} - k_1 \cdot q_{1D}}{k_2} \tag{4.300}$$

Else if $(A_{e1D} \cdot k_1 \cdot q_{1D}) < \left(0.9 \cdot {k_1}^2 \cdot {q_{1D}}^2 \cdot (k_1 \cdot q_{1D} + f_{qctD})\right)$:

$$q_{iD} = \frac{A_{e1D}}{k_1 \cdot q_{1D} + f_{qctD}} \tag{4.301}$$

$$q_{2D} = \frac{q_{iD} - k_1 \cdot q_{1D}}{k_2} \tag{4.302}$$

Else:

$$q_{2D} = x_{g2x} - x_{g1x} + q_{1D} + 2 \cdot \ln(k_1 \cdot q_{1D} + f_{qctD}) - f_{lnD} \tag{4.303}$$

$$q_{iD} = k_1 \cdot q_{1D} + k_2 \cdot q_{2D} \tag{4.304}$$

Finally, some required quantities, including the drift electrostatic potential, are computed:

$$A_{e2D} = A_0 \cdot \exp(x_{g2x} - q_{2D} - x_{Deff}) \tag{4.305}$$

If $q_{iS} > 10^{-6}$:

$$b_{1D} = A_{e1D}/k_1 \tag{4.306}$$

$$b_{2D} = A_{e2D}/k_2 \tag{4.307}$$

$$a_{1D} = b_{1D} + 2 \cdot k_1 \cdot q_{1D} \tag{4.308}$$

$$a_{2D} = b_{2D} + 2 \cdot k_2 \cdot q_{2D} \tag{4.309}$$

$$\Sigma_D = 2 \cdot q_{iD} + b_{1D} + b_{2D} \tag{4.310}$$

$$\begin{cases} Z_D = \frac{-4 \cdot f_{qsqD} \cdot \Sigma_D}{q_{iD} \cdot (a_{1D} \cdot a_{2D} + 2 \cdot a_{1D} \cdot (q_{2D}+2) + 2 \cdot a_{2D} \cdot (q_{1D}+2))} & \text{if } |f_{qsqD}| > 0.005 \\ Z_D = \frac{A_{e1D} \cdot A_{e2D} \cdot \Sigma_D}{q_{iD} \cdot (a_{1D} \cdot A_{e1D} + a_{2D} \cdot A_{e2D} + a_{1D} \cdot a_{2D} \cdot q_{iD} \cdot \Gamma_D)} & \text{else} \\ \text{with } \Gamma_D = 1 + \frac{q_{iD}}{6} \cdot \left(1 - \frac{f_{qsqD}}{30} \cdot \left(1 - \frac{f_{qsqD}}{28} \cdot \left(1 - \frac{f_{qsqD}}{30}\right)\right)\right) & \end{cases} \tag{4.311}$$

Finally:

$$x_{driftD} = x_{Deff} + \ln(q_{iD}) \tag{4.312}$$

4.2.10 Mid-point inversion charge

$$q_{im} = \frac{q_{iS} + q_{iD}}{2} \tag{4.313}$$

$$\delta x_{drift} = x_{driftD} - x_{driftS} \tag{4.314}$$

4.2.11 Gate poly-depletion calculation

The equations in this section are only calculated when **SWPDEP** $= 1$; $ratio_{pd} = 1$ by default.
Physical background is similar to [5]. Some quantities are modified (charge, current ...) through application of folowing correction ratio:

$$ratio_{pd} = 1 - \frac{\left(\sqrt{\phi_T \cdot qim_{pd} + \frac{K_p^2}{4}} - \frac{K_p}{2}\right)^2}{\phi_T \cdot qim_{pd}} \tag{4.315}$$

Where charge used in this ratio comes from surface potential calculation without poly-depletion:

$$qim_{pd} = \frac{k_1 \cdot q_{1S} + k_1 \cdot q_{1D}}{2 \cdot k_1} \tag{4.316}$$

4.2.12 Mobility attenuation and series resistance

Front and back transverse effective fields, normalized to $C'_{Si}\phi_T/\epsilon_{ch}$:

$$e_{surf1D} = 2 \cdot \ln(1 + \exp(k_1 \cdot q_{1D}/2)) \tag{4.317}$$

$$e_{surf2D} = 2 \cdot \ln(1 + \exp(k_2 \cdot q_{2D}/2)) \tag{4.318}$$

$$e_{cpl1D} = e_{surf2D} - k_2 \cdot q_{2D} \tag{4.319}$$

$$e_{cpl2D} = e_{surf1D} - k_1 \cdot q_{1D} \tag{4.320}$$

$$e_{eff1D} = \eta_\mu \cdot e_{surf1D} + (1 - \eta_\mu) \cdot e_{cpl1D} \tag{4.321}$$

$$e_{eff2D} = \eta_\mu \cdot e_{surf2D} + (1 - \eta_\mu) \cdot e_{cpl2D} \tag{4.322}$$

Mid-values of surface and effective fields:

$$e_{surf1} = \frac{e_{surf1S} + e_{surf1D}}{2} \tag{4.323}$$

$$e_{surf2} = \frac{e_{surf2S} + e_{surf2D}}{2} \tag{4.324}$$

$$e_{cpl1} = \frac{e_{cpl1S} + e_{cpl1D}}{2} \tag{4.325}$$

$$e_{cpl2} = \frac{e_{cpl2S} + e_{cpl2D}}{2} \tag{4.326}$$

$$e_{eff1} = \frac{e_{eff1S} + e_{eff1D}}{2} \tag{4.327}$$

$$e_{eff2} = \frac{e_{eff2S} + e_{eff2D}}{2} \tag{4.328}$$

Non-universality correction factor:

$$f_{cor} = \frac{\text{MAX\_FUNC}(1 + X_{cor} \cdot (e_{cpl1} + \textbf{XCORB} \cdot e_{cpl2}), 0, 0.01)}{\text{MAX\_FUNC}(1 + 0.2 \cdot X_{cor} \cdot (e_{cpl1} + \textbf{XCORB} \cdot e_{cpl2}), 0, 0.01)} \tag{4.329}$$

Coulomb scattering term:

$$q_{i1m} = \frac{e_{surf1}}{e_{surf1} + e_{surf2}} \cdot q_{im} \tag{4.330}$$

$$q_{i2m} = \frac{e_{surf2}}{e_{surf1} + e_{surf2}} \cdot q_{im} \tag{4.331}$$

$$G_{CS} = \frac{C_S \cdot (1 + \textbf{CSFI} \cdot e_{cpl1} + \textbf{CSBI} \cdot e_{cpl2})}{(1 + q_{i1m}/q_{i1th,CS} + q_{i2m}/q_{i2th,CS})^{\theta_{CS}}} \tag{4.332}$$

Series resistance term:

$$\begin{cases} f_{RSG} = 1 - \textbf{RSG} \cdot {q_{im}}^{\textbf{THERSG}} & \text{if } \textbf{RSG} < 0 \\ f_{RSG} = 1/(1 + \textbf{RSG} \cdot {q_{im}}^{\textbf{THERSG}}) & \text{else} \end{cases} \tag{4.333}$$

$$G_{RS} = f_{RS} \cdot C'_{Si} \cdot (q_{im} \cdot f_{RSG} + \mathbf{RSIG}) \cdot \text{MAX\_FUNC}(1 - \mathbf{RSB} \cdot x_{g20}, 0, 0.01) \tag{4.334}$$

Total mobility degradation term, including high field mobility effect:

$$G_{mob1} = 1 + (f_{\mu E} \cdot e_{eff1})^{\theta_\mu} + G_{CS} + \mathbf{FACTUO} \cdot \beta_{N1} G_{RS} \tag{4.335}$$

$$G_{mob2} = 1 + (f_{\mu E} \cdot e_{eff2})^{\theta_\mu} + G_{CS} + \mathbf{FACTUO} \cdot \beta_{N2} G_{RS} \tag{4.336}$$

$$c_1 = e_{surf1} \cdot \beta_{N1} \cdot ratio_{pd} \tag{4.337}$$

$$c_2 = e_{surf2} \cdot \beta_{N2} \tag{4.338}$$

$$c_{sum} = c_1 + c_2 \tag{4.339}$$

$$G_{mob} = f_{cor} \cdot \frac{c_{sum}}{c_1/G_{mob1} + c_2/G_{mob2}} \tag{4.340}$$

4.2.13 Channel length modulation

$$q^*_{im1} = q_{im} + 4 \tag{4.341}$$

$$\begin{cases} r_1 = \frac{q_{im}}{q^*_{im1}} \cdot 1/(1 + \mathbf{ALPB} \cdot q_{im2}) & \text{if } \mathbf{ALPB} > 0 \\ r_1 = \frac{q_{im}}{q^*_{im1}} \cdot (1 - \mathbf{ALPB} \cdot q_{im2}) & \text{else} \end{cases} \tag{4.342}$$

$$f_{clm} = \ln\left(1 + \frac{x_D - x_{Deff}}{\mathbf{VP}/\phi_T + \mathbf{VPG} \cdot {q_{im}}^2}\right) \tag{4.343}$$

$$\Delta L/L = \mathbf{ALP} \cdot f_{clm} \cdot r_1 \tag{4.344}$$

$$G_{\Delta L} = \frac{1}{1 + \Delta L/L \cdot (1 + \Delta L/L)} \tag{4.345}$$

4.2.14 Velocity saturation

$$w_{sat1} = \frac{100 \cdot e_{surf1}}{100 + e_{surf1}} \tag{4.346}$$

$$\begin{cases} sat_{fact1} = 1/(1 - w_{sat1} \cdot \mathbf{THESATG}) & \text{if } \mathbf{THESATG} < 0 \\ sat_{fact1} = 1 + w_{sat1} \cdot \mathbf{THESATG} & \text{else} \end{cases} \tag{4.347}$$

$$w_{sat2} = \frac{100 \cdot e_{surf2}}{100 + e_{surf2}} \tag{4.348}$$

$$\begin{cases} sat_{fact2} = 1/(1 - w_{sat2} \cdot \mathbf{THESATB}) & \text{if } \mathbf{THESATB} < 0 \\ sat_{fact2} = 1 + w_{sat2} \cdot \mathbf{THESATB} & \text{else} \end{cases} \tag{4.349}$$

$$G_\gamma = f_{vsat} \cdot \delta x_{drift} \cdot \frac{sat_{fact1} + sat_{fact2}}{2} \tag{4.350}$$

$$z_{sat} = \left(\frac{G_\gamma}{G_{mob} \cdot G_{\Delta L}}\right)^2 \tag{4.351}$$

4.2.15 Quantum confinement

Here is detailed the second quantum confinement correction that complements the correction described in paragraph 4.2.3.

If **QMC** > 0:

$$qm_{fact1} = \frac{1 + k_1 \cdot qq \cdot 0.6/({e_{surf1}}^2 + 60)^{1/6}}{t_{ox1,fact}} \tag{4.352}$$

$$qm_{fact2} = \frac{1 + k_2 \cdot qq \cdot 0.6/({e_{surf2}}^2 + 60)^{1/6}}{t_{ox2,fact}} \tag{4.353}$$

Else:

$$qm_{fact1} = 1 \tag{4.354}$$

$$qm_{fact2} = 1 \tag{4.355}$$

4.2.16 Normalized channel current

If $q_{iS} > 10^{-6}$:

$$Z_{iD} = \frac{Z_D \cdot \Sigma_D}{a_{1D} \cdot a_{2D}} - \frac{1}{q_{iD}} \cdot \left(\frac{A_{e1D}}{a_{1D}} + \frac{A_{e2D}}{a_{2D}}\right) \tag{4.356}$$

If $\left(q_{iD} > 10^{-6}\right)$:

If $(|a_{2D}| < 0.01)$:

$$\left| \begin{array}{l} t_1 = \frac{2+q_{1D}+a_{1D}/2}{(2.0+q_{2D})} \cdot \frac{a_{2D}}{a_{1D}} \\ t_2 = 1 - t_1 + {t_1}^2 - {t_1}^3 \\ t_3 = \frac{Z_D \cdot q_{iD} - A_{e1D}}{a_{1D} \cdot q_{iD}} - \frac{1}{q_{iD}} \cdot \frac{k_2 \cdot q_{2D} - 2 \cdot f_{qsqD} \cdot (t_1/a_{2D} - 1/a_{1D}) \cdot t_2}{2+q_{2D}} \\ d_D = \frac{t_3 \cdot q_{iD}}{t_3+1} \end{array} \right. \tag{4.357}$$

Else:

$$\left| \; d_D = \frac{Z_{iD} \cdot q_{iD}}{Z_{iD}+1} \right. \tag{4.358}$$

Else:

$$\left| \; d_D = d_\infty \right. \tag{4.359}$$

If $|d_D - d_S| > 10^{-3}$:

$$\left| \begin{array}{l} L_S = q_{iD} - q_{iS} - d_D \cdot \delta x_{drift} \\ L_D = q_{iD} - q_{iS} - d_S \cdot \delta x_{drift} \\ U_S = \sqrt{{L_S}^2 + 1 + 36 \cdot (d_D - d_S)^2} \\ U_D = \sqrt{{L_D}^2 + 1 + 36 \cdot (d_D - d_S)^2} \\ \delta i_{drift} = \frac{L_S \cdot U_D - L_D \cdot U_S + \left(1+36 \cdot (d_D - d_S)^2\right) \ln((L_D+U_D)/(L_S+U_S))}{4 \cdot (d_D - d_S)} \end{array} \right. \tag{4.360}$$

Else:

$$\delta i_{drift} = -\frac{\delta x_{drift}{}^3 \cdot (d_D - d_S)^2}{24 \cdot \sqrt{1 + 36 \cdot (d_D - d_S)^2}} \tag{4.361}$$

Else:

$$d_D = d_\infty \tag{4.362}$$

$$\delta x_{drift} = 0 \tag{4.363}$$

$$i_{DS,norm} = q_{im} \cdot \delta x_{drift} + \delta i_{drift} + q_{iS} - q_{iD} \tag{4.364}$$

4.2.17 Effective gate charge at front and back interfaces

If $q_{iS} > 10^{-6}$:

$$\hat{q}_{1S} = \frac{a_{1S}}{A_{e1S}/q_{iS} - Z_S} \tag{4.365}$$

$$\hat{q}_{1D} = \frac{a_{1D}}{A_{e1D}/q_{iD} - Z_D} \tag{4.366}$$

$$k_1 h_{10} = \frac{i_{DS,norm}}{\hat{q}_{1S} - \hat{q}_{1D}} \tag{4.367}$$

$$\hat{q}_{2S} = \frac{a_{2S}}{A_{e2S}/q_{iS} - Z_S} \tag{4.368}$$

$$\hat{q}_{2D} = \frac{a_{2D}}{A_{e2D}/q_{iD} - Z_D} \tag{4.369}$$

$$k_2 h_{20} = \frac{i_{DS,norm}}{\hat{q}_{2S} - \hat{q}_{2D}} \tag{4.370}$$

Else:

$$\zeta_1 = -2 \cdot s_1 \cdot \left(\frac{1}{k_1 \cdot \hat{q}_{1\infty}} + \frac{1}{d_\infty} \right) \tag{4.371}$$

$$\zeta_2 = -2 \cdot s_2 \cdot \left(\frac{1}{k_2 \cdot \hat{q}_{2\infty}} + \frac{1}{d_\infty} \right) \tag{4.372}$$

$$\xi_1 = \frac{\zeta_2/k_2 + (\zeta_2 - \zeta_1)/d_\infty - (\zeta_1/k_1 + \zeta_2/k_2)/\hat{q}_{1\infty}}{3 + 2 \cdot (s_1/k_1 + s_2/k_2)} \tag{4.373}$$

$$\xi_2 = \frac{\zeta_1/k_1 + (\zeta_1 - \zeta_2)/d_\infty - (\zeta_2/k_2 + \zeta_1/k_1)/\hat{q}_{2\infty}}{3 + 2 \cdot (s_2/k_2 + s_1/k_1)} \tag{4.374}$$

$$k_1 h_{10} = -\frac{1}{\hat{q}_{1\infty} \cdot (\xi_1 \cdot \hat{q}_{1\infty} + 1/d_\infty)} \tag{4.375}$$

$$k_2 h_{20} = -\frac{1}{\hat{q}_{2\infty} \cdot (\xi_2 \cdot \hat{q}_{2\infty} + 1/d_\infty)} \tag{4.376}$$

$$k_1h_1 = k_1h_{10} \cdot \frac{\sqrt{1+z_{sat}}}{1+3\cdot z_{sat}/2} \tag{4.377}$$

$$k_2h_2 = k_2h_{20} \cdot \frac{\sqrt{1+z_{sat}}}{1+3\cdot z_{sat}/2} \tag{4.378}$$

$$\Delta_{k1q1} = \frac{k_1 \cdot q_{1D} - k_1 \cdot q_{1S}}{2} \tag{4.379}$$

$$\Delta_{k2q2} = \frac{k_2 \cdot q_{2D} - k_2 \cdot q_{2S}}{2} \tag{4.380}$$

$$P_1 = \frac{\Delta_{k1q1}}{k_1h_1} \tag{4.381}$$

$$P_2 = \frac{\Delta_{k2q2}}{k_2h_2} \tag{4.382}$$

4.3 Channel current

In the remainder of this document, some variables (e.g., x_{g10}) are labeled 'dc' or 'ac' (e.g., $x_{g10,dc}$ or $x_{g10,ac}$). For the channel current calculation, the variables labeled 'dc' result from the *first* evaluation of Eqs. (4.113)–(4.382).

4.3.1 Mobility attenuation and series resistance

$$\beta_{Neff} = \mathbf{FACTUO} \cdot \frac{e_{sum,dc}}{e_{surf1,dc} + e_{surf2,dc}} \tag{4.383}$$

4.3.2 Channel length modulation

$$\Delta L_1/L = \left(\mathbf{ALP} + \frac{f_{alp1}}{q^*_{im1,dc}}\right) \cdot f_{clm,dc} \cdot r_{1,dc} \tag{4.384}$$

$$F_{\Delta L} = \frac{1 + \Delta L_1/L \cdot (1 + \Delta L_1/L)}{1 + \Delta L/L \cdot (1 + \Delta L/L)} \tag{4.385}$$

4.3.3 Velocity saturation

$$G_{vsat} = G_{mob,dc} \cdot G_{\Delta L_{dc}} \cdot \sqrt{1 + z_{sat,dc}} \tag{4.386}$$

4.3.4 Quantum mechanical corrections

If $\mathbf{QMC} > 0$:

$$qm_{fact} = \frac{e_{surf1,dc} + e_{surf2,dc}}{e_{surf1}/qm_{fact1,dc} + e_{surf2,dc}/qm_{fact2,dc}} \tag{4.387}$$

Else:

$$qm_{fact} = 1 \tag{4.388}$$

4.3.5 Channel current

$$I_{DS} = \frac{F_{\Delta L}}{G_{vsat} \cdot qm_{fact}} \cdot \beta_{Neff} \cdot {\phi_T}^2 \cdot C'_{Si,dc} \cdot i_{DS,norm,dc} \tag{4.389}$$

4.4 Gate current, intrinsic charges and overlap related variables

In this paragraph are defined some variables that will be used for gate current, GIDL/GISL current and intrinsic charge models.

4.4.1 Voltages for overlap currents and charges

$$x_{gs,ov} = -\frac{V_{GS}}{\phi_{T0}} \tag{4.390}$$

$$x_{gd,ov} = -\frac{V_{GD}}{\phi_{T0}} \tag{4.391}$$

$$x_{gs,ovcv} = -\frac{V_{GS} - \textbf{TYPE} \cdot \textbf{DVFBOV} - E_g/2}{\phi_{T0}} \tag{4.392}$$

$$x_{gd,ovcv} = -\frac{V_{GD} - \textbf{TYPE} \cdot \textbf{DVFBOV} - E_g/2}{\phi_{T0}} \tag{4.393}$$

4.4.2 Surface potential and gate dielectric voltage drop in gate-source overlap region

$$G_{ov} = \frac{\sqrt{2 \cdot q \cdot \epsilon_{ch} \cdot \textbf{NOV} \cdot 10^6}}{C'_{ox1} \cdot \sqrt{\phi_{T0}}} \tag{4.394}$$

$$\xi_{ov} = 1 + \frac{G_{ov}}{\sqrt{2}} \tag{4.395}$$

$$\xi_{mrg,ov} = 10^{-5} \cdot \xi_{ov} \tag{4.396}$$

$$x_{g1,ov} = 1.25 + 0.7324648775 \cdot G_{ov} \tag{4.397}$$

$$x_{S,ov} = \text{SP\_FDSOI\_OV}(x_{gs,ov}) \tag{4.398}$$

$$x_{S,ovcv} = \text{SP\_FDSOI\_OV}(x_{gs,ovcv}) \tag{4.399}$$

$$V_{ovS} = -\phi_{T0} \cdot (x_{gs,ov} + x_{S,ov}) \tag{4.400}$$

$$V_{ovS,cv} = -\phi_{T0} \cdot (x_{gs,ovcv} + x_{S,ovcv}) \tag{4.401}$$

4.4.3 Surface potential and gate dielectric voltage drop in gate-drain overlap region

$$G_{ov} = \frac{\sqrt{2 \cdot q \cdot \epsilon_{ch} \cdot \textbf{NOVD} \cdot 10^6}}{C'_{ox1} \cdot \sqrt{\phi_{T0}}} \tag{4.402}$$

$$\xi_{ov} = 1 + \frac{G_{ov}}{\sqrt{2}} \tag{4.403}$$

$$\xi_{mrg,ov} = 10^{-5} \cdot \xi_{ov} \tag{4.404}$$

$$x_{g1,ov} = 1.25 + 0.7324648775 \cdot G_{ov} \tag{4.405}$$

$$x_{D,ov} = \text{SP\_FDSOI\_OV}(x_{gd,ov}) \tag{4.406}$$

$$x_{D,ovcv} = \text{SP\_FDSOI\_OV}(x_{gd,ovcv}) \tag{4.407}$$

$$V_{ovD} = -\phi_{T0} \cdot (x_{gd,ov} + x_{D,ov}) \tag{4.408}$$

$$V_{ovD,cv} = -\phi_{T0} \cdot (x_{gd,ovcv} + x_{D,ovcv}) \tag{4.409}$$

4.5 Gate current

In this section is detailed the calculation of gate current components. This calculation is not carried out when the flag **SWIGATE** is set to 0.

4.5.1 Gate to source overlap component

$$\psi_t = \text{MIN\_FUNC}(V_{ovS} + D_{ov}, 0, 0.01) \tag{4.410}$$

$$G_{C2,oveff} = \frac{\mathbf{GC2OVACC} + \mathbf{GC2OVINV} \cdot \exp(x_{gs,ov}/2)}{1 + \exp(x_{gs,ov}/2)} \tag{4.411}$$

$$G_{C3,oveff} = \frac{\mathbf{GC3OVACC} + \mathbf{GC3OVINV} \cdot \exp(x_{gs,ov}/2)}{1 + \exp(x_{gs,ov}/2)} \tag{4.412}$$

$$G_{CQ,oveff} = \frac{G_{CQ,ovacc} + G_{CQ,ovinv} \cdot \exp(x_{gs,ov}/2)}{1 + \exp(x_{gs,ov}/2)} \tag{4.413}$$

$$I_{G,oveff} = \frac{I_{G,ovacc} + I_{G,ovinv} \cdot \exp(x_{gs,ov}/2)}{1 + \exp(x_{gs,ov}/2)} \tag{4.414}$$

$$I_{G,ovefffowler} = \frac{I_{G,fnovinv} \cdot \exp(x_{gs,ov}/2)}{1 + \exp(x_{gs,ov}/2)} \cdot 10^{-6} \tag{4.415}$$

$$\begin{cases} z_g = \text{MIN\_FUNC}\left(\frac{\sqrt{V_{ovS}{}^2 + 10^{-4}}}{\mathbf{CHIB}}, G_{CQ,oveff}, 10^{-6}\right) & \text{if } G_{C3,oveff} < 0 \\ z_g = \frac{\sqrt{V_{ovS}{}^2 + 10^{-4}}}{\mathbf{CHIB}} & \text{else} \end{cases} \tag{4.416}$$

$$\Delta_{Si} = \exp\left(3 + x_{S,ov} + \frac{\psi_t}{\phi_{T0}}\right) \tag{4.417}$$

$$\Delta_{gate} = \exp\left(3 + x_{S,ov} + \frac{\psi_t - V_{GS}}{\phi_{T0}}\right) \tag{4.418}$$

$$f_{GS,ov} = \frac{1 + \exp(\mathbf{GCDOV} \cdot (V_{GD} - \mathbf{GCVDOV}))}{1 + \exp(\mathbf{GCDOV} \cdot (V_{GD} - \mathbf{GCVDOV})) \cdot \exp(\mathbf{GCDOV} \times V_{SD})} \tag{4.419}$$

$$\begin{aligned} I_{GS,ov} =& I_{G,oveff} \cdot \ln\left(\frac{1 + \Delta_{Si}}{1 + \Delta_{gate}}\right) \cdot \exp\left(B_{ov} \cdot \left(z_g \cdot (G_{C2,oveff} + z_g \cdot G_{C3,oveff}) - \frac{3}{2}\right)\right) \cdot f_{GS,ov} \\ & - I_{G,ovefffowler} \cdot \exp\left(-B_{ov} \cdot \left(\frac{\mathbf{GCOVINVFN}}{z_g}\right)\right) \cdot f_{GS,ov} \end{aligned} \tag{4.420}$$

4.5.2 Gate to drain overlap component

$$\psi_t = \text{MIN\_FUNC}(V_{ovD} + D_{ov}, 0, 0.01) \tag{4.421}$$

$$G_{C2,oveff} = \frac{\textbf{GC2OVACC} + \textbf{GC2OVINV} \cdot \exp(x_{gd,ov}/2)}{1 + \exp(x_{gd,ov}/2)} \tag{4.422}$$

$$G_{C3,oveff} = \frac{\textbf{GC3OVACC} + \textbf{GC3OVINV} \cdot \exp(x_{gd,ov}/2)}{1 + \exp(x_{gd,ov}/2)} \tag{4.423}$$

$$G_{CQ,oveff} = \frac{G_{CQ,ovacc} + G_{CQ,ovinv} \cdot \exp(x_{gd,ov}/2)}{1 + \exp(x_{gd,ov}/2)} \tag{4.424}$$

$$I_{G,oveff} = \frac{I_{G,ovaccD} + I_{G,ovinvD} \cdot \exp(x_{gd,ov}/2)}{1 + \exp(x_{gd,ov}/2)} \tag{4.425}$$

$$I_{G,oveff fowler} = \frac{I_{G,fnovinvD} \cdot \exp(x_{gd,ov}/2)}{1 + \exp(x_{gd,ov}/2)} \cdot 10^{-6} \tag{4.426}$$

$$\begin{cases} z_g = \text{MIN\_FUNC}\left(\frac{\sqrt{V_{ovD}{}^2+10^{-4}}}{\textbf{CHIB}}, G_{CQ,oveff}, 10^{-6}\right) & \text{if } G_{C3,oveff} < 0 \\ z_g = \frac{\sqrt{V_{ovD}{}^2+10^{-4}}}{\textbf{CHIB}} & \text{else} \end{cases} \tag{4.427}$$

$$\Delta_{Si} = \exp\left(3 + x_{D,ov} + \frac{\psi_t}{\phi_{T0}}\right) \tag{4.428}$$

$$\Delta_{gate} = \exp\left(3 + x_{D,ov} + \frac{\psi_t - V_{GD}}{\phi_{T0}}\right) \tag{4.429}$$

$$f_{GD,ov} = \frac{1 + \exp(\textbf{GCDOV} \cdot (V_{GS} - \textbf{GCVDOV}))}{1 + \exp(\textbf{GCDOV} \cdot (V_{GS} - \textbf{GCVDOV})) \cdot \exp(\textbf{GCDOV} \cdot V_{DS})} \tag{4.430}$$

$$\begin{aligned} I_{GD,ov} =& I_{G,oveff} \cdot \ln\left(\frac{1+\Delta_{Si}}{1+\Delta_{gate}}\right) \cdot \exp\left(B_{ov} \cdot \left(z_g \cdot (G_{C2,oveff} + z_g \cdot G_{C3,oveff}) - \frac{3}{2}\right)\right) \cdot f_{GD,ov} \\ & - I_{G,oveff fowlerD} \cdot \exp\left(-B_{ov} \cdot \left(\frac{\textbf{GCOVINVFND}}{z_g}\right)\right) \cdot f_{GD,ov} \end{aligned} \tag{4.431}$$

4.5.3 Gate to channel component

For the gate-channel current calculation, the variables labeled 'dc' result from the *first* evaluation of Eqs. (4.113)–(4.382).

$$x_{DS} = -2 \cdot \Delta_{k1q1,dc}/k_{1,dc} \tag{4.432}$$

$$V_m = \phi_T \cdot \left(\frac{x_{DS}}{2} - \ln\left(\frac{1 + \exp(x_{DS} - x_{Deff,dc})}{2}\right)\right) \tag{4.433}$$

$$q_{1m} = \frac{q_{1S,dc} + q_{1D,dc}}{2} \tag{4.434}$$

$$V_{oxm} = \phi_T \cdot q_{1m,dc} \tag{4.435}$$

$$\psi_t = \text{MIN\_FUNC}(V_{oxm} + D_{ch}, 0, 0.01) \tag{4.436}$$

$$\begin{cases} z_g = \text{MIN\_FUNC}\left(\frac{\sqrt{V_{oxm}{}^2+10^{-4}}}{\mathbf{CHIB}}, G_{CQ,ch}, 10^{-6}\right) & \text{if } \mathbf{GC3CH} < 0 \\ z_g = \frac{\sqrt{V_{oxm}{}^2+10^{-4}}}{\mathbf{CHIB}} & \text{else} \end{cases} \tag{4.437}$$

$$x_{g1xshift} = x_{g1x,dc} + \frac{E_g}{2 \cdot \phi_{T0}} \tag{4.438}$$

$$\Delta_{Si} = \exp\left(\left(x_{g1xshift} - q_{1m} + \frac{\psi_t - \alpha_b - V_m}{\phi_{T0}}\right) \cdot n_{iginv}\right) \tag{4.439}$$

$$\Delta_{gate} = \Delta_{Si} \cdot \exp\left(-\frac{V_{GS} - V_m}{\phi_T} \cdot n_{iginv}\right) \tag{4.440}$$

$$I_{GC0} = I_{G,inv} \cdot \ln\left(\frac{1 + \Delta_{Si}}{1 + \Delta_{gate}}\right) \cdot \exp\left(B_{ch} \cdot \left(z_g(\mathbf{GC2CH} + z_g \cdot \mathbf{GC3CH}) - \frac{3}{2}\right)\right) \tag{4.441}$$

4.5.4 Source/drain partitioning of gate to channel current

If $x_{g1xshift} > 0$:

$$u_0 = \frac{\mathbf{CHIB}/\phi_T}{B_{ch} \cdot (\mathbf{GC2CH} + 2 \cdot z_g \cdot \mathbf{GC3CH})} \tag{4.442}$$

$$x = \frac{x_{DS,dc}}{2 \cdot u_0} \tag{4.443}$$

$$b = \frac{u_0}{k_1 h_{1,dc}/k_{1,dc}} \tag{4.444}$$

$$B_g = \frac{b \cdot (1 - b)}{2} \tag{4.445}$$

$$A_g = \frac{1}{2} - 3 \cdot B_g \tag{4.446}$$

$$p_{GC} = (1 - b) \cdot \frac{\sinh(x)}{x} + b \cdot \cosh(x) \tag{4.447}$$

$$p_{GD} = \frac{p_{GC}}{2} - B_g \cdot \sinh(x) - \frac{A_g}{x} \cdot \left(\cosh(x) - \frac{\sinh(x)}{x}\right) \tag{4.448}$$

Else:

$$p_{GC} = 1 \tag{4.449}$$

$$p_{GD} = \frac{1}{2} \tag{4.450}$$

$$I_{GC} = I_{GC0} \cdot p_{GC} \tag{4.451}$$

$$I_{GCD} = I_{GC0} \cdot p_{GD} \tag{4.452}$$

$$I_{GCS} = I_{GC} - I_{GCD} \tag{4.453}$$

4.5.5 Gate to source and gate to drain total currents

$$I_{GS} = I_{GCS} + I_{GS,ov} \tag{4.454}$$

$$I_{GD} = I_{GCD} + I_{GD,ov} \tag{4.455}$$

4.6 Gate Induced Drain/Source Leakage (GIDL/GISL)

This paragraph details the calculation of GIDL and GISL currents. Notice that the model, coming from Leti-UTSOI1, has been simplified. In particular, GIDL/GISL currents are no longer proportional to **LOV**.

4.6.1 Gate induced source leakage

$$V_{tovS} = \sqrt{{V_{ovS}}^2 + \mathbf{CGIDL}^2 \cdot {V_{SB}}^2 + 10^{-6}} \tag{4.456}$$

$$I_{GISL} = -A_{GIDL} \cdot V_{SD} \cdot V_{ovS} \cdot V_{tovS} \cdot \exp\left(-\frac{B_{GIDL}}{V_{tovS}}\right) \cdot \frac{1 + \exp(\mathbf{DGIDL} \cdot V_{SD})}{2} \tag{4.457}$$

4.6.2 Gate induced drain leakage

$$V_{tovD} = \sqrt{{V_{ovD}}^2 + \mathbf{CGIDLD}^2 \cdot {V_{DB}}^2 + 10^{-6}} \tag{4.458}$$

$$I_{GIDL} = -A_{GIDLD} \cdot V_{DS} \cdot V_{ovD} \cdot V_{tovD} \cdot \exp\left(-\frac{B_{GIDLD}}{V_{tovD}}\right) \cdot \frac{1 + \exp(\mathbf{DGIDLD} \cdot V_{DS})}{2} \tag{4.459}$$

4.7 Edge Transistor Current

The equations in this Section are only calculated when **SWEDGE** $= 1$ and **BETNEDGE** > 0.
Voltages for channel current and intrinsic charge models for edge transistor:

$$x_{d,edge} = \frac{V_{DS}}{\phi_{T,edge}} \tag{4.460}$$

$$x_{dsx,edge} = \frac{\sqrt{{V_{DS}}^2 + 0.01} - 0.1}{\phi_{T,edge}} \tag{4.461}$$

$$x_{g10,edge} = \frac{V_{GS} - V_{FB1,edge}}{\phi_{T,edge}} - \frac{x_{d,edge} - x_{dsx,edge}}{2} - \frac{E_g}{2 \cdot \phi_{T0}} \tag{4.462}$$

$$x_{g20,edge} = \frac{-V_{SB} - V_{FB2,edge}}{\phi_{T,edge}} - \frac{x_{d,edge} - x_{dsx,edge}}{2} - \frac{E_g}{2 \cdot \phi_{T0}} \tag{4.463}$$

Short channel effects for edge transistor:

$$c_{sce1,edge} = \frac{1}{1 + PSCE_{1,edge}} \tag{4.464}$$

$$c_{sce2,edge} = \frac{1}{1 + PSCE_{2,edge}} \tag{4.465}$$

$$x_{d0,edge} = \frac{\mathbf{CFDEDGE}}{\phi_{T,edge}} \tag{4.466}$$

$$\delta x_{g1,DIBL,edge} = 2 \cdot CF_{1,edge} \cdot x_{d0,edge} \cdot \left(\sqrt{1 + x_{dsx,edge}/x_{d0,edge}} - 1 \right) \tag{4.467}$$

$$\delta x_{g2,DIBL,edge} = 2 \cdot CF_{2,edge} \cdot x_{d0,edge} \cdot \left(\sqrt{1 + x_{dsx,edge}/x_{d0,edge}} - 1 \right) \tag{4.468}$$

$$x_{g1,edge} = (x_{g10,edge} + \delta x_{g1,DIBL,edge}) \cdot c_{sce1,edge} + \frac{x_{d,edge} - x_{dsx,edge}}{2} \tag{4.469}$$

$$x_{g2,edge} = (x_{g20,edge} + \delta x_{g2,DIBL,edge}) \cdot c_{sce2,edge} + \frac{x_{d,edge} - x_{dsx,edge}}{2} \tag{4.470}$$

$$x_{g1x,edge} = \text{MIN\_FUNC}(x_{g2,edge} + \mathbf{CICFEDGE} \cdot (x_{g1,edge} - x_{g2,edge}), x_{satmax}, 0.01) \tag{4.471}$$

$$x_{g2x,edge} = \text{MIN\_FUNC}(x_{g1,edge} + \mathbf{CICEDGE} \cdot (x_{gZ,edge} - x_{g1,edge}), x_{satmax}, 0.01) \tag{4.472}$$

Interface coupling in subthreshold regime for edge transistor:

$$k_{1,edge} = \frac{k_{1,1D}}{c_{sce1,edge}} \tag{4.473}$$

$$k_{2,edge} = \frac{k_{2,1D}}{c_{sce2,edge}} \tag{4.474}$$

$$k_{eq,edge} = \frac{1}{1 + 1/k_{1,edge} + 1/k_{2,edge}} \tag{4.475}$$

$$A_{0,edge} = \frac{f_{A0,edge}}{{C'_{Si,dc}}^2} \tag{4.476}$$

$$\delta x_{WI,edge} = k_{eq,edge} \cdot (x_{g1x,edge} - x_{g2x,edge}) \tag{4.477}$$

Inversion charge in weak inversion:
If $|x_{g2x,edge} - x_{g1x,edge}| <= 10^{-12}$:

$$t_1 = \left(1.0 - \frac{k_{eq,edge}}{k_{1,edge}} - \frac{k_{eq,edge}}{k_{2,edge}} \right) \tag{4.478}$$

$$t_2 = \delta x_{WI,edge} \cdot \left(\frac{1}{k_{2,edge}} + \frac{k_{eq,edge}}{2 \cdot k_{1,edge}^2} - \frac{k_{eq,edge}}{2 \cdot k_{2,edge}^2} - \frac{1}{k_{eq,edge}} \right) \tag{4.479}$$

$$q_{ilow,edge} = \frac{1}{2} \cdot (t_1 - t_2) \cdot \frac{A_{0,edge}}{k_{eq,edge}} \cdot \exp\left(x_{g1x,edge}\right) \tag{4.480}$$

Else:

$$t_1 = \exp\left(\frac{-\delta x_{WI,edge}}{k_{1,edge}} \right) \tag{4.481}$$

$$t_2 = \exp\left(\left(\frac{1}{k_{2,edge} \cdot k_{eq,edge}} - \frac{1}{k_{eq,edge}} \right) \cdot \delta x_{WI,edge} \right) \tag{4.482}$$

$$q_{ilow,edge} = \frac{1}{2} \cdot (t_1 - t_2) \cdot \frac{A_{0,edge}}{\delta x_{WI,edge}} \cdot \exp\left(x_{g1x,edge}\right) \tag{4.483}$$

Inversion charge at source and drain side:

$$q_{is,edge} = \mathcal{W}_0(q_{ilow,edge}) \tag{4.484}$$

$$q_{id,edge} = \mathcal{W}_0(q_{ilow,edge} \cdot \exp\left(-x_{Deff,dc}\right)) \tag{4.485}$$

Edge transistors current:

$$i_{DS,norm,edge} = \left(\frac{1}{2} \cdot (q_{is,edge} + q_{id,edge}) + 1.0\right)(q_{is,edge} - q_{id,edge}) \tag{4.486}$$

$$I_{DS,edge} = \frac{C'_{ox1} \cdot \beta_{N,edge} \cdot {\phi_{T,edge}}^2}{G_{mob}} \cdot i_{DS,norm,edge} \tag{4.487}$$

4.8 Impact ionization current

The equations in this section are only calculated when **SWIMPACT** $= 1$.

$$\Delta V_{sat} = (x_{d,dc} - \mathbf{A3} \cdot x_{Deff,dc}) \cdot \phi_T \tag{4.488}$$

$$M_{\text{avl}} = \begin{cases} 0 & \text{for } \Delta V_{sat} \leq 0 \\ \mathbf{A1} \cdot \Delta V_{sat} \cdot \exp\left(-\dfrac{a_2}{\Delta V_{sat}}\right) & \text{for } \Delta V_{sat} > 0 \end{cases} \tag{4.489}$$

$$I_{impact} = M_{avl} \cdot I_{DS} \tag{4.490}$$

4.9 Charge model

This part is dedicated to the calculations of intrinsic and parasitic charges in the different electrodes. The intrinsic charge model is derived from [10].
In this section, some variables are labeled 'ac'. These ones are calculated by following two possibilities. If **SWQMOD** $= 1$, their values result from the *second* evaluation of Eqs. (4.113)–(4.382). In other case, their values are equal to their 'dc'-counterpart.

4.9.1 Quantum mechanical corrections

$$k_1q_{1m} = \frac{k_1q_{1S,ac} + k_1q_{1D,ac}}{2} + \mathbf{FSCEAC} \cdot \frac{\delta x_{WI,1D,ac} - \delta x_{WI,ac}}{1 + q_{im,ac}/4} \tag{4.491}$$

$$k_2q_{2m} = \frac{k_2q_{2S,ac} + k_2q_{2D,ac}}{2} - \mathbf{FSCEAC} \cdot \frac{\delta x_{WI,1D,ac} - \delta x_{WI,ac}}{1 + q_{im,ac}/4} \tag{4.492}$$

$$\begin{cases} k_1q_{1eff} = k_1q_{1m} - \frac{qm_{fact1,ac}-1}{qm_{fact1,ac}} \cdot q_{i1m,ac} & \text{if } \mathbf{QMC} > 0 \\ k_1q_{1eff} = k_1q_{1m} & \text{else} \end{cases} \tag{4.493}$$

$$\begin{cases} k_2q_{2eff} = k_2q_{2m} - \frac{qm_{fact2,ac}-1}{qm_{fact2,ac}} \cdot q_{i2m,ac} & \text{if } \mathbf{QMC} > 0 \\ k_2q_{2eff} = k_2q_{2m} & \text{else} \end{cases} \tag{4.494}$$

4.9.2 Intrinsic charge model

$$Q_G = C'_{Si,ac} \cdot f_{area} \cdot \left(k_1 q_{1eff} \cdot ratio_{pd,ac} + \frac{\Delta_{k1q1,ac}}{3} \cdot P_{1,ac} \right) \tag{4.495}$$

$$Q_B = C'_{Si,ac} \cdot f_{area} \cdot \left(k_2 q_{2eff} + \frac{\Delta_{k2q2,ac}}{3} \cdot P_{2,ac} \right) \tag{4.496}$$

$$Q_D = -\frac{1}{2} \cdot C'_{Si,ac} \cdot f_{area} \cdot \left(k_1 q_{1eff} \cdot ratio_{pd,ac} + \frac{\Delta_{k1q1,ac}}{3} \cdot \left(1 + P_{1,ac} - \frac{{P_{1,ac}}^2}{5} \right) + k_2 q_{2eff} + \frac{\Delta_{k2q2,ac}}{3} \cdot \left(1 + P_{2,ac} - \frac{{P_{2,ac}}^2}{5} \right) \right) \tag{4.497}$$

4.9.3 Parasitic charges

Inner fringe charges, computed if **FIF** > 0

$$x_{effS} = x_{driftS,ac} + x_{th,1D} + 2 \cdot \ln(2) \tag{4.498}$$

$$x_{effD} = x_{driftD,ac} + x_{th,1D} + 2 \cdot \ln(2) \tag{4.499}$$

$$x_S^* = \text{MIN\_FUNC}(x_{effS}, x_{th,1D}, 9.0) \tag{4.500}$$

$$x_D^* = \text{MIN\_FUNC}(x_{effD}, x_{th,1D} + x_D, 9.0) \tag{4.501}$$

$$\lambda_f = \lambda_{2D} \cdot \sqrt{k_{eq,ac} \cdot \left(\frac{1}{2} + \frac{1}{k_{2,ac}} \right)} \tag{4.502}$$

$$\lambda_b = \lambda_{2D} \cdot \sqrt{k_{eq,ac} \cdot \frac{k_{1,ac}}{k_{2,ac}} \cdot \left(\frac{1}{2} + \frac{1}{k_{1,ac}} \right)} \tag{4.503}$$

$$x_{\alpha f} = {\lambda_f}^2 \cdot f_{SD,inner} \tag{4.504}$$

$$x_{\alpha b} = {\lambda_b}^2 \cdot f_{SD,inner} \tag{4.505}$$

$$x_{edge,fS} = x_S^* + 2 \cdot x_{\alpha f} \cdot \left(\sqrt{1 + \frac{x_{SD} - x_S^*}{x_{\alpha f}}} - 1 \right) \tag{4.506}$$

$$x_{edge,fD} = x_D^* + 2 \cdot x_{\alpha f} \cdot \left(\sqrt{1 + \frac{x_{SD} + x_D - x_D^*}{x_{\alpha f}}} - 1 \right) \tag{4.507}$$

$$x_{edge,bS} = x_S^* + 2 \cdot x_{\alpha b} \cdot \left(\sqrt{1 + \frac{x_{SD} - x_S^*}{x_{\alpha b}}} - 1 \right) \tag{4.508}$$

$$x_{edge,bD} = x_D^* + 2 \cdot x_{\alpha b} \cdot \left(\sqrt{1 + \frac{x_{SD} + x_D - x_D^*}{x_{\alpha b}}} - 1 \right) \tag{4.509}$$

$$Q_{GS,if} = -f_{if} \cdot C'_{Si,ac} \cdot \lambda_f \cdot k_{1,ac} \cdot c_{sce1,ac} \cdot \frac{\text{MAX\_FUNC}(x_{edge,fS} - x_{effS}, 0.0, 1.0)^2}{x_{edge,fS} - x_S^*} \tag{4.510}$$

$$Q_{GD,if} = -f_{if} \cdot C'_{Si,ac} \cdot \lambda_f \cdot k_{1,ac} \cdot c_{sce1,ac} \cdot \frac{\text{MAX\_FUNC}(x_{edge,fD} - x_{effD}, 0.0, 1.0)^2}{x_{edge,fD} - x^*_D} \tag{4.511}$$

$$Q_{BS,if} = -f_{if} \cdot C'_{Si,ac} \cdot \lambda_b \cdot k_{2,ac} \cdot c_{sce2,ac} \cdot \frac{\text{MAX\_FUNC}(x_{edge,bS} - x_{effS}, 0.0, 1.0)^2}{x_{edge,bS} - x^*_S} \tag{4.512}$$

$$Q_{BD,if} = -f_{if} \cdot C'_{Si,ac} \cdot \lambda_b \cdot k_{2,ac} \cdot c_{sce2,ac} \cdot \frac{\text{MAX\_FUNC}(x_{edge,bD} - x_{effD}, 0.0, 1.0)^2}{x_{edge,bD} - x^*_D} \tag{4.513}$$

Outer fringe and overlap charges

$$Q_{GS} = \mathbf{CFR} \cdot V_{GS} \tag{4.514}$$

$$Q_{GD} = \mathbf{CFRD} \cdot V_{GD} \tag{4.515}$$

$$Q_{ovS} = \mathbf{COV} \cdot V_{ovS,cv} \cdot \text{MAX\_FUNC}(1 - \mathbf{COVDL} \cdot \delta l_{eff,ac} \cdot (1 - \mathbf{COVDLB} \cdot x_{g20shift,ac}), 0, 0.01) \tag{4.516}$$

$$Q_{ovD} = \mathbf{COVD} \cdot V_{ovD,cv} \cdot \text{MAX\_FUNC}(1 - \mathbf{COVDL} \cdot \delta l_{eff,ac}(1 - \mathbf{COVDLB} \cdot x_{g20shift,ac}), 0, 0.01) \tag{4.517}$$

$$Q_{GB} = \mathbf{CGBOV} \cdot V_{GB} \tag{4.518}$$

Drain to source direct coupling

$$Q_{DS} = \mathbf{CSD} \cdot V_{DS} \tag{4.519}$$

Substrate extrinsic charge model

$$Q_{BS} = -\left(\frac{\epsilon_{ox}}{\mathbf{TBOX}} \cdot A_{source,f} + \mathbf{CSDBP} \cdot P_{source,f}\right) \cdot V_{SB} \tag{4.520}$$

$$Q_{BD} = -\left(\frac{\epsilon_{ox}}{\mathbf{TBOX}} \cdot A_{drain,f} + \mathbf{CSDBP} \cdot P_{drain,f}\right) \cdot V_{DB} \tag{4.521}$$

4.10 Self-heating

As in Leti-UTSOI1, a temperature node named “Tnode” is used to compute the channel temperature elevation induced by the self-heating effect.

Note that, from version Leti-UTSOI 2.1.0, this node is accessible.

While this node voltage represented one hundredth of the temperature elevation in Leti-UTSOI 2.0.0, it corresponds, from Leti-UTSOI 2.1.0, to the actual temperature elevation (i.e. there is no more 1/100 factor).

This node is linked to the ground node through a simple parallel RC circuit, as illustrated in Figure 4.1.

$$\Delta T_C = \text{Temp(Tnode)} \tag{4.522}$$

$$I_{th} = \frac{\Delta T_C}{R_{th}} - I_{DS} \cdot V_{DS} \tag{4.523}$$

$$Q_{th} = \mathbf{CTH} \cdot \Delta T_C \tag{4.524}$$

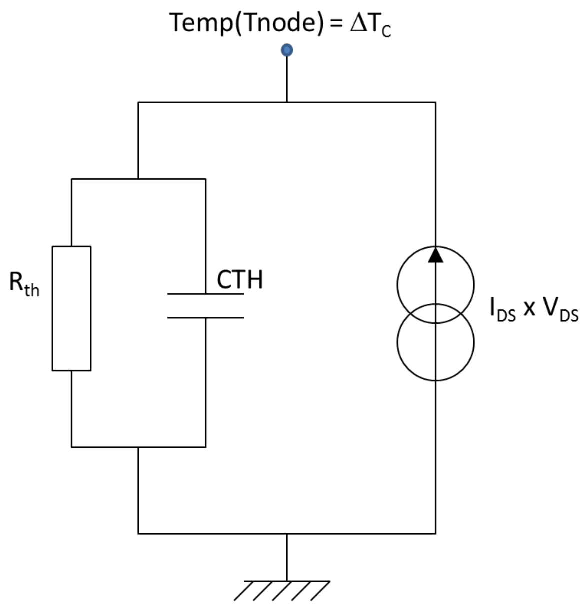

Figure 4.1: Description of the self-heating equivalent circuit used to compute channel temperature elevation.

4.11 Noise model

In this section are described the noise sources and the calculation of their power spectral densities. The way thermal induced gate noise is implemented in the VerilogA code is also indicated.

4.11.1 Channel thermal noise

$$d_m = -\frac{d_{S,dc} + d_{D,dc}}{2} \tag{4.525}$$

$$q_{im}^* = q_{im,dc} + d_m \tag{4.526}$$

$$t_1 = \frac{q_{im,dc}}{q_{im}^*} \tag{4.527}$$

$$t_2 = \left(-\frac{\Delta_{k1q1,dc}}{6 \cdot k_1 h_{10,dc}}\right)^2 \tag{4.528}$$

$$R = \frac{1 + 3 \cdot z_{sat,dc}/2}{\sqrt{1 + z_{sat,dc}}} - 1 \tag{4.529}$$

$$l_C = 1 - 12 \cdot t_2 \cdot R \tag{4.530}$$

$$g_{ideal} = \beta_{Neff} \cdot C'_{Si,dc} \cdot \phi_T \cdot q_{im}^* \cdot \frac{F_{\Delta L}}{G_{vsat} \cdot qm_{fact}} \tag{4.531}$$

$$\begin{cases} S_{ids,exc} = f_{exc} \cdot \frac{I_{DS} \cdot x_{Deff,dc} \cdot \phi_{T0}}{l_C{}^2 \cdot (1 + G_{\gamma,dc}{}^2 / G_{mob,dc}{}^2)} & \text{if } \mathbf{FNTEXC} > 0 \\ S_{ids,exc} = 0 & \text{else} \end{cases} \tag{4.532}$$

$$g_{Sid} = \frac{g_{ideal}}{l_C{}^2} \cdot (t_1 + 12 \cdot t_2 - 24 \cdot (1 + t_1) \cdot t_2 \cdot R) + \frac{S_{ids,exc}}{n_{T0}} \tag{4.533}$$

$$S_{ids,th} = \mathbf{MULT\_I} \cdot n_T \cdot g_{Sid} \tag{4.534}$$

4.11.2 Induced gate noise

Induced gate noise and its correlation with drain thermal noise are computed only when **SWIGN** flag is set to 1.

$$C_{Geff} = (1 + z_{sat,ac}) \cdot \frac{k_{1,ac} \cdot C'_{Si,ac}}{qm_{fact1,ac}} \mathbf{AREAQ} \tag{4.535}$$

$$g_{Sig} = \frac{g_{ideal} \cdot {l_C}^2}{t_1/12 - (1/5 + t_1 - 12 \cdot t_2) \cdot t_2 - 8/5 \cdot (1 + t_1 - 12 \cdot t_2) \cdot t_2 \cdot R} \tag{4.536}$$

$$S_{ig,th} = n_T \cdot g_{Sig} \cdot \frac{(2 \cdot \pi \cdot f_{op} \cdot C_{Geff}/g_{Sig})^2}{1 + (2 \cdot \pi \cdot f_{op} \cdot C_{Geff}/g_{Sig})^2} \tag{4.537}$$

The gate induced noise current is finally partitioned between the gate-source and the gate-drain branches, with a V_{DS} dependent fraction equal to $1/2 + \sqrt{t_2}/4$ and $1/2 - \sqrt{t_2}/4$, respectively. The noise currents in these two branches are thus perfectly correlated and correspond to the following spectral densities:

$$S_{igs,th} = \left(\frac{1}{2} + \frac{\sqrt{t_2}}{4}\right)^2 \cdot S_{ig,th} \tag{4.538}$$

$$S_{igd,th} = \left(\frac{1}{2} - \frac{\sqrt{t_2}}{4}\right)^2 \cdot S_{ig,th} \tag{4.539}$$

4.11.3 Drain and gate thermal noise correlation

$$m_{igid} = \frac{\sqrt{t_2}}{{l_C}^2} (1 - 12 \cdot t_2 - (t_1 + 96 \cdot t_2/5 - 12 \cdot t_1 \cdot t_2) \cdot t_2 \cdot R) \tag{4.540}$$

$$S_{igid,th} = n_T \cdot \frac{j \cdot 2 \cdot \pi \cdot f_{op} \cdot C_{Geff} \cdot m_{igid}}{1 + j \cdot 2 \cdot \pi \cdot f_{op} \cdot C_{Geff}/g_{Sig}} \tag{4.541}$$

4.11.4 VerilogA implementation of induced gate noise and correlation

Since there is no noise function with frequency dependence suitable for the induced gate noise in VerilogA, an internal node NSIG linked to the ground through a parallel RC circuit is used. The equivalent circuit is described in Figure 4.2 and the currents in the different branches are given by:

$$I(\text{NOIR}) = g_{Sig} \cdot V(\text{NSIG}) \tag{4.542}$$

$$I(\text{NOIC}) = \frac{d}{dt} \left(C_{Geff} \cdot V(\text{NSIG})\right) \tag{4.543}$$

$$I(\text{NOII}) = \text{white\_noise}(Mult_f \cdot n_T \cdot g_{Sig}) \tag{4.544}$$

The noise current in the gate-source and gate-drain branches is obtained by:

$$I_{gs,th} = \frac{d}{dt} \left(-\left(\frac{1}{2} + \frac{\sqrt{t_2}}{4}\right) \cdot C_{Geff} \cdot V(\text{NSIG})\right) \tag{4.545}$$

$$I_{gd,th} = \frac{d}{dt} \left(-\left(\frac{1}{2} - \frac{\sqrt{t_2}}{4}\right) \cdot C_{Geff} \cdot V(\text{NSIG})\right) \tag{4.546}$$

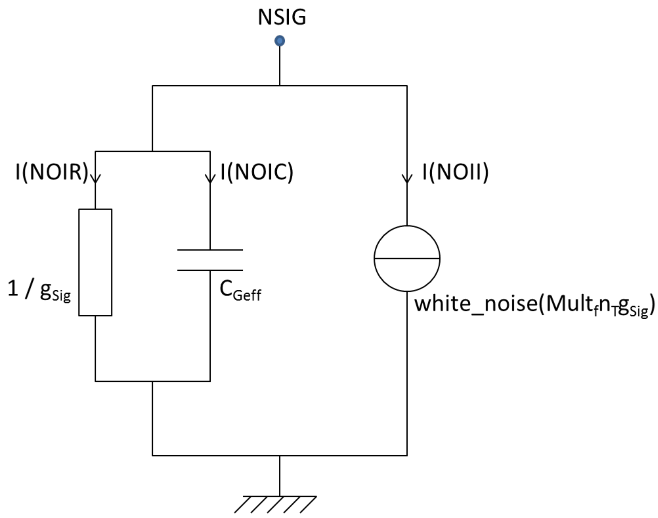

Figure 4.2: Description of the equivalent circuit used to define induced gate noise.

A variable c_{igid} is then defined to obtain the correct correlation between the drain thermal noise current and the gate induced one:

$$c_{igid} = m_{igid} \cdot \sqrt{\frac{g_{Sig}}{g_{Sid}}} \tag{4.547}$$

The channel thermal noise current is finally given as the sum of an uncorrelated part and a correlated one:

$$I_{ds,th} = \text{white\_noise}\left(Mult_f \cdot S_{ids,th} \cdot (1 - {c_{igid}}^2)\right) + m_{igid} \cdot I(\text{NOII}) \tag{4.548}$$

4.11.5 Channel flicker noise

$$N_{unit} = \frac{C'_{Si,dc} \cdot \phi_T}{q} \tag{4.549}$$

$$N^* = N_{unit} \tag{4.550}$$

$$N^*_m = N_{unit} \cdot (q_{im,dc} + 1) \tag{4.551}$$

$$\Delta N = N_{unit} \cdot (q_{iS,dc} - q_{iD,dc}) \tag{4.552}$$

$$f_{NFE} = \text{MAX\_FUNC}\left(1 + \frac{e_{surf1,dc} \cdot \textbf{NFE} + e_{surf2,dc} \cdot \textbf{NFEB}}{q_{im,dc} + 1}, 0.01, 0.0001\right) \tag{4.553}$$

$$S_{ids,fl} = \frac{q \cdot \beta_{Neff} \cdot {\phi_T}^2 \cdot I_{DS}}{{f_{op}}^{\textbf{EF}} \cdot G_{vsat,dc} \cdot N^*} \cdot f_{NFE} \cdot \left(\left(\textbf{NFA} - \textbf{NFB} \cdot N^* + \textbf{NFC} \cdot {N^*}^2\right) \cdot \ln\left(\frac{N^*_m + \Delta N/2}{N^*_m - \Delta N/2}\right) + (\textbf{NFB} + \textbf{NFC} \cdot (N^*_m - 2 \cdot N^*)) \cdot \Delta N\right) \tag{4.554}$$

4.11.6 Shot noises

Gate current shot noise

$$S_{igs,sh} = \textbf{MULT\_I} \cdot 2 \cdot q \cdot |I_{GS}| \tag{4.555}$$

$$S_{igd,sh} = \textbf{MULT\_I} \cdot 2 \cdot q \cdot |I_{GD}| \tag{4.556}$$

GIDL/GISL current shot noise

$$S_{ids,sh} = \textbf{MULT\_I} \cdot 2 \cdot q \cdot |I_{GIDL} - I_{GISL}| \tag{4.557}$$

Impact ionization current shot noise

$$S_{avl} = \textbf{MULT\_I} \cdot 2 \cdot q \cdot (M_{avl} + 1) \cdot |I_{impact}| \tag{4.558}$$

4.11.7 Thermal noise for parasitic resistances

Noise related to gate resistance.

$$S_{\mathrm{R_G}} = \textbf{MULT\_I} \cdot 4 \cdot k_\mathrm{B} \cdot T_\mathrm{KD}/\textbf{RG} \tag{4.559}$$

Noise related to source resistance.

$$S_{\mathrm{R_S}} = \textbf{MULT\_I} \cdot 4 \cdot k_\mathrm{B} \cdot T_\mathrm{KD}/\textbf{RSE} \tag{4.560}$$

Noise related to drain resistance.

$$S_{\mathrm{R_D}} = \textbf{MULT\_I} \cdot 4 \cdot k_\mathrm{B} \cdot T_\mathrm{KD}/\textbf{RDE} \tag{4.561}$$

Noise related to well resistance.

$$S_{\mathrm{R_WELL}} = \textbf{MULT\_I} \cdot 4 \cdot k_\mathrm{B} \cdot T_\mathrm{KD}/\textbf{RWELL} \tag{4.562}$$

4.12 Total currents and charges

According to Figure 1.3, the total currents in the branches and the total node charges are obtained as follows.

4.12.1 Static currents

$$I_{DS} = \textbf{MULT\_I} \cdot Mult_f \cdot \textbf{TYPE} \cdot (I_{DS} + I_{GIDL} - I_{GISL} + I_{impact} + I_{DS,edge}) \tag{4.563}$$

$$I_{GS} = \textbf{MULT\_I} \cdot Mult_f \cdot \textbf{TYPE} \cdot I_{GS} \tag{4.564}$$

$$I_{GD} = \textbf{MULT\_I} \cdot Mult_f \cdot \textbf{TYPE} \cdot I_{GD} \tag{4.565}$$

$$I_{BS} = 0 \tag{4.566}$$

$$I_{BD} = 0 \tag{4.567}$$

$$I_{GB} = 0 \tag{4.568}$$

4.12.2 Total charges

$$Q_{G,tot} = \textbf{MULT\_Q} \cdot Mult_f \cdot \textbf{TYPE} \cdot (Q_G + Q_{GS,if} + Q_{GD,if} + Q_{GS} + Q_{GD} + Q_{ovS} + Q_{ovD} + Q_{GB}) \tag{4.569}$$

$$Q_{D,tot} = \textbf{MULT\_Q} \cdot Mult_f \cdot \textbf{TYPE} \cdot (Q_D + Q_{DS} - Q_{GD,if} - Q_{GD} - Q_{ovD} - Q_{BD,if} - Q_{BD}) \tag{4.570}$$

$$Q_{B,tot} = \textbf{MULT\_Q} \cdot Mult_f \cdot \textbf{TYPE} \cdot (Q_B + Q_{BS,if} + Q_{BD,if} + Q_{BS} + Q_{BD} - Q_{GB}) \tag{4.571}$$

$$Q_{S,tot} = -Q_{G,tot} - Q_{D,tot} - Q_{B,tot} \tag{4.572}$$

4.12.3 Dynamic currents in Quasi-Static

$$i_{DS} = \textbf{MULT\_Q} \cdot Mult_f \cdot \textbf{TYPE} \cdot d(Q_D + Q_{DS})/dt \tag{4.573}$$

$$i_{GS} = \textbf{MULT\_Q} \cdot Mult_f \cdot \textbf{TYPE} \cdot d(Q_G + Q_{GS,if} + Q_{GS} + Q_{ovS})/dt \tag{4.574}$$

$$i_{GD} = \textbf{MULT\_Q} \cdot Mult_f \cdot \textbf{TYPE} \cdot d(Q_{GD,if} + Q_{GD} + Q_{ovD})/dt \tag{4.575}$$

$$i_{BS} = \textbf{MULT\_Q} \cdot Mult_f \cdot \textbf{TYPE} \cdot d(Q_B + Q_{BS,if} + Q_{BS})/dt \tag{4.576}$$

$$i_{BD} = \textbf{MULT\_Q} \cdot Mult_f \cdot \textbf{TYPE} \cdot d(Q_{BD,if} + Q_{BD})/dt \tag{4.577}$$

$$i_{GB} = \textbf{MULT\_Q} \cdot Mult_f \cdot \textbf{TYPE} \cdot dQ_{GB}/dt \tag{4.578}$$

4.12.4 Dynamic currents with Non-Quasi-Static model

NQS model

This following model is directly inspired from references [11] and [12] which propose Relaxation-Time-Approximation (RTA) based nonquasi-static. Internal nodes are used to define RTA model as illustrated in Figure 4.3.

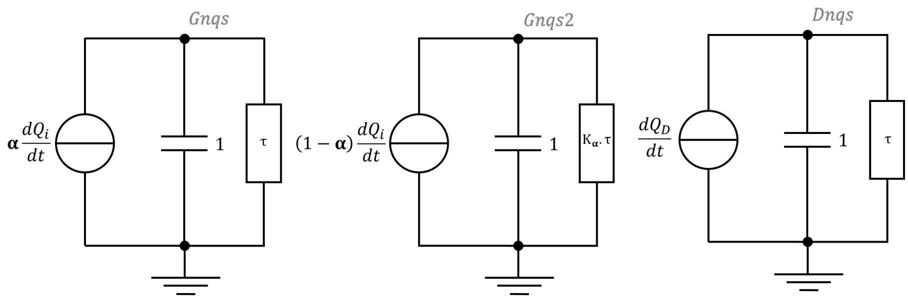

Figure 4.3: Description of the equivalent circuit used to emulate NQS model.

$$Q_{i,wo,mult} = Q_{G,wo,mult} + Q_{B,wo,mult} \tag{4.579}$$

$$\frac{1}{\tau} = \textbf{MULT\_I} \cdot \left(\frac{\beta_{Neff}}{G_{vsat} \cdot \textbf{AREAQ}}\right) \cdot \left(\textbf{KDRIFT} \cdot \frac{Q_{i,wo,mult}}{\textbf{AREAQ} \cdot C'_{ox1}} + \textbf{KDIFF} \cdot \phi_T\right) \tag{4.580}$$

FRACINV parameter is set to 1 by default and use for fine tuning only. In practice, it is the fraction of inversion charge with different cut off frequencies (using **KFRACINV**). The corresponding parameter of Figure 4.3 are:

$$\alpha = \textbf{FRACINV} \tag{4.581}$$

$$K_\alpha = \textbf{KFRACINV} \tag{4.582}$$

Dynamic currents

$$i_{DS} = \textbf{MULT\_Q} \cdot Mult_f \cdot \textbf{TYPE} \cdot \left(d(Q_{DS})/dt - \frac{1}{\tau} \cdot V(Dnqs, gndd)\right) \tag{4.583}$$

If **FRACINV** < 1, a second pole is introduced where the frequency is controlled by KFRACINV.

$$\begin{aligned} i_{GS} = \textbf{MULT\_Q} \cdot Mult_f \cdot \textbf{TYPE} \cdot (d(Q_{GS,if} + Q_{GS} + Q_{ovS} - Q_B)/dt \\ - \frac{1}{\tau} \cdot \left(V(Gnqs, gndg) + \frac{V(Gnqs2, gndg2)}{\textbf{KFRACINV}}\right)\right) \end{aligned} \tag{4.584}$$

Else, with un single pole.

$$i_{GS} = \textbf{MULT\_Q} \cdot Mult_f \cdot \textbf{TYPE} \cdot \left(d(Q_{GS,if} + Q_{GS} + Q_{ovS} - Q_B)/dt - \frac{1}{\tau} \cdot V(Gnqs, gndg) \right) \tag{4.585}$$

$$i_{GD} = \textbf{MULT\_Q} \cdot Mult_f \cdot \textbf{TYPE} \cdot d(Q_{GD,if} + Q_{GD} + Q_{ovD})/dt \tag{4.586}$$

$$i_{BS} = \textbf{MULT\_Q} \cdot Mult_f \cdot \textbf{TYPE} \cdot d(Q_B + Q_{BS,if} + Q_{BS})/dt \tag{4.587}$$

$$i_{BD} = \textbf{MULT\_Q} \cdot Mult_f \cdot \textbf{TYPE} \cdot d(Q_{BD,if} + Q_{BD})/dt \tag{4.588}$$

$$i_{GB} = \textbf{MULT\_Q} \cdot Mult_f \cdot \textbf{TYPE} \cdot dQ_{GB}/dt \tag{4.589}$$

Section 5

Operating Point output

In this section are described the quantities of DC operating point output, defined with **nMOSFET sign and positive V_{DS} convention**. These values give information on the device state at its current operation point.

5.1 Voltages

First, device threshold voltage is calculated. This requires re-computing of several quantities in order to get device state (temperature, effective channel length, effective back gate depletion effect, quantum confinement correction) at gate to source voltage equal to threshold voltage. Therefore, an initial value of threshold voltage is computed from (5.1) to (5.8). Then, required expressions of sections 4.1 to 4.2.4 are computed with T_{KD} instead of T_{KC}, and with $V_{GS} = v_{th}$ as given by (5.8) as input gate to source voltage, to get updated values of k_1, k_2, $diff_{min}$,... Finally, equation sequence (5.1) to (5.8) is re-computed to obtain final threshold voltage value.

$$r_{1,op} = \frac{k_{2,dc}}{k_{1,dc} \cdot (1 + k_{2,dc})} \tag{5.1}$$

$$r_{2,op} = \frac{k_{1,dc}}{k_{2,dc} \cdot (1 + k_{1,dc})} \tag{5.2}$$

$$x_{1sat,op} = \ln\left(k_{1,dc} \cdot (1 + r_{1,op}) \cdot \frac{diff_{min,dc}}{A_{0,dc}}\right) + 2 + \frac{E_g}{2 \cdot \phi_{T0}} \tag{5.3}$$

$$x_{2sat,op} = \ln\left(k_{2,dc} \cdot (1 + r_{2,op}) \cdot \frac{diff_{min,dc}}{A_{0,dc}}\right) + 2 + \frac{E_g}{2 \cdot \phi_{T0}} \tag{5.4}$$

$$x_{th1,op} = x_{1sat,op} \cdot (1 + r_{1,op}) - x_{g2x,dc} \cdot r_{1,op} \tag{5.5}$$

$$x_{th2,op} = x_{2sat,op} \cdot (1 + 1/r_{2,op}) - x_{g2x,dc}/r_{2,op} \tag{5.6}$$

$$x_{g1th,op} = \frac{\text{MIN\_FUNC}(x_{th1,op}, x_{th2,op}, 38) - x_{g2,dc}}{\textbf{CICF}} + x_{g2,dc} \tag{5.7}$$

$$v_{th} = \phi_T \cdot \left(\frac{x_{g1th,op} - x_{edge,dc}}{c_{sce1,dc}} - \delta x_{g1,DIBL,dc} + x_{edge,dc}\right) + V_{FB1} \tag{5.8}$$

If SWQMOD = 1, $v_{th,ac} = v_{th}$ where v_{th} is calculated using same equation with dedicated AC parameters and variables (ie. V_{FB1}, $k_{1,dc}$, $k_{2,dc}$, $A_{0,dc}$, $diff_{min,dc}$, $x_{g2x,dc}$, $x_{g2,dc}$, $x_{edge,dc}$, $c_{sce1,dc}$ and $\delta x_{g1,DIBL,dc}$ are replaced by $V_{FB1,ac}$, $k_{1,dc}$, $k_{2,dc}$, $A_{0,dc}$, $diff_{min,dc}$, $x_{g2x,dc}$, $x_{g2,dc}$, $x_{edge,dc}$, $c_{sce1,dc}$ and $\delta x_{g1,DIBL,dc}$, respectively).

Drain saturation voltage is also calculated:

$$v_{Dsat} = \phi_T \cdot x_{nDS,sat} \tag{5.9}$$

Name	Unit	Description	Value
Vds	V	Internal drain-source DC voltage	V_{DS}
Vsb	V	Internal source-bulk DC voltage	V_{SB}
Vgs	V	Internal gate-source DC voltage	V_{GS}
Vth	V	Threshold voltage	v_{th}
Vthac	V	Threshold voltage of the charge model	$v_{th,ac}$
Vth_drive	V	Effective gate drive voltage	$V_{GS} - v_{th}$
Vdsat	V	Drain saturation voltage at the given bias	v_{Dsat}
Vdsat_marg	V	V_{DS} voltage margin	$V_{DS} - v_{Dsat}$

5.2 Current components

Name	Unit	Description	Value
Id	A	Total DC current flowing into drain terminal	**TYPE** $\cdot (I_{DS} - I_{GD})$
Ig	A	Total DC current flowing into gate terminal	**TYPE** $\cdot (I_{GS} + I_{GD})$
Is	A	Total DC current flowing into source terminal	**TYPE** $\cdot (-I_{DS} - I_{GS})$
Ib	A	Total DC current flowing into bulk terminal	0
Ids	A	DC channel current, excl. tunnel, GISL and GIDL	$Mult_f \cdot I_{DS}$
Igidl	A	DC Gate Induced Drain Leakage current	$Mult_f \cdot I_{GIDL}$
Igisl	A	DC Gate Induced Source Leakage current	$Mult_f \cdot I_{GISL}$
Igs	A	DC gate-source leakage current	$Mult_f \cdot I_{GS}$
Igd	A	DC gate-drain leakage current	$Mult_f \cdot I_{GD}$
Isb	A	DC source-bulk current	0
Idb	A	DC drain-bulk current	0
Iavl	A	Substrate current due to weak avelanche	$Mult_f \cdot I_{impact}$
Idsedge	A	Drain current of edge transistors	$Mult_f \cdot I_{DS,edge}$

5.3 Conductances and transconductances

Here, V_G, V_B and V_D refer to the electrode potentials with the nMOSFET sign convention already applied.

Name	Unit	Description	Value
Gm	S	Internal DC transconductance	$\partial\mathbf{Id}/\partial V_G$
Gmb	S	Internal DC bulk transconductance	$\partial\mathbf{Id}/\partial V_B$

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Name	Unit	Description	Value
Gds	S	Internal DC output conductance	∂**Id**$/\partial V_D$

5.4 Quasi-static Capacitances and transcapacitances

Here, V_G, V_B and V_D refer to the electrode potentials with the nMOSFET sign convention already applied.

Name	Unit	Description	Value
Cgg	F	Internal AC gate capacitance	**TYPE** $\cdot\, \partial Q_{G,tot}/\partial V_G$
Cgd	F	Internal AC gate-drain transcapacitance	$-$**TYPE** $\cdot\, \partial Q_{G,tot}/\partial V_D$
Cgb	F	Internal AC gate-bulk transcapacitance	$-$**TYPE** $\cdot\, \partial Q_{G,tot}/\partial V_B$
Cgs	F	Internal AC gate-source transcapacitance	**Cgg** $-$ **Cgd** $-$ **Cgb**
Cdd	F	Internal AC drain capacitance	**TYPE** $\cdot\, \partial Q_{D,tot}/\partial V_D$
Cdg	F	Internal AC drain-gate transcapacitance	$-$**TYPE** $\cdot\, \partial Q_{D,tot}/\partial V_G$
Cdb	F	Internal AC drain-bulk transcapacitance	$-$**TYPE** $\cdot\, \partial Q_{D,tot}/\partial V_B$
Cds	F	Internal AC drain-source transcapacitance	**Cdd** $-$ **Cdg** $-$ **Cdb**
Cbb	F	Internal AC bulk capacitance	**TYPE** $\cdot\, \partial Q_{B,tot}/\partial V_B$
Cbg	F	Internal AC bulk-gate transcapacitance	$-$**TYPE** $\cdot\, \partial Q_{B,tot}/\partial V_G$
Cbd	F	Internal AC bulk-drain transcapacitance	$-$**TYPE** $\cdot\, \partial Q_{B,tot}/\partial V_D$
Cbs	F	Internal AC bulk-source transcapacitance	**Cbb** $-$ **Cbg** $-$ **Cbd**
Csg	F	Internal AC source-gate transcapacitance	**Cgg** $-$ **Cdg** $-$ **Cbg**
Csb	F	Internal AC source-bulk transcapacitance	**Cbb** $-$ **Cgb** $-$ **Cdb**
Csd	F	Internal AC source-drain transcapacitance	**Cdd** $-$ **Cgd** $-$ **Cbd**
Css	F	Internal AC source capacitance	**Csg** $+$ **Csd** $+$ **Csb**

5.5 Miscellaneous

Name	Unit	Description	Value
TYPE	–	MOSFET type	**TYPE**
Tk	K	MOSFET channel temperature	T_{KC}
DTsh	K	Channel temperature increase due to self-heating	ΔT_C
Self_gain	–	Internal self gain	**Gm**$/$**Gds**
Rout	Ω	AC output resistance	$1/$**Gds**
Beff	A/V^2	Gain factor in saturation	$2 \cdot$ **Id**$/$**Vth_drive**^{2}
Ft	Hz	Unity gain frequency	**Gm**$/(2 \cdot \pi \cdot$ **Cgg**$)$
Rgate	Ω	MOSFET gate resistance	**RG**
Rsource	Ω	MOSFET source resistance	**RSE**
Rdrain	Ω	MOSFET drain resistance	**RDE**
Rbwell	Ω	MOSFET well resistance	**RWELL**
Gmoverid	$1/V$	Transconductance over drain current ratio	**Gm**$/$**Id**

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Name	Unit	Description	Value
Vearly	V	Equivalent Early voltage	$\mathbf{Id/Gds}$

Section 6

Parameter extraction

The following tables summarize the main steps of a local parameter extraction sequence, first for a long channel transistor, and then for short channel devices. Note that accounting for self heating requires temperature dependence extraction at each step of the flow.

6.1 Long channel device

Step	Physical effect	Curves	Parameters
1	Self heating	DC self heating measurement	**RTH**
2	Process	Cgc(Vg) at various Vb	**TOXE, VFB, (TSI)**
3	Process	Cgb(Vg) in subthreshold regime	**TBOX, NSUB, (CGBOV)**
4	Process	Log(Id)(Vg) at low Vd and various Vb	**VFB, CT, (NSUB)**
5	Process	Log(Id)(Vg) at low Vd and various temp.	**STVFB**
6	Mobility	Id(Vg), Gm(Vg), Gm2(Vg) at low Vd and Vb=0V	**BETN, CS, THECS, MUE, THEMU**
7	Mobility	Id(Vg), Gm(Vg), Gm2(Vg) at low Vd and various Vb	**BETN, CS, THECS, MUE, THEMU, XCOR, BETNB**
8	Mobility	Id(Vg), Gm(Vg), Gm2(Vg) at low Vd and various temperatures	**STBET, STCS, STTHECS, STMUE, STTHEMU, STXCOR**
9	Saturation velocity	Id(Vd), Log(Gd)(Vd) at high Vg	**AX, THESAT**
10	Gate current	Ig(Vg)	**IGINV, IGOVINV, IGOVACC, GC2CH, GC3CH, GC2OV, GC3OV, CHIB**
11	Gate current	Ig(Vg) at various temperatures	**STIG**
12	GIDL	Log(Id)(Vg) at high Vd and various Vb	**AGIDL, BGIDL, CGIDL**
13	GIDL	Log(Id)(Vg) at high Vd and various temp.	**STBGIDL**

6.2 Short channel device

Step	Physical effect	Curves	Parameters
1	Self heating	DC self heating measurement	**RTH**
2	Process	Log(Id)(Vg) at low Vd and various Vb	**VFB, VFBB, PSCE, PSCEB**
3	Process	Log(Id)(Vg) at low Vd and various temp.	**STVFB**
4	Mobility, Rseries	Id(Vg), Gm(Vg), Gm2(Vg) at low Vd and Vb=0V	**BETN, CS, RS, RSG, THERSG**
5	Mobility, Rseries	Id(Vg), Gm(Vg), Gm2(Vg) at low Vd and various Vb	**BETN, CS, RS, RSG, THERSG, XCOR**
6	Mobility, Rseries	Id(Vg), Gm(Vg), Gm2(Vg) at low Vd and various temperatures	**STBET, STRS, STXCOR**
7	DIBL	Log(Id)(Vg) at high Vd and various Vb	**CF, CFB**
8	DIBL	Log(Id)(Vg) at high Vd and various temp.	**STCF**
9	DIBL	Log(Id)(Vd), Log(Gd)(Vd) at low Vg	**CF, CFD**
10	Sat. velocity, CLM	Id(Vd), Log(Gd)(Vd) at various Vg, Gm(Vg) at various Vd	**AX, ALP, ALP1, VP, THESAT, THESATG**
11	Sat. velocity, CLM	Id(Vd) at high Vg and various Vb	**THESAT, THESATB**
12	Sat. velocity, CLM	Id(Vd), Log(Gd)(Vd) at high Vg and various temperatures	**STTHESAT**
13	Capacitance	Cgb(Vg)	**CGBOV**
14	Capacitance	Cgc(Vg)	**AREAQ, COV, NOV, CFR**
15	Gate current	Ig(Vg)	**IGINV, IGOVINV, IGOVACC**
16	GIDL	Log(Id)(Vg) at high Vd	**AGIDL**

An accurate charge model requires 2-ports RF-measurements. The accuracy can be obtained by using **SWQMOD** = 1. In this case, the model parameters **VFBAC**, **VFBBAC**, **PSCEAC**, **CFAC**, **THESATAC**, **AXAC** and **ALPAC** can be extracted from S-parameters.

Section 7

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Section 8

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Section 9

Model history

This section details all modifications brought to the code from Leti-UTSOI2.0.0 (former model name version) to L-UTSOI102.4.

9.1 Leti-UTSOI2.0.0 to Leti-UTSOI2.1.0

Important change: Channel temperature elevation node is now accessible from the circuit netlist and can be declared in transistor instantiation. Thus, with Leti-UTSOI2.1.0, transistor instantiation accepts 5 nodes (D, G, S, B, Tnode), while 4 nodes were available with previous versions.

9.1.1 Bug fixes

- ***BF1***: Modification of **CSD** default value for consistency with default global parameter values.
- ***BF2***: Correction of a bug concerning temperature dependence of DIBL. **CF** scaling law is now applied to **STCF**, which suppresses apparition of undesirable DIBL on long channel devices at low and/or high temperature when **STCF** is non-null. Hard clamp previously applied to **CF** has been removed to avoid discontinuities on derivatives that could appear with some parameter configurations and activated self-heating.
- ***BF3***: Correction of a bug concerning the conditionnal clipping of **SWSHE** flag. In previous version, condition was defined for local scale mode only.
- ***BF4***: Correction of **CFB** clipping. In previous version, clipping was done on **CF** $\times$ **CFB**, which could lead to uncorrect clipping in some cases (**CF** < 0 and **CFB** < 0).
- ***BF5***: Correction of a bug concerning the recalculation of DIBL related variable “xd0” when self-heating is activated. Line was missing.
- ***BF6***: Modification of **CIC** and **CICF** implementation to avoid negative transcapacitances that could be observed for **CIC** and/or **CICF** different from 1.
- ***BF7***: Improvement of numerical robustness in the subthreshold regime: corrects a bug in thermal noise calculation and avoids divisions by zero that could occur in some extreme bias/temperature conditions.
- ***BF8***: Adding of a protection in the calculation of effective VDS.
- ***BF9***: Correction of GIDL/GISL component output.
- ***BF10***: Modification of the operating point section to account properly for source/drain interchange.

9.1.2 Accuracy and predictability improvements

- ***AP1***: The effect of effective channel length dependence on front and back gate biases has been introduced in the 2D electrostatic part of the model (subthreshold slope, DIBL) through local parameters **PSCED**L and **CFDL**, and in overlap capacitance model through parameter **COVDL**.
- ***AP2***: The impact of narrow channel effect on front/back gate to interface couplings has been introduced through parameter **PNCE**.
- ***AP3***: To improve the description of gate to source/drain overlap tunnelling currents, **GC2OV** and **GC3OV** parameters have been split into **GC2OVINV**, **GC3OVINV**, **GC2OVACC** and **GC3OVACC**.
- ***AP4***: Gate to channel tunnelling current model has been improved for a better description in the subthreshold regime, with introduction of local parameter **NIGINV**.
- ***AP5***: Predictability of the 2D electrostatic part of the model has been improved, by linking subthreshold slope and DIBL related parameters (**PSCE**, **CF**) to process parameters (**TSI**, **TOXE**, **TBOX**) in scaling laws. In addition, introduction of subthreshold slope degradation parameter in model equations has been modified, so that threshold voltage roll-off predictability is also improved.
- ***AP6***: Scaling law of **BETN** has been changed for better accuracy. PSP-like scaling law has been adopted.
- ***AP7***: Introduction of narrow-short channel effect in **VFB** scaling law has been modified for better description.
- ***AP8***: The impact of 2D electrostatic effect on back gate depletion has been introduced through calculation of an effective back gate doping level.
- ***AP9***: Adding of a flat-band voltage adjustment parameter for overlap capacitances (**DVFBOV**).

9.1.3 Changes in model inputs and outputs

- ***IO1***: Warnings about clipping of instance, global and local parameters, as well as channel temperature elevation, have been added. Display is controlled through **SWCLIPCHK** flag.
- ***IO2***: Device temperature node (Tnode) was internal in previous version, and was corresponding to 1/100th of channel temperature elevation due to self-heating. This node is now accessible (declared as “inout”) and represents the actual channel temperature elevation. Device instantiation requires now 5 nodes.
- ***IO3***: Modification of some parameter clipping: **MULT** min value from 1 to 0, **PSCE** min value from -0.5 to 0 and **AX** max value from 12 to 16. Change of **RTHL**, **RTHW** and **RTHLW** default values.
- ***IO4***: Update of description and notation of operating point outputs. Adding of MOSFET type and device temperature as new outputs.

9.2 Leti-UTSOI2.1.0 to Leti-UTSOI2.1.1

9.2.1 Bug fixes

- ***BF1***: Modification of smoothing in drain saturation voltage calculation to improve smoothness of third order derivatives.
- ***BF2***: Modification of smoothing in gate current model to improve smoothness of third order derivatives.

9.2.2 Accuracy and predictability improvements

- ***AP1***: Improvement of threshold voltage calculation in Operating Point section.

9.3 Leti-UTSOI2.1.1 to Leti-UTSOI2.2.0

9.3.1 Bug fixes

- ***BF1***: Parameters **DLQ** and **DWQ** were involved in device effective length and width, respectively, for computation of external parasitic capacitances, such as external fringe capacitance or gate to substrate one. This is no longer the case, and these capacitances are now calculated with the physical length and width of the device.
- ***BF2***: In the scaling law of velocity saturation parameter **THESAT**, **THESATO** and **THESATL** were not treated similarly. This has been modified for more consistent description of this scaling law.

9.3.2 Accuracy and predictability improvements

- ***AP1***: Additional flexibility has been introduced in channel length modulation model for more accurate description of current and conductances over applied biases.
- ***AP2***: Model has been extended to the case of doped thin film transistors, with introduction of channel doping level local parameter **NCH**.
- ***AP3***: Channel length dependence over front and back bias in subthreshold and moderate inversion regimes has been introduced in Leti-UTSOI 2.1. This part of the model has been improved for better accuracy, with a physical description of source/drain depletion effect.
- ***AP4***: Mobility model accuracy has been improved, in particular with the introduction of transverse field dependence of Coulomb scattering component.
- ***AP5***: Series resistance model flexibility has been increased, with introduction of source/drain extension component and explicit back bias dependence.
- ***AP6***: Gate-overlap tunnelling current description at high drain voltage has been improved.
- ***AP7***: Introduction high longitudinal field dependence of GISL/GIDL currents.
- ***AP8***: Introduction of a physical inner fringe capacitance model, including source/drain depletion effect (see AP3), for better description of short channel capacitances.
- ***AP9***: Overlap capacitance model flexibility has been increased to account for source/drain depletion effect introduced in DC model (see AP3).
- ***AP10***: Global scale parameters **VFBL2** and **VFBLEXP2** have been added for more flexibility in **VFB** scaling description.
- ***AP11***: Global scale parameters **ALPLEXP2** and **ALP1LEXP2** have been added for more flexibility in **ALP** and **ALP1** scaling description, respectively.
- ***AP12***: Description of edge effect in gate to substrate parasitic capacitance has been introduced through global scale parameter **CGBOVO**.
- ***AP13***: Geometrical dependences of thermal resistance and thermal capacitance have been extended to multifinger transistors, by accounting for thermal coupling between fingers.
- ***AP14***: A new strain relaxation model, dedicated to strained-SOI technologies, has been introduced besides existing STI-based stress model.
- ***AP15***: High order derivability of the model around null source-drain voltage has been improved.

9.3.3 Changes in model inputs and outputs

- ***IO1***: Warnings about clipping of instance, global and local parameters, as well as channel temperature elevation, have been updated. Display is still controlled through **SWCLIPCHK** flag.

9.4 Leti-UTSOI2.2.0 to Leti-UTSOI2.3.0

9.4.1 Bug fixes

- ***BF1***: Correction of an undesired W-scaling of Cbs and Cbd capacitances that was induced in Leti-UTSOI 2.2 for negative values of the narrow channel parameter **PNCEW**.
- ***BF2***: Correction of a small undesired scaling of the flicker noise in the subthreshold regime as a function of gate length and back bias.

9.4.2 Accuracy and predictability improvements

- ***AP1***: Simplification of the implementation of the self-heating.
- ***AP2***: Introduction of a model for the excess noise.
- ***AP3***: Global scale parameters **AXL2** and **AXLEXP2** added for more flexibility in **AX** scaling description.
- ***AP4***: Global scale parameters **AGIDLO** and **DGIDLO** added for more scaling flexibility of the GIDL model.

9.4.3 Changes in model inputs and outputs

- ***IO1***: Warnings about clipping of instance, global and local parameters, as well as channel temperature elevation, has been updated. Display is still controlled through **SWCLIPCHK** flag.

9.5 Leti-UTSOI2.3.0 to L-UTSOI102.4

9.5.1 Accuracy and predictability improvements

- ***AP1***: Introduction of impact ionization model inspired from PSP model (introduction of **SWIMPACT** flag). **A1**, **A2**, **STA2** and **A3** are the related model parameters.
- ***AP2***: Introduction of extra gate to overlap current pre-factor in inversion. **FNOVINV**, **FNOVINVD**, **GCOVINVFN** and **STIGFN** are the related model parameters.
- ***AP3***: Modification of scaling law for noise model **NFA** parameter.

9.5.2 Changes in model inputs and outputs

- ***IO1***: Warnings about clipping of instance (corresponding to SWCLIPCHK), global and local parameters, as well as channel temperature elevation, have been removed.
- ***IO2***: Parameter and .OP declarations have been aligned on CMC declaration by using IPRxx/MPRxx/OPPxx macros.
- ***IO3***: Noise source naming has been updated.
- ***IO4***: GMIN is now obtained from the simulator, following the GMIN subcommittee's recommendation.

9.6 L-UTSOI102.4 to L-UTSOI102.5

9.6.1 Bug fixes

- ***BF1***: Improvement of Cgb-Cbg in strong inversion to avoid negative values.
- ***BF2***: Correction of possible division by zero for TMAX parameter.
- ***BF3***: Correction of Flicker noise implementation.
- ***BF4***: Correction of "q zero" variable conditioning.

9.6.2 Accuracy and predictability improvements

- ***AP1***: Introduction of Gate Poly-Depletion model (introduction of **SWPDEP** flag). **NP**, **NPO** and **NPL** are the related model parameters (in local and global mode).

9.6.3 Changes in model inputs and outputs

- ***IO1***: Device temperature offset **DTEMP** has been added.
- ***IO2***: **TREF** has been added and can be used instead of **TR**.

9.7 L-UTSOI102.5 to L-UTSOI102.6

9.7.1 Bug fixes

- ***BF1***: OP-variables: change OPP functions by OPM or OPD ones.

9.7.2 Accuracy and predictability improvements

- ***AP1***: Introduction of RTA NQS model. **KDRIFT**, **KDRIFTO**, **KDRIFTL**, **KDIFF**, **KDIFFO**, **KDIFFL**, **FRACINV**, **FRACINVO**, **KFRACINV** and **KFRACINVO** are the related model parameters.

9.7.3 Changes in model inputs and outputs

- ***IO1***: 4 nodes declaration has been removed and replaced by port connected Verilog-A function.

9.7.4 Comment about this version

- ***C1***: As is done in PSP for NQS, the code has been split in 2 to avoid any confusion on node collapsing.

9.8 L-UTSOI102.6 to L-UTSOI102.7

9.8.1 Bug fixes

- ***BF1***: Noise calculation: missing Mult in impact ionization noise calculation.

9.8.2 Accuracy and predictability improvements

- ***AP1***: Potential reference shift from the source intrinsic energy level to the source conduction band energy level. This shift allows simulations down to 1K.
- ***AP2***: Introduction of the switch **SWCRYO** to activate low temperature effect model and low temperature accuracy improvement.
- ***AP3***: Introduction of low temperature subthreshold slope saturation model. **TMIN**, **ATMIN** and **BTMIN** are the related model parameters.
- ***AP4***: Extra global error correction calculation for low temperature simulation.

9.8.3 Changes in model inputs and outputs

- ***IO1***: Substrate current due to weak avalanche has been added as OP-variables.

9.9 L-UTSOI102.7 to L-UTSOI102.8

9.9.1 Bug fixes

- ***BF1***: IV discontinuity issue: d_D has been revised to avoid any channel current discontinuity.
- ***BF2***: NQS calculation: missing NF in τ calculation.
- ***BF3***: NQS calculation: Add scale factor in calculation of each quantity of equivalent NQS-RTA circuit which changes capacitors from 1F to 1nF. The second pole implementation has been updated to reduce large stamping values also.
- ***BF4***: Add parenthesis in $x_{DS} - x_{Deff,dc}$ of Gate-channel component of gate current calculation. Parentheses on macro in all mathematical functions has been updated.
- ***BF5***: **FACTUO** bug fix has been corrected.

9.9.2 Accuracy and predictability improvements

- ***AP1***: Introduction of edge transistor (introduction of **SWEDGE** flag). **CTEDGE**, **VFBEDGE**, **VFBBEDGE**, **STVFBEDGE**, **CICFEDGE**, **CICEDGE**, **PSCEEDGE**, **PSCEBEDGE**, **CFEDGE**, **CFBEDGE**, **CFDEDGE**, **BETNEDGE** and **STBETEDGE** are the related local model parameters. **WEDGE**, **WEDGEW**, **CTEDGEO**, **VFBEDGEO**, **VFBEDGEL**, **VFBEDGELEXP**, **VFBEDGEW**, **VFBEDGELW**, **VFBBEDGEO**, **STVFBEDGEO**, **STVFBEDGEL**, **STVFBEDGEW**, **STVFBEDGELW**, **CICFEDGEO**, **CICEDGEO**, **PSCEEDGEL**, **PSCEEDGELEXP**, **PSCEEDGEW**, **PSCEBEDGEO**, **CFEDGEL**, **CFEDGELEXP**, **CFEDGEW**, **CFBEDGEO**, **CFDEDGEO**, **FBETEDGE**, **LPEDGE**, **BETEDGEW**, **STBETEDGEO**, **STBETEDGEL**, **STBETEDGEW** and **STBETEDGELW** are the related global model parameters.
- ***AP2***: Introduction of the switch **SWQMOD** which adds Q-model decoupling. **VFBAC**, **VFBBAC**, **PSCEAC**, **CFAC**, **THESATAC**, **AXAC**, **ALPAC** are the related local model parameters. **VFBACL**, **VFBACLEXP**, **VFBACL2**, **VFBACLEXP2**, **VFBACW**, **VFBACLW**, **VFBBACO**, **VFBLBACO**, **PSCEACL**, **PSCEACLEXP**, **PSCEACW**, **CFACL**, **CFACLEXP**, **CFACW**, **THESATACO**, **THESATACL**, **THESATACLEXP**, **THESATACW**, **THESATACLW**, **AXACO**, **AXACL**, **AXACLEXP**, **AXACL2**, **AXACLEXP2**, **ALPACL1**, **ALPACLEXP**, **ALPACL2**, **ALPACLEXP2**, **ALPACW** are the related global model parameters.

9.9.3 Changes in model inputs and outputs

- ***IO1***: threshold voltage $Vthac$ due to charge decoupling has been added as OP-variables.
- ***IO2***: Drain current of edge transistors has been added as OP-variables.

9.10 L-UTSOI102.8 to L-UTSOI102.9

9.10.1 Bug fixes

- ***BF1***: NQS calculation: The second pole implementation has been updated to reduce large stamping values.
- ***BF2***: Update the descriptions of the parameters and the unused macro.

9.10.2 Accuracy and predictability improvements

- ***AP1***: Introduction of gate resistance (introduction of **RG** for local model and **RGO**, **RINT**, **RVPOLY**, **RSHG**, **DLSIL** for global mode.
- ***AP2***: Introduction of source, drain and well resistance (introduction of **RSE**, **RDE** and **RWELL** for local model and **RSH**, **RSHD** and **RWELLO** for global mode).

9.10.3 Changes in model inputs and outputs

- ***IO1***: MOS gate, source, drain and well resistance (intrinsic input resistance) OP-variables has been updated in accordance with introduction of gate resistance.
- ***IO2***: Introduction of **MULT_I**, **MULT_Q** and **MULT_FN**.
- ***IO3***: Introduction of **NGCON**, **XGW**, **NRS** and **NRD**.

Appendix A

Surface potential calculation routines

In this appendix is detailed the calculation of the front interface surface potential for a given normalized quasi-Fermi level δ_n. More precisely, it gives the calculation of the front gate charge density. The first paragraph gives the calculation sequence and the following ones detail the routines used.

A.1 Surface potential calculation sequence

With x_{g1} and x_{g2} the normalized front gate and back gate voltages respectively, δ_n the normalized quasi-Fermi level, and q_1 and q_2 the front gate and back gate charge densities respectively, the function is called as follows:

$$q_1 = \text{CHARGE\_DENSITY}(x_{g1}, x_{g2}, \delta_n) \tag{A.1}$$

The calculation sequence is composed of the following routines:

$$\begin{cases} \text{initial\_guess}(x_{g1}, x_{g2}, \delta_n, q_1, q_2) \\ \text{global\_correction}(x_{g1}, x_{g2}, \delta_n, q_1, \epsilon_2) \\ \text{strong\_inversion\_1st\_correction}(x_{g1}, \delta_n, q_1, q_2) \\ \text{strong\_inversion\_2nd\_correction}(x_{g1}, \delta_n, q_1, q_2) \\ \text{global\_correction}(x_{g1}, x_{g2}, \delta_n, q_1, \epsilon_2) \\ \text{global\_correction}(x_{g1}, x_{g2}, \delta_n, q_1, \epsilon_2) \\ \text{If } \textbf{SWCRYO} = 1 \text{ and } |\epsilon_2| > .01\text{:} \\ \qquad \text{global\_correction}(x_{g1}, x_{g2}, \delta_n, q_1, \epsilon_2) \end{cases} \tag{A.2}$$

At the end of this sequence, the gate charge density q_1 and, thus, its corresponding surface potential (simply given by $x_1 = x_{g1} - q_1$) is known with a very high accuracy.

A.2 Initial guess

Here is calculated the initial guess on the gate charge densities q_1 and q_2 as a function of the gate voltages x_{g1} and x_{g2} and of the quasi-Fermi level δ_n. This computation uses previously defined quantities: k_1, k_2, A_0, $diff_{min}$, $x_{1,WI0}$, $x_{2,WI0}$.

The routine is called as follows: initial_guess$(x_{g1}, x_{g2}, \delta_n, q_1, q_2)$. It returns values for q_1 and q_2.

$$x_{1,sat} = \ln\left(\left(k_1 + \frac{k_2}{1+k_2}\right) \cdot \frac{diff_{min}}{A_0}\right) + \delta_n + 3 \tag{A.3}$$

$$x_{2,sat} = \ln\left(\left(k_2 + \frac{k_1}{1+k_1}\right) \cdot \frac{diff_{min}}{A_0}\right) + \delta_n + 3 \tag{A.4}$$

$$x_{1,0} = x_{1,sat} - 3 \cdot \ln\left(1 + \exp\left(\frac{x_{1,sat} - x_{1,WI0}}{3}\right)\right) \tag{A.5}$$

$$x_{2,0} = x_{2,sat} - 3 \cdot \ln\left(1 + \exp\left(\frac{x_{2,sat} - x_{2,WI0}}{3}\right)\right) \tag{A.6}$$

$$x_{1,WI} = \frac{x_{2,0} + k_1 \cdot x_{g1}}{1 + k_1} \tag{A.7}$$

$$x_{2,WI} = \frac{x_{1,0} + k_2 \cdot x_{g2}}{1 + k_2} \tag{A.8}$$

$$x_1 = x_{1,sat} - 3 \cdot \ln\left(1 + \exp\left(\frac{x_{1,sat} - x_{1,WI}}{3}\right)\right) \tag{A.9}$$

$$x_2 = x_{2,sat} - 3 \cdot \ln\left(1 + \exp\left(\frac{x_{2,sat} - x_{2,WI}}{3}\right)\right) \tag{A.10}$$

$$q_1 = x_{g1} - x_1 \tag{A.11}$$

$$q_2 = x_{g2} - x_2 \tag{A.12}$$

A.3 Global correction

In the following routine, corrections are brought to the front gate charge density q_1 from its initial value.
The routine is called as follows: global_correction($x_{g1}, x_{g2}, \delta_n, q_1, \epsilon_2$). It returns new values for q_1 and ϵ_2.

$$f_{qsq} = {k_1}^2 \cdot {q_1}^2 - A_0 \cdot \exp(x_{g1} - q_1 - \delta_n) \tag{A.13}$$

$$d_{1,qsq} = 2 \cdot {k_1}^2 \cdot q_1 + A_0 \cdot \exp(x_{g1} - q_1 - \delta_n) \tag{A.14}$$

$$d_{2,qsq} = 2 \cdot {k_1}^2 - A_0 \cdot \exp(x_{g1} - q_1 - \delta_n) \tag{A.15}$$

If $f_{qsq} < -0.005$:

$$f_{qct} = \sqrt{|f_{qsq}|} \cdot \cot\left(\frac{\sqrt{|f_{qsq}|}}{2}\right) \tag{A.16}$$

$$d_{1,qct} = \frac{f_{qsq} + f_{qct} \cdot (2 - f_{qct})}{4 f_{qsq}} \cdot d_{1,qsq} \tag{A.17}$$

$$d_{2,qct} = \frac{d_{1,qsq} - 2 \cdot d_{1,qct} \cdot (1 + f_{qct})}{4 \cdot f_{qsq}} \cdot d_{1,qsq} + \frac{d_{1,qct} \cdot d_{2,qsq}}{d_{1,qsq}} \tag{A.18}$$

$$f_{sh} = -\frac{f_{qsq}}{\sin\left(\sqrt{|f_{qsq}|}/2\right)^2} \tag{A.19}$$

$$f_{ln} = \ln(f_{sh}) \tag{A.20}$$

$$d_{1,ln} = \left(1 - \frac{f_{qct}}{2}\right) \cdot \frac{d_{1,qsq}}{f_{qsq}} \tag{A.21}$$

$$d_{2,ln} = \left(1 - \frac{f_{qct}}{2}\right) \cdot \frac{d_{2,qsq}}{f_{qsq}} - \left(d_{1,ln} - \frac{d_{1,qct}}{2}\right) \cdot \frac{d_{1,qsq}}{f_{qsq}} \tag{A.22}$$

Else if $f_{qsq} > 0.005$:

$$\zeta = \exp\left(-\sqrt{|f_{qsq}|}\right) \tag{A.23}$$

$$f_{qct} = \sqrt{|f_{qsq}|} \cdot \frac{1+\zeta}{1-\zeta} \tag{A.24}$$

$$d_{1,qct} = \frac{f_{qsq} + f_{qct} \cdot (2 - f_{qct})}{4 \cdot f_{qsq}} \cdot d_{1,qsq} \tag{A.25}$$

$$d_{2,qct} = \frac{d_{1,qsq} - 2 \cdot d_{1,qct} \cdot (1 + f_{qct})}{4 \cdot f_{qsq}} \cdot d_{1,qsq} + \frac{d_{1,qct} \cdot d_{2,qsq}}{d_{1,qsq}} \tag{A.26}$$

$$f_{sh} = \frac{4 \cdot f_{qsq}}{1 - \zeta \cdot (2 - \zeta)} \cdot \zeta \tag{A.27}$$

$$f_{ln} = \ln\left(\frac{4 \cdot f_{qsq}}{1 - \zeta \cdot (2 - \zeta)}\right) - \sqrt{|f_{qsq}|} \tag{A.28}$$

$$d_{1,ln} = \left(1 - \frac{f_{qct}}{2}\right) \cdot \frac{d_{1,qsq}}{f_{qsq}} \tag{A.29}$$

$$d_{2,ln} = \left(1 - \frac{f_{qct}}{2}\right) \cdot \frac{d_{2,qsq}}{f_{qsq}} - \left(d_{1,ln} - \frac{d_{1,qct}}{2}\right) \cdot \frac{d_{1,qsq}}{f_{qsq}} \tag{A.30}$$

Else:

$$f_{qct} = 2 + \frac{f_{qsq}}{6} \cdot \left(1 - \frac{f_{qsq}}{60} \cdot \left(1 - \frac{f_{qsq}}{42} \cdot \left(1 - \frac{f_{qsq}}{40}\right)\right)\right) \tag{A.31}$$

$$d_{1,qct} = \frac{d_{1,qsq}}{6} \cdot \left(1 - \frac{f_{qsq}}{30} \cdot \left(1 - \frac{f_{qsq}}{28} \cdot \left(1 - \frac{f_{qsq}}{30}\right)\right)\right) \tag{A.32}$$

$$\begin{aligned} d_{2,qct} = {} & \frac{d_{2,qsq}}{6} \cdot \left(1 - \frac{f_{qsq}}{30} \cdot \left(1 - \frac{f_{qsq}}{28} \cdot \left(1 - \frac{f_{qsq}}{30}\right)\right)\right) \\ & - \frac{{d_{1,qsq}}^2}{180} \cdot \left(1 - \frac{f_{qsq}}{14} \cdot \left(1 - \frac{f_{qsq}}{20} \cdot \left(1 - \frac{25 f_{qsq}}{594}\right)\right)\right) \end{aligned} \tag{A.33}$$

$$f_{sh} = 4 - \frac{f_{qsq}}{3} \cdot \left(1 - \frac{f_{qsq}}{20} \cdot \left(1 - \frac{5 f_{qsq}}{126}\right)\right) \tag{A.34}$$

$$f_{ln} = \ln(f_{sh}) \tag{A.35}$$

$$d_{1,ln} = -\frac{d_{1,qsq}}{12} \cdot \left(1 - \frac{f_{qsq}}{60} \cdot \left(1 - \frac{f_{qsq}}{42} \cdot \left(1 - \frac{f_{qsq}}{40}\right)\right)\right) \tag{A.36}$$

$$d_{2,ln} = -\frac{d_{2,qsq}}{12} \cdot \left(1 - \frac{f_{qsq}}{60} \cdot \left(1 - \frac{f_{qsq}}{42} \cdot \left(1 - \frac{f_{qsq}}{40}\right)\right)\right) + \frac{{d_{1,qsq}}^2}{720} \cdot \left(1 - \frac{f_{qsq}}{42} \cdot \left(2 - \frac{3f_{qsq}}{40}\right)\right) \quad (A.37)$$

If $1.01k_1q_1 + f_{qct} > 0$:

$$f_{en} = k_1q_1 + f_{qct} \quad (A.38)$$

$$d_{1,en} = k_1 + d_{1,qct} \quad (A.39)$$

$$d_{2,en} = d_{2,qct} \quad (A.40)$$

Else:

$$f_{en} = \frac{A_0 \cdot \exp(x_{g1} - q_1 - \delta_n) - f_{sh}}{k_1 \cdot q_1 - f_{qct}} \quad (A.41)$$

$$d_{1,en} = \frac{(d_{1,qct} - k_1) \cdot f_{en} - A_0 \cdot \cdot \exp(x_{g1} - q_1 - \delta_n) - d_{1,ln} \cdot f_{sh}}{k_1 \cdot q_1 - f_{qct}} \quad (A.42)$$

$$d_{2,en} = \frac{d_{2,qct} \cdot f_{en} + 2 \cdot (d_{1,qct} - k_1) \cdot d_{1,en} + A_0 \cdot \exp(x_{g1} - q_1 - \delta_n) - (d_{2,ln} + {d_{1,ln}}^2) \cdot f_{sh}}{k_1 \cdot q_1 - f_{qct}} \quad (A.43)$$

If $f_{en} > 0$:

$$f_{len} = \ln(f_{en}) \quad (A.44)$$

$$d_{1,len} = \frac{d_{1,en}}{f_{en}} \quad (A.45)$$

$$d_{2,len} = \frac{d_{2,en}}{f_{en}} - {d_{1,len}}^2 \quad (A.46)$$

Else:

$$f_{len} = k_1 \cdot q_1 + \ln(-2 \cdot k_1 \cdot q_1) \quad (A.47)$$

$$d_{1,len} = k_1 + \frac{1}{q_1} \quad (A.48)$$

$$d_{2,len} = -\frac{1}{{q_1}^2} \quad (A.49)$$

$$q_{2,int} = x_{g2} - x_{g1} + q_1 + 2 \cdot f_{len} - f_{ln} \quad (A.50)$$

$$d_{1,q2} = 1 + 2 \cdot d_{1,len} - d_{1,ln} \quad (A.51)$$

$$d_{2,q2} = 2 \cdot d_{2,len} - d_{2,ln} \quad (A.52)$$

$$q_{i,int} = k_1 \cdot q_1 + k_2 \cdot q_{2,int} \quad (A.53)$$

$$d_{1,qi} = k_1 + k_2 \cdot d_{1,q2} \tag{A.54}$$

$$d_{2,qi} = k_2 \cdot d_{2,q2} \tag{A.55}$$

$$f_{zero} = q_{i,int} \cdot f_{en} - A_0 \cdot \exp(x_{g1} - q_1 - \delta_n) \tag{A.56}$$

$$d_{1,zero} = d_{1,qi} \cdot f_{en} + q_{i,int} \cdot d_{1,en} + A_0 \cdot \exp(x_{g1} - q_1 - \delta_n) \tag{A.57}$$

$$d_{2,zero} = d_{2,qi} \cdot f_{en} + 2 \cdot d_{1,qi} \cdot d_{1,en} + q_{i,int} \cdot d_{2,en} - A_0 \cdot \exp(x_{g1} - q_1 - \delta_n) \tag{A.58}$$

$$d_{temp} = d_{1,zero}^2 - 0.5 \cdot f_{zero} \cdot d_{2,zero} \tag{A.59}$$

$$\epsilon_2 = -\frac{f_{zero} \cdot d_{1,zero} \cdot d_{temp}}{d_{temp}^2 + 10^{-200}} \tag{A.60}$$

$$q_1 = q_1 + \epsilon_2 \tag{A.61}$$

A.4 First correction in strong inversion

The computation detailed here brings correction to q_1 by using the value of q_2 coming from the initial guess. The routine is called as follows: strong_inversion_1st_correction$(x_{g1}, \delta_n, q_1, q_2)$. It returns a new value for q_1.

$$q_{i,int} = k_1 \cdot q_1 + k_2 \cdot q_2 \tag{A.62}$$

$$a = 1 + \left(\frac{1}{6} - \frac{1}{\pi^2}\right) \cdot q_{i,int} \tag{A.63}$$

$$b = 4\pi^2 + k_1 \cdot q_1 \cdot k_2 \cdot q_2 + 2 \cdot \left(1 + \frac{\pi^2}{3}\right) q_{i,int} \tag{A.64}$$

$$c = 4\pi^2 \cdot (2 \cdot q_{i,int} + k_1 \cdot q_1 \cdot k_2 \cdot q_2) \tag{A.65}$$

$$f_{qsq} = \frac{\sqrt{b^2 - 4 \cdot a \cdot c} - b}{2 \cdot a} \tag{A.66}$$

$$\delta = {k_1}^2 \cdot {q_1}^2 - f_{qsq} \tag{A.67}$$

If $\delta > 0$:

$$f_{zero} = (\ln(\delta/A_0) + \delta_n - x_{g1} + q_1) \cdot \delta \tag{A.68}$$

$$d_{1,zero} = 2 \cdot {k_1}^2 \cdot q_1 + \delta \tag{A.69}$$

If ($f_{zero} < 0$ and $d_{1,zero} > 0$ and $x_{g1} - q_1 > x_{1,sat} - \ln(10k_1)$) or ($x_{g1} - q_1 > x_{1,sat} + 1$):

$$q_1 = q_1 - \frac{f_{zero}}{d_{1,zero}} \tag{A.70}$$

A.5 Second correction in strong inversion

The computation detailed here brings a second correction to q_1 in strong inversion by using the value of q_2 coming from the initial guess.

The routine is called as follows: strong_inversion_2nd_correction$(x_{g1}, \delta_n, q_1, q_2)$. It returns a new value for q_1.

$$q_{i,int} = k_1 \cdot q_1 + k_2 \cdot q_2 \tag{A.71}$$

$$a = 1 + \left(\frac{1}{6} - \frac{1}{\pi^2}\right) \cdot q_{i,int} \tag{A.72}$$

$$b = 4 \cdot \pi^2 + k_1 \cdot q_1 \cdot k_2 \cdot q_2 + 2 \cdot \left(1 + \frac{\pi^2}{3}\right) \cdot q_{i,int} \tag{A.73}$$

$$c = 4 \cdot \pi^2 \cdot (2 \cdot q_{i,int} + k_1 \cdot q_1 \cdot k_2 \cdot q_2) \tag{A.74}$$

$$f_{qsq0} = \frac{\sqrt{b^2 - 4 \cdot a \cdot c} - b}{2 \cdot a} \tag{A.75}$$

If $f_{qsq0} < -0.005$:

$$f_{qct} = \sqrt{|f_{qsq0}|} \cdot \cot\left(\frac{\sqrt{|f_{qsq0}|}}{2}\right) \tag{A.76}$$

$$d_{1,qct} = \frac{f_{qsq0} + f_{qct} \cdot (2 - f_{qct})}{4 \cdot f_{qsq0}} \tag{A.77}$$

Else if $f_{qsq0} > 0.005$:

$$\zeta = \exp\left(-\sqrt{|f_{qsq0}|}\right) \tag{A.78}$$

$$f_{qct} = \sqrt{|f_{qsq0}|} \cdot \frac{1 + \zeta}{1 - \zeta} \tag{A.79}$$

$$d_{1,qct} = \frac{f_{qsq0} + f_{qct} \cdot (2 - f_{qct})}{4 \cdot f_{qsq0}} \tag{A.80}$$

Else:

$$f_{qct} = 2 + \frac{f_{qsq0}}{6} \cdot \left(1 - \frac{f_{qsq0}}{60} \cdot \left(1 - \frac{f_{qsq0}}{42} \cdot \left(1 - \frac{f_{qsq}}{40}\right)\right)\right) \tag{A.81}$$

$$d_{1,qct} = \frac{1}{6} \cdot \left(1 - \frac{f_{qsq0}}{30} \cdot \left(1 - \frac{f_{qsq0}}{28} \cdot \left(1 - \frac{f_{qsq0}}{30}\right)\right)\right) \tag{A.82}$$

$$f_{qsq} = f_{qsq0} - \frac{q_{i,int} \cdot f_{qct} + k_1 \cdot q_1 \cdot k_2 \cdot q_2 + f_{qsq0}}{q_{i,int} \cdot d_{1,qct} + 1} \tag{A.83}$$

$$\delta = {k_1}^2 \cdot {q_1}^2 - f_{qsq} \tag{A.84}$$

If $\delta > 0$:

$$f_{zero} = (\ln(\delta / A_0) + \delta_n - x_{g1} + q_1) \cdot \delta \tag{A.85}$$

$$d_{1,zero} = 2 \cdot {k_1}^2 \cdot q_1 + \delta \tag{A.86}$$

If ($f_{zero} < 0$ and $d_{1,zero} > 0$ and $x_{g1} - q_1 > x_{1,sat} - \ln(10 \cdot k_1)$) or ($x_{g1} - q_1 > x_{1,sat} + 1$):

$$q_1 = q_1 - \frac{f_{zero}}{d_{1,zero}} \tag{A.87}$$

Appendix B

Surface potential calculation in overlap regions

The surface potential in gate to source or drain overlap region comes from PSP. It is returned by the function:

$$sp_{ov} = \text{SP\_FDSOI\_OV}(x_{gf,ov}) \tag{B.1}$$

where the argument $x_{gf,ov}$ is the electrostatic potential difference between the electrodes, normalized to ϕ_{T0}, and where some previously calculated variables are used: G_{ov}, ξ_{ov}, $\xi_{mrg,ov}$, $x_{g1,ov}$. The calculation sequence inside the SP_FDSOI_OV function is:

If $|x_{gf,ov}| \leq \xi_{mrg,ov}$:

$$sp_{ov} = -\frac{x_{gf,ov}}{\xi_{ov}} \tag{B.2}$$

Else if $x_{gf,ov} < -\xi_{mrg,ov}$:

$$y_{gf} = -x_{gf,ov} \tag{B.3}$$

$$z = 1.25 \cdot \frac{y_{gf}}{\xi_{ov}} \tag{B.4}$$

$$\eta = \frac{z + 10 - \sqrt{(z-6)^2 + 64}}{2} \tag{B.5}$$

$$a = (y_{gf} - \eta)^2 + {G_{ov}}^2 \cdot (\eta + 1) \tag{B.6}$$

$$c = 2 \cdot (y_{gf} - \eta) - {G_{ov}}^2 \tag{B.7}$$

$$\tau = -\eta + \ln\left(a/{G_{ov}}^2\right) \tag{B.8}$$

$$y_0 = \sigma_3(a, c, \tau, \eta) \tag{B.9}$$

$$\Delta_0 = \exp(y_0) \tag{B.10}$$

$$p = 2 \cdot (y_{gf} - y_0) - {G_{ov}}^2 \cdot (\Delta_0 - 1) \tag{B.11}$$

$$q = (y_{gf} - y_0)^2 - {G_{ov}}^2 \cdot (y_0 - \Delta_0 + 1) \tag{B.12}$$

$$sp_{ov} = y_0 + \frac{2 \cdot q}{p + \sqrt{p^2 - 2 \cdot q \cdot (2 - {G_{ov}}^2 \cdot \Delta_0)}} \tag{B.13}$$

Else:

$$\frac{1}{\widetilde{x}_{g1}} = \frac{1.25 \cdot \xi_{ov}/x_{g1,ov} - 1}{x_{g1,ov}} \tag{B.14}$$

$$\bar{x} = \frac{x_{gf,ov}}{\xi_{ov}} \cdot \left(1 + \frac{x_{gf,ov}}{\widetilde{x}_{g1}}\right) \tag{B.15}$$

$$w = 1 - \exp(-\bar{x}) \tag{B.16}$$

$$x_0 = x_{gf,ov} + \frac{{G_{ov}}^2}{2} - G_{ov} \cdot \sqrt{x_{gf,ov} + \frac{{G_{ov}}^2}{4} - w} \tag{B.17}$$

$$\Delta_0 = \exp(-x_0) \tag{B.18}$$

$$p = 2 \cdot (x_{gf,ov} - x_0) + {G_{ov}}^2 \cdot (1 - \Delta_0) \tag{B.19}$$

$$q = (x_{gf,ov} - x_0)^2 - {G_{ov}}^2 \cdot (x_0 - 1 + \Delta_0) \tag{B.20}$$

$$sp_{ov} = -x_0 - \frac{2 \cdot q}{p + \sqrt{p^2 - 2 \cdot q \cdot (2 - {G_{ov}}^2 \cdot \Delta_0)}} \tag{B.21}$$

Appendix C

Some useful functions

Here are defined some functions used in the model equations.

$$\text{MIN\_FUNC}(x, y, a) = \frac{1}{2} \cdot \left(x + y - \sqrt{(x-y)^2 + a}\right) \tag{C.1}$$

$$\text{MAX\_FUNC}(x, y, a) = \frac{1}{2} \cdot \left(x + y + \sqrt{(x-y)^2 + a}\right) \tag{C.2}$$

Hereunder are defined the sigma functions used for the backplane surface potential calculation:

$$\nu = a + c \tag{C.3}$$

$$\mu_3 = \frac{\nu^2}{\tau} + \frac{c^2}{2} - a \tag{C.4}$$

$$\sigma_3(a, c, \tau, \eta) = \eta + \frac{a\nu}{\mu_3 + c \cdot \left(\frac{c^2}{3} - a\right) \cdot \frac{\nu}{\mu_3}} \tag{C.5}$$

$$\mu_2 = \frac{\nu^2}{\tau} + \frac{c^2}{2} - a \cdot b \tag{C.6}$$

$$\sigma_2(a, b, c, \tau, \eta) = \eta + \frac{a \cdot \nu}{\mu_2 + c \cdot \left(\frac{c^2}{3} - ab\right) \cdot \frac{\nu}{\mu_2}} \tag{C.7}$$

A good analytical approximation of the asymptotic Lambert function is:

$$\mathcal{W}_0(x) \simeq \ln(1+x) \cdot \left[1 - \frac{\ln(1 + \ln(1+x))}{2 + \ln(1+x)}\right] \tag{C.8}$$